

Learning When to Fight: Technological Change and Conflict in the Gunpowder Revolution*

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Abstract

Does technological change make conflict more likely? I argue that new technologies can generate conflict by creating uncertainty about the balance of power, but demonstrations of the effectiveness of new technology in combat subsequently diminish this uncertainty. During the Siege of Constantinople in 1453, gunpowder artillery—then a new and unproven technology—devastated the most sophisticated fortifications in Europe. I examine how this informational shock affected the frequency of siege warfare in Western Europe. Using a difference-in-differences design and an original dataset of siege locations, I find that areas more exposed to information about the Ottomans’ use of cannons at Constantinople experienced fewer sieges in the decade after the city’s capture. Contemporary sources support the claim that the decline in siege warfare was due to a convergence in beliefs about the usefulness of gunpowder artillery.

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1 Introduction

Advances in artificial intelligence, robotics, and other fields have inspired a wave of research aimed at understanding the implications of technological change for international security.¹ The resulting scholarly output has yielded no consensus. Some studies portray emerging technologies as exerting a destabilizing influence on the international system. They claim that new technologies enable rapid shifts in military capabilities at low cost, which can incite arms races, disrupt the balance of power, and erode deterrence (Ellman, Samp and Coll, 2017; Johnson, 2019; Schneider, 2021; Volpe, 2021). Others are more sanguine, arguing that most new technologies have an incremental effect on capabilities that is insufficient to alter existing interstate power dynamics (Biddle, 2004; Rid, 2012). A few conclude that some new technologies limit escalation and promote peace (Kroenig, 2021; Lin-Greenberg, 2022).

I argue that uncertainty about the implications of a new technology is itself a cause of conflict. Disagreements about a new technology's combat effectiveness create contrasting beliefs about the balance of power, magnifying the risk of bargaining failure and disputes escalating into war (Blainey, 1988; Fearon, 1995). However, war can resolve debates about the effectiveness of new military technologies. During a conflict, states witness the combat performance of new technology and converge on the doctrine required to use it effectively (Biddle, 2004; Hoffman, 2015). Wartime demonstrations provide common reference points for future assessments of military capabilities, including for states not directly involved in the conflict (Goldman and Eliason, 2003; Horowitz, 2010). Conflicts are therefore public signals of a new technology's effectiveness that facilitate a convergence in beliefs about the balance of power.

In this article, I test if better information about the combat effectiveness of new military technology makes conflict less likely. To do so, I examine whether the demonstration of gunpowder

¹I use "technology" to refer to physical equipment (i.e., weapons or vehicles) produced via the application of scientific knowledge. The term "technological change" refers to the introduction of such new equipment.

artillery's ability to destroy fortifications at the successful Ottoman siege of Constantinople in 1453 influenced the frequency of siege warfare. Like interstate wars, sieges can be viewed as the products of bargaining failure. Sieges were almost always preceded by negotiations between garrisons and besiegers intended to yield a fortress without a blockade or assault (Mallett, 2009; Purton, 2009). These negotiations were characterized by informational asymmetries: besiegers often did not know how much stored food garrisons held and thus how long they could withstand a siege, while garrisons were unaware of besiegers' resolve to directly assault a fortification (Bradbury, 1992). Such asymmetries often led to bargaining failure and long blockades of well-fortified castles that wooden siege engines were unable to damage.

Siege warfare would eventually be transformed by the introduction of gunpowder artillery in the 14th and 15th centuries, which rendered formerly stalwart fortifications vulnerable (Bradbury, 1992). However, early cannons were crude weapons that were perceived as inferior to older siege engines (DeVries, 2013; Hale, 1983). Indeed, historians have argued that "for more than a century after 1326, catapults continued to surpass anything a gun could do, except when it came to making noise" (McNeill, 2013, 83). Although gunpowder artillery underwent technical improvements in the 1430s that enabled it to batter down walls, this fact was recognized by only a few early adopters, while many continued to believe that existing fortifications remained impregnable (Duffy, 2013; Rogers, 2018). As a result, there were contrasting beliefs about the ability of existing fortifications to withstand a siege by forces equipped with cannons, contributing to bargaining failures that produced sieges.

Attitudes shifted in 1453, when the Siege of Constantinople demonstrated the ability of gunpowder artillery to destroy existing defenses. Employing a large array of bombards, the forces of the Ottoman Empire breached the walls of Constantinople—fortifications that were considered the most sophisticated in Europe—in an accomplishment that historians have characterized as "the most spectacular early achievement" of gunpowder artillery (Ágoston, 2014; Pepper and Adams, 1986, 8). The capture of the city was closely watched by Europeans, and accounts of the siege spread across the continent (Bisaha, 2017; Schwoebel, 1967). These accounts universally empha-

sized the decisive role played by artillery and described the tactics that the Ottomans employed, setting off a period of European-Ottoman military acculturation (Christensen, 1987; Philippides and Hanak, 2011). Europeans who had believed their fortifications remained secure from assault were confronted with evidence that this was not the case. Contemporary sources suggest that after 1453, siege defenders became more inclined to surrender when faced with a besieging army equipped with cannons (Rogers, 2018). Importantly, the era's slow travel times resulted in settlements located further from Constantinople obtaining detailed reports of the siege later than settlements located nearer.² For a short time after the Fall of Constantinople there was geographic variation in awareness of the power of gunpowder artillery, which can be approximated and validated using primary sources (Pertusi, 1976).

The explicit bargaining structure of sieges, high salience of gunpowder artillery, and slow pace of information diffusion make the period around the Fall of Constantinople an unusually clean setting to test whether battlefield demonstrations of new technology reduced bargaining failure. Using a difference-in-differences design, I exploit proximity-driven variation in when different locations learned about the Fall of Constantinople to study how information about artillery's effectiveness influenced decisions to fight in siege warfare. As proximity to Constantinople is primarily determined by geography, this provides plausibly exogenous variation in information about the effectiveness of cannons. Comparing the number of sieges occurring in locations of varying proximity to Constantinople before and after its capture therefore permits an evaluation of the causal effect of information transmission on conflict. I carry out this empirical strategy on a new dataset of 431 sieges that occurred in Western and Central Europe between 1443 and 1463, constituting the most comprehensive account of sieges during this period. I proxy for exposure to information about Constantinople using estimates of transportation costs, which I construct using historical shipping records and a shortest-path algorithm.

I find that in the decade following the Fall of Constantinople, places closer to Constantinople experienced fewer sieges on average. Consistent with a theory of learning, the effect is

²For example, some eyewitnesses to the siege, such as Greek refugees, arrived in Northern Europe years after they had reached Italy or the Balkans (Harris, 2022).

strongest immediately after 1453 and dissipates later on, as information about the siege fully diffused throughout Europe. Using historical evidence and placebo tests, I rule out alternative explanations, such as balancing by European powers against the Ottoman Empire, for the shift in the geography of siege warfare. In addition, I show that the results are robust to alternative measures of proximity to Constantinople, corrections for spatial correlation, the use of different functional forms, and the presence of outliers. Documentary evidence corroborates the claim that the decline in siege warfare resulted from greater agreement about the power of gunpowder artillery.

This article contributes to three active areas of research. First, it offers and tests a new theory linking technological change to the frequency of conflict via uncertainty about military capabilities. Most prior scholarship on technological change and conflict has been conducted through the lens of offense-defense theory, which argues that new technologies affect the likelihood of conflict by modifying the relative ease of capturing territory with military force compared to defending it (Glaser and Kaufmann, 1998; Hopf, 1991; Lynn-Jones, 1995; Quester, 2002; Van Evera, 1998). In the offense-defense framework, technologies that amplify offensive advantage make international conflict more likely by increasing first-strike advantage and exacerbating security dilemmas (Jervis, 1978; Van Evera, 1999). Yet evaluating the predictions of offense-defense theory has proven difficult, as empirical tests require measuring how new technologies shift the offense-defense balance, which is not directly observable (Levy, 1984). Critics of offense-defense theory also argue that militaries can develop doctrine to apply technologies to many ends, complicating attempts to determine the overall effect of a new technology on offense or defense (Biddle, 2004).

Rather than attempting to identify attributes of technologies that affect conflict, the theoretical framework presented in this paper emphasizes that the significance of a new technology is subject to contestation. Building on work by scholars of military innovation, my theory argues that assessments of unproven technology can be influenced by bureaucratic competition within militaries, doctrinal differences, the external security environment, and the availability of technical resources, all of which differ across states (Grissom, 2006; Posen, 1984; Rosen, 1991). These

domestic debates can lead to contrasting perceptions of new technologies and military capabilities. Disagreement about the implications of new technologies, rather than any property of the technologies themselves, is the cause of conflict. Moreover, the empirical results of this paper contrast with the predictions of offense-defense theory. New tactics that enable cannons to better destroy walls are clearly innovations that enhance offensive advantage. Yet locations that learned of these new tactics via the Fall of Constantinople experienced fewer rather than more sieges, the opposite of what offense-defense theory would predict. These results suggest that understanding the causes of uncertainty about new technologies may be a profitable direction for future research.

Second, this article provides new evidence for the claim that uncertainty about the balance of power can generate conflict.³ Bargaining theorists have argued that states may misperceive the balance of power because of incentives to conceal military strength or noisy private assessments of military capabilities (Blainey, 1988; Debs, 2022; Fearon, 1995, 2021; Slantchev, 2010). This can lead to war if states in a dispute jointly believe that more can be achieved through fighting than yielding to the other's demands. Despite bargaining theory's prominence, convincing empirical tests of its predictions have proven elusive, in part due to the difficulty of measuring private information held by states. This article takes a novel approach to the problem of private information by exploiting a historical case that offers plausibly exogenous variation in the degree of potential disagreement about the balance of power. In doing so, this paper illustrates the promise of using historical data to test general theories of international politics, joining a growing quantitative literature on historical international conflict (Lindsey, 2019).

Third, this paper articulates a new mechanism linking battlefield learning to conflict that unifies disparate strands of research on wartime demonstrations of new technologies. Scholars studying the diffusion of military technology have shown that wartime demonstration of an innovation's effectiveness instigates its wider adoption by states and have outlined the cultural, financial, and organizational factors that facilitate adoption (Eyre, Suchman et al., 1996; Goldman

³Early arguments of this type are due to Blainey (1988) and Fearon (1995). See Bas and Schub (2017), Ramsay (2017), and Powell (2002) for surveys of this expansive body of research.

and Eliason, 2003; Horowitz, 2010). Separately, recent work has argued that states may intentionally employ new technology in ways that will influence postwar perceptions of the balance of power (Byun and Carson, 2025). The theory presented in this paper unifies and extends such work by arguing that wartime demonstrations also foster a consensus about technologies’ usefulness that directly affects the likelihood of future conflict.

2 New technology and perceptions of the balance of power

2.1 Technological change and bargaining

When states adopt a new military technology, there is often no consensus on how the new technology should be used in combat. This arises because militaries’ pursuit of combat advantage means that new military technologies diffuse far more quickly through the international system than the knowledge needed to use them effectively (Goldman and Eliason, 2003; Horowitz, 2010). Indeed, for truly novel technologies, such knowledge does not exist yet—technologies do not spring into existence with a fully developed doctrine for their use. Consequently, states must independently assess the effect of new technologies on capabilities and determine tactics prior to battle.⁴ Yet the process of evaluating a new technology can be subject to systematic biases and pressures, which can result in states arriving at contrasting expectations about how a new technology will affect the outcomes of future conflicts. Diverging private assessments can produce conflict if two states in a dispute both believe that more can be achieved through fighting than negotiating. In other words, technological change can generate bargaining failure.

2.2 Mechanisms of uncertainty

I outline three channels through which new military technology can influence perceptions of the balance of power: technical challenges in assessing new technologies, differences in doctrine, and

⁴In other words, states receive noisy private signals of how new technology affects the balance of power. Examples of formal models of crisis bargaining that use such a method to generate uncertainty about the balance of power include Debs (2022) and Fearon (2021).

bureaucratic competition between military services.⁵

First, assessing the combat potential of new technologies is inherently difficult (Miller, 1985; Sechser, Narang and Talmadge, 2019). Evaluating new military technologies requires major engineering efforts and tests approximating real combat. At each step, variation in state security goals, differences in technical capacity, and even luck may influence perceptions of a new technology's usefulness in combat. For instance, states with different security concerns may envision distinct uses for the technology and evaluate it on different standards (Rosen, 1991; Talmadge, 2015). A country that lacks scientific capital may suffer technical setbacks that generate pessimism about the new technology and discourage further investment (Jones and Hildreth, 1984). This evaluation process in turn influences perceptions of the balance of power: a state that views a technology as promising and adopts it will likely have a more favorable view of its own capabilities than a state that dismissed the technology as ineffective.

Second, states adopting a new technology may develop different doctrines for the technology's use in the field. Scholars have argued that victory in war depends less on the particular tools and technologies available to a state than on how a state employs them (Biddle, 2004; Horowitz, 2010, 2020). Yet new technologies lack an established doctrine, which militaries must create independently. Such doctrine is incorporated into military planning, which can become unreliable if another state develops a different doctrine. When planning for future conflicts, a state may assume that rivals will use new technology following the same doctrine. However, this can leave them vulnerable if an adversary employs a technology in an unexpected way. Indeed, as Biddle (2004) points out, innovative new tactics can render a new technology far more effective in battle and transform perceptions of the technology's potency. Therefore, technological change can cause two states to disagree about the balance of power when one believes that it has developed a doctrine for the technology that provides it with combat advantage, of which the other state is unaware or dismisses.

Finally, technological change can also create disagreements about the balance of power via ac-

⁵Examples of these mechanisms are in Appendix A.1.

tions taken by bureaucratic actors to promote self-serving perceptions of new technologies. Past research has found that the reception of an innovation is mediated by the effect that adoption will have on existing distributions of power and wealth (Juma, 2016; Mokyr, 1998; Solstad, 2023). Interest groups may attempt to block the use of new technologies that could reduce their economic or political power (Acemoglu and Robinson, 2000; Frieden and Silve, 2023; Mokyr, 1990). Militaries are not immune to such dynamics, and organizations within militaries have not hesitated to present slanted assessments in support of self-interested goals (Evangelista, 1988; Grissom, 2006; Kier, 2017; Posen, 1984; Rosen, 1991). For example, Snyder (1984) argues that European military staff deliberately promoted perceptions of offensive advantage prior to 1914 to enhance their budgets, prestige, and autonomy. New innovations can affect the budget, mandate, and prestige of groups within the military by rendering their role obsolete or creating competition for resources. To maintain their status, groups may attempt to influence the perception of new technologies or the doctrine for their use. Biased views of technological capabilities can emerge from this internal competition.

2.3 Battlefield demonstrations, doctrinal convergence, and bargaining

In peacetime, the three mechanisms of uncertainty generate variation in beliefs across states about the combat potential and operational use of new technologies. This variation can contribute to diverging expectations about the balance of power that drive bargaining failure. But the onset of war introduces evolutionary pressures that push states toward similar doctrines for the use of new technology (Biddle, 2004). Ineffective tactics and technologies can stymie combat effectiveness, putting slow-to-adapt militaries at risk of defeat. Militaries that recognize their tactics are failing may experiment and undergo a process of “learning by doing,” discarding ineffective tactics and developing best practices for the new technologies they deploy in combat (Farrell, 2010; Hoffman, 2015; Hunzeker and Harkness, 2021). Experimentation can generate new doctrine for applying technologies that improves combat effectiveness (Rosen, 1991). One example is the German Army’s development of integrated artillery doctrine under Georg Bruchmüller

in the First World War. Organizational changes to artillery use, such as shorter but more targeted bombardments combined with infantry support, are credited with helping Germany capture significant territory in a March 1918 offensive (Lupfer, 1981). The process of doctrinal learning can also induce militaries to discard technologies once thought promising. An example is the U.S. Army tank destroyer in World War II, which was seldom effective against mass armor as envisioned by planners' original doctrine, leading to the closing of the U.S. Tank Destroyer Center and the assumption of anti-tank functions by other units (Gabel, 1985, 2009).

The evolutionary pressures introduced by war have implications for postwar bargaining beyond convergence in doctrine. Demonstrations of new technologies can cause states to reassess their beliefs about how the technologies influence the balance of power. Wars are never private affairs—conflicts therefore serve as “demonstration points” that reveal the capabilities of new technologies and the tactics needed to apply them successfully, leading states throughout the international system to adopt the newly discovered best practices (Biddle, 2004; Horowitz, 2010). With the diffusion of doctrine also comes a consensus about how the new technology should be used in combat. Prior conflict therefore serves as a common reference point to evaluate the technology, giving states a mutual basis for assessing military capabilities. While the mechanisms that cause uncertainty about initial assessments of technology—such as bureaucratic infighting among military services—can also affect learning after war, battlefield demonstrations serve as evidence to counteract such influences. Altogether, combat demonstrations reduce disagreements about how a new technology will be employed. Common beliefs in turn ameliorate uncertainty about the balance of power caused by the introduction of the new technology and make it more likely for states in a dispute to arrive at a negotiated settlement. In other words, conflict involving a new technology provides a public signal about the effect of the technology on the balance of power.

Not all signals from combat have an equal effect on states' perceptions of new technology. The most effective demonstrations will be those that clearly produced combat success or failure and can be linked to a doctrinal innovation. A battlefield success or failure, particularly one that was strategically significant or unexpected, is more likely to be noticed and studied by analysts.

Moreover, if the outcome can clearly be linked to a doctrinal innovation employing a new technology (e.g., a success in a bombing raid after changing only doctrine), then observers will be more inclined to view the new technology as playing a central role in the outcome. In contrast, it is harder to ascribe outcomes to the effectiveness of new technology when the result was expected or one of many small changes implemented before combat. These forces are also mediated by organizational culture and a willingness to adapt, as emphasized in prior work on military learning (Nagl, 2002; Levy, 1984; Posen, 1984).

2.4 Observable implications

The theory generates four observable implications relating to military doctrine, perceptions of technology, and the frequency of conflict. First, in the period between a new military technology's invention and its first salient battlefield demonstration, states should vary in their appraisals of the technology's combat effectiveness and appropriate operational use. Second, bargaining failure should be more frequent when states hold divergent beliefs about a new military technology. Third, states should converge in their assessments of a new technology's effectiveness and the doctrine required to realize it after a battlefield demonstration. Finally, bargaining failure should be less frequent among states who witnessed a battlefield demonstration of the new technology.

The observable implications of my theory differ from those derived from offense-defense theory, which also makes predictions about the effects of technological change on the frequency of conflict (Glaser and Kaufmann, 1998). In offense-defense theory, implications for conflict focus on security dilemmas. The severity of security dilemmas is determined by two variables: the relative ease of attack and defense (the offense-defense balance) and whether offensive and defensive military buildups are distinguishable (Jervis, 1978). Greater (perceptions of) offensive advantage and indistinguishability of forces are both theorized to increase conflict by incentivizing first strikes and prompting concern about the intentions of other states. New technology therefore influences conflict by either changing the offense-defense balance or the distinguishability of forces.

While my theory and offense-defense theory share a focus on beliefs about new technology,

they diverge on whether these beliefs operate primarily in a bargaining setting or a security dilemma. In my theory, states' beliefs about technology affect perceptions of the balance of power, and thus bargaining leverage, in a potential dispute with another state. Agreement about the effectiveness of new technologies makes peaceful settlement more likely, regardless of whether the new technology benefits offense or defense—though such shifts may produce more favorable settlement terms for one side. In offense-defense theory, states' beliefs about technology focus on underlying shifts in offensive or defensive advantage that condition interactions between all states in the international system. Offense-defense theory therefore predicts that states that agree a new technology benefits the offense may nevertheless be at greater risk of fighting, due to incentives to strike first and uncertainty about intentions. Different observable implications for offense-enhancing technologies generated by each theory permit comparative testing of each theory's predictive power.

2.5 Application to gunpowder artillery and siege warfare

In the following sections, I test the observable implications of my theory by studying how information conveyed by the Fall of Constantinople affected the frequency of siege warfare in Europe. By the middle of the 15th century, technological advances had produced gunpowder artillery capable of piercing existing fortifications. I first present qualitative evidence that disagreement existed among Europeans about the usefulness of cannons in siege warfare even after these technical developments, which could contribute to informational asymmetries in pre-siege bargaining. Then, using primary sources, I show that the Fall of Constantinople constituted a singular demonstration of the effective use of cannons against even very strong fortifications. These sources also illustrate how the Siege of Constantinople created recognition among Europeans about the advantages provided by properly utilized cannons, leading to a documented shift in siege tactics. Finally, I employ an original dataset of sieges and a difference-in-differences design to show that locations more exposed to information about artillery at the Fall of Constantinople experienced fewer sieges, consistent with more successful bargaining. I conclude by addressing

alternative explanations and contrasting the empirical results with the implications of offense-defense theory.

3 Gunpowder, siege warfare and the Fall of Constantinople

3.1 Siege warfare and firearms in Medieval Europe

Siege warfare dominated military conflict in medieval Europe due to the importance of fortresses for both offensive and defensive military operations. Castles and fortified cities provided a base from which soldiers could venture out to pacify rebels, harry an invading army, or stage an attack (Bartlett, 1989; Pounds, 1993). Territorial control therefore depended on possession of fortifications (Cappelen and Hariri, 2026). But the defenses that made fortresses useful for force projection also made them costly to attack. Before the 15th century, stout walls and weak artillery meant that taking a well-fortified castle or town by force required a siege consisting of a long blockade, a direct assault, mining of walls, or some combination thereof (Bradbury, 1992; Purton, 2009). Hence, sieges were dangerous affairs—it was not unheard of for attrition rates among siege combatants to exceed 50% (Manning, 2019; Pépin, 2023). Yet the necessity of sieges for territorial conquest made them common, in extreme cases comprising up to 60% of all engagements during wars of the Late Middle Ages (Desierto and Koyama, 2026).

Mutual dread over the prospect of a siege encouraged bargaining between garrisons and besiegers, with hopes of reaching a mutually acceptable settlement that would yield the fortification without a siege, generally in return for promises of autonomy, money, or safety (Mallett, 2009). These pre-siege negotiations were characterized by information asymmetries between the besiegers and garrison that made them micro-level analogues of the bargaining process that Fearon (1995) and others argue characterize interstate disputes. The besieging army was often unaware how much stored food and munitions were held by the defenders, or the degree of deprivation besieged civilians would tolerate before revolting against the garrison and opening the gates to the besiegers. Attackers also had to consider the risk of a relief army arriving before the fortification

could be taken, forcing the besieging army to scatter or face battle from two sides. Defenders also faced uncertainty, confronting the possibility of relief not arriving, variation in quality and quantity of the attacker's siege engines, and whether the besiegers would try to take the fortification by assault or blockade.⁶ Attackers would attempt to awe defenders into surrendering with threats of storming the defenders' position (Purton, 2009). In the face of clear defeat, defenders would prefer to negotiate a handover of the fortification to avoid a sack, but the initial technological balance favored defensive fortifications, and 15th-century garrisons generally opted to endure long blockades (Bradbury, 1992; Rogers, 2018).

While sieges lasting months or years were typical for centuries, the early 14th century brought a development that would transform siege warfare: gunpowder. Early guns were short, squat things that were unable to launch projectiles with greater force than existing catapults and trebuchets, on top of being inaccurate, slow to fire, and prone to exploding (Cipolla, 1965; DeVries, 2024). Nevertheless, primitive cannons spread rapidly throughout Europe due to their low cost compared to other siege engines, whose complex mechanisms required skilled labor to construct (Rogers, 2018). During sieges, cannons were used in conjunction with existing siege engines, as evinced by the fact that early cannons were primarily utilized to fire *over* the walls of fortifications rather than at them. Europeans saw the cannon as a new variant of existing siege weapons, and early cannons had little impact on the course or outcome of sieges for over a century after their first appearance in 1326 (Heuser, 2012).

Beginning in the 1410s and 1420s, gunsmiths made technical advancements that improved the effectiveness of gunpowder artillery. First, the innovation of "corning" the ingredients of gunpowder by mixing them with water to form coarse granules yielded a propellant that gave off greater energy after ignition, permitting artillery to fire projectiles at greater speeds (Gray, Marsh and McLaren, 1982). Second, by the 1430s, gunsmiths began manufacturing bombards with longer barrels, which extended the amount of time projectiles were accelerated by the ignition of gunpowder (Davies, 2019). The resulting increase in muzzle velocity improved the accuracy,

⁶See Piccolomini (1889) for an eyewitness account of deliberations by Holy Roman Emperor Frederick III and his advisors, who were besieged in Wiener Neustadt by the Austrian Estates in 1452.

range, and power of artillery. Greater accuracy had the compound effect of enabling new tactics, such as the concentration of fire by multiple bombards, that had previously been impossible. As a result of these innovations, by the 1440s, gunsmiths could produce artillery that reliably pierced the curtain walls that had frustrated besieging armies for the past two centuries (McNeill, 2013; Rogers, 2018).

Not all European military leaders recognized the significance of the technical advances in artillery and munition manufacture. Opinion on the usefulness of gunpowder artillery remained mixed and there is little evidence of widespread European acceptance of the idea that bombards heralded a revolution in the conduct of siege warfare in the first half of the 15th century (Hale, 1983; Heuser, 2012). A common attitude was indifference: a survey of pre-1450 military and fortification manuals by De la Croix (1963) finds only a single work that mentions cannons. Yet there were some enthusiastic adopters of artillery, such as the French artillery officers Gaspard and Jean Bureau, the Dukes of Burgundy, James II of Scotland, and Ottoman Sultan Mehmed II (Purton, 2009). These leaders saw the potential of the new bombards and expanded their use in military campaigns. But even artillery optimists hamstrung the power of their guns by using poor tactics. There are accounts of overconfident commanders who brought insufficient cannonballs to sieges and were forced to revert to traditional tactics (DeVries, 2024). Consequently, by the early 1440s, Europeans held a wide range of contrasting views on the effectiveness of cannons, creating the potential for mutual optimism.

3.2 Cannons at the Siege of Constantinople

European attitudes about gunpowder artillery began to evolve after May 29, 1453, when an Ottoman army led by Mehmed II captured Constantinople, the capital and final remnant of the Byzantine Empire, after a 53-day siege. During the siege, the Ottomans made extensive use of gunpowder artillery, employing as many as 70 cannons to fire over 5,000 projectiles (Ágoston, 2014). The cannons the Ottomans deployed were not more sophisticated than those used in Western Europe at the time; in fact, the Ottomans employed European gunsmiths to cast some of their

artillery (Ágoston, 2014; Runciman, 1965). The Ottomans' innovation was employing artillery tactics in ways that took full advantage of the technical advancements made in artillery construction over the preceding half century. For instance, the Ottomans created batteries of three or more cannons that fired simultaneously at a single point to maximize damage inflicted (Philipides and Hanak, 2011). In addition, the Ottomans brought enough artillery and ammunition to allow for near-continuous bombardment, preventing the defenders from repairing damage to the walls between shots. These tactics ensured the Ottomans' cannons had a devastating effect on the fortifications of Constantinople, which had been regarded by contemporaries as among the most formidable in Europe. The city was defended by multiple sets of land walls, which had not been breached by a besieging army in the millennium since their construction in 413 A.D., resisting at least 25 sieges (Runciman, 1965).

The Ottomans' successful application of gunpowder artillery to pierce the formerly impregnable defense of Constantinople was not lost on European contemporaries. Due to the religious significance of an Ottoman conquest of Constantinople, many European observers were present at the siege of the city, with some fighting on behalf of the Byzantines. These observers noted the Ottomans' novel artillery tactics and their effectiveness against the city's fortifications. Some eyewitnesses produced detailed written accounts of the siege, in which a universal emphasis was the power of the Ottoman artillery. For instance, one account records how the walls of Constantinople were unable to resist sustained bombardment:

And the stone, borne with tremendous force and velocity, hit the wall, which it immediately shook and knocked down, and was itself broken into many fragments and scattered, hurling the pieces everywhere and killing those who happened to be nearby. Sometimes, it demolished a whole section, and sometimes a half-section, and sometimes a larger or smaller section of a tower or turret or battlement. And there was no part of the wall strong enough or resistant enough or thick enough to be able to withstand it, or to wholly resist such force and such a blow of the stone cannon-ball (DeVries, 1997, 357-358).

Reports of the Fall of Constantinople began to spread almost as soon as the Ottomans entered the city. The religious implications of the conquest, along with the fact that Constantinople's defenses were considered nearly impregnable, meant that the city's fall was taken with shock. News of the siege was spread orally via European eyewitnesses and Greek refugees, as well as through the publication of written accounts. The accounts of the siege produced by eyewitnesses were some of the first texts mass-produced by the newly invented printing press (Bisaha, 2017). At least four narrative accounts of the Siege of Constantinople were published in the years after 1453, each emphasizing the power of the Ottomans' cannons and the tactics used to wield them (Philippides and Hanak, 2011).

Documentary evidence from primary sources suggests that the Fall of Constantinople constituted a singular informational shock about gunpowder artillery. At least 17 major 15th-century chronicles from 11 polities spanning all of Europe discuss the Ottoman siege. No other contemporaneous event received comparable attention in so wide a geographic area.⁷ Moreover, the contents of the chronicles evince that Europeans recognized the militarily significant role cannons played in the siege, emphasizing the strength of Constantinople's fortifications and the innovations employed by the Ottomans. For instance, Pope Pius II (2013, 95-96) describes how the Ottomans were able to use cannons to "shatter" the "main walls of the city[, which] were famous throughout the world for their height and thickness." Likewise, the Hungarian chronicler János Thuróczy (1991, 172) claims Sultan Mehmed II was able to take the city by "devising a new kind of assault" that included a variety of artillery. Though some have suggested that other events, such as the rapid French conquest of Normandy and Gascony, were also demonstrations of the power of gunpowder artillery, the lack of discussion of the Hundred Years' War in chronicles written outside of England and France suggests that contemporaries neither regarded France's success as notable nor ascribed it to cannons. Indeed, Rogers (2018) argues that the significance accorded to artillery in the Normandy and Gascony campaigns is a modern assessment.

⁷Appendix A.2 describes primary source references to the Fall of Constantinople and includes excerpts describing cannon use and the strength of Constantinople's walls.

3.3 Shifts in artillery perceptions and doctrine after 1453

As news of the Fall of Constantinople spread across Europe, military leaders took note of the Ottomans' effective application of cannons and changed their assessments of the utility of gunpowder artillery. Evidence of a shift comes from the actions of Philip the Good, the Duke of Burgundy. Upon hearing reports of the Ottomans' guns at Constantinople, Philip expanded the artillery train for a planned campaign to include "five or six hundred gunners" (DeVries, 2017; Smith and DeVries, 2005). Philip was not the only one to recognize that the Siege of Constantinople had demonstrated how gunpowder artillery could be used to destroy existing fortifications. After 1453, European soldiers and military theorists penned treatises urging rulers to adopt Ottoman artillery and field tactics.⁸ While the goal of these writers was to prepare Europeans to wage a crusade against the Ottomans, their acknowledgement of the validity of these tactics indicates a shift in European opinions about artillery.

Furthermore, there is evidence that the information about guns' capabilities revealed in the Siege of Constantinople influenced the conduct of sieges in the latter half of the 15th century. Written accounts of the Siege of Constantinople described how the Ottomans would form batteries of cannons that would concentrate fire at a single point to maximize damage to walls (Philippides and Hanak, 2011). Primary source accounts of sieges taking place after 1453 show that Europeans began imitating such tactics. For instance, at the 1464 siege of Bamburgh Castle in Scotland, the besieging army coordinated the first shot of its cannons so that they fired in unison, dealing so much damage to the walls that "stones flew into the sea" (Giles, 1845, 87).

The siege of Bamburgh also demonstrated a second shift in artillery use after 1453: the use of cannons and bombards as negotiating tools. The Earl of Warwick, leader of the besiegers at Bamburgh, arrayed his cannons in full view of the garrison and challenged them to bet their lives against the strength of the artillery, threatening that for each cannon ball that breached the walls of the castle, "it shall cost you the chieftain's head, and so proceeding for every gun shot to the least head of any person within said place" (Giles, 1845, 87). While sieges were always preceded

⁸See Christensen (1987) for a list of primary sources in this genre.

by a mixture of promises, bribes, and threats meant to achieve a peaceful outcome, sieges after 1453 are distinguished by the inclusion of cannons in these negotiations. Increasingly, attackers in pre-siege and pre-battle negotiations would focus on demonstrating to defenders the power of their artillery in order to awe the defenders into surrender, thus averting a fight (Purton, 2009). Displays of artillery power seem to have been used as statements of resolve to bring an opponent to the negotiating table.

These examples reveal that a greater awareness of effective tactics for the use of gunpowder artillery emerged among Europeans in the aftermath of the Fall of Constantinople. Evidence for this shift is corroborated by changes in the ways that Europeans discussed artillery in written sources following 1453. According to De la Croix (1963), starting in the mid-1450s, there is a sharp increase in the mentions of gunpowder artillery in military manuals and treatises on fortification. Italian mercenary captains, who had previously been silent on the use of cannons, began devoting chapters of manuals to the proper use of artillery in siege warfare, even offering guidance on the number of cannons and projectiles needed to successfully capture fortresses with varying levels of defenses. The articulation of such artillery tactics, as well as their uniformity across different manuals, suggests that after 1453, Europeans experienced a convergence in both attitudes toward gunpowder artillery and tactics for its use.

3.4 Implications for bargaining before sieges

The preceding discussion has supplied some qualitative evidence for variation in opinions about artillery doctrine and efficacy, informational asymmetries and bargaining failure in sieges, and convergence in artillery assessments and tactics following the Fall of Constantinople. If my theory is true, lower incidence of bargaining failure in siege warfare should follow. The Constantinople-induced convergence in artillery doctrine should manifest in military leaders updating their expectations about their likelihood of winning a siege. I theorize that a public signal about the power of artillery induced medieval military leaders to update their beliefs about the balance of power in two ways.

First, the diffusion of artillery tactics used at the Siege of Constantinople created a consensus among the attackers and defenders about how cannons would perform in a siege. This common reference point gave both sides in a siege similar information to forecast how they would perform in combat. A smaller dispersion of beliefs about the balance of power reduced the probability that adversaries would arrive at mutually optimistic beliefs about their chances of winning. Second, convergence on an artillery doctrine that provided advantages to attackers diminished the influence of other private information that contributes to bargaining failure and encouraged settlement terms that benefitted besiegers, such as total surrender. When negotiating before a siege, attackers were generally not aware of how much food the defenders had stored or how many soldiers served in the garrison, both factors that influence defenders' beliefs about how long they could resist a siege. However, private information would be less important in the face of overwhelming artillery power—having an extra month of food stored matters little if the walls can be breached after a day of bombardment. Formal models of crisis bargaining have shown that both of these shifts predict a higher probability of negotiated settlement (Fearon, 2021; Reed, 2003).

To test whether information about artillery conveyed by the Fall of Constantinople produced fewer sieges, I exploit the fact that slow travel in the Middle Ages meant that the realization news of the siege came later in locations farther from Constantinople. Drawing from the theory, I hypothesize that locations more proximate to Constantinople experienced fewer sieges than those less proximate after 1453. I now describe and carry out an empirical test of this hypothesis.

4 Data on siege warfare and proximity to Constantinople

An ideal dataset to study how exposure to information about the use of cannons at Constantinople influenced decisions to fight in siege warfare would consist of all threatened sieges in the mid-15th century and an indicator of whether the defenders surrendered, as well as the precise timing of when combatants updated their assessments of gunpowder artillery. With such data, one could

examine how frequently confrontations before fortifications escalated into sieges before and after the combatants learned of artillery tactics employed at Constantinople.

Unfortunately, such data do not exist. One problem is that “potential” sieges where defenders immediately surrendered upon arrival of an attacking army are rarely attested. Even perfect data on potential sieges would produce an undercount, as sometimes multiple fortifications were exchanged in an agreement to avoid or end a single siege. In such cases, it is difficult to determine whether the additional fortifications would have been besieged in a counterfactual scenario where the agreement was rejected. Likewise, detailed records of when individuals learned of the artillery tactics used at Constantinople are unavailable. Given the relative disorganization of militaries at the time, changes in thinking about artillery may not have been written down at all.

I therefore collect data on the raw number of confirmed sieges and construct multiple proxies—transport costs, distance, and travel time—for when Europeans learned about the Fall of Constantinople. Using the raw count of sieges as the outcome requires caution, as changes in the number of sieges may be due to other factors correlated with when a location learned about the Fall of Constantinople, such as changes in broader patterns of warfare, alliances, or even natural disasters. To account for these potential confounders, I also collect data on additional conflict, political, and economic variables for use in robustness checks.

4.1 Outcome: sieges

The dependent variable of the study is the number of sieges occurring in a Roman Catholic diocese per year. To construct this variable, I built a new dataset of sieges that occurred in Western and Central Europe between 1443 A.D. and 1463 A.D. I focus on sieges that took place between 1443 and 1463 in order to isolate a period when military technology was constant. As described in Section 3, prior to the 1440s cannons and bombards did not have enough power to pierce the walls of most fortifications. Conversely, beginning in the 1460s, rulers invested in new forms of fortification, such as the *trace italienne*, as countermeasures to resist artillery (Bradbury, 1992; Mangini and Petroff, 2022). As technological changes could have influenced decisions to fight,

restricting attention to the decades around 1453 helps ensure that the effect of technology on conflict operates primarily through new information on how to use gunpowder artillery.

Most sieges are drawn from a collection of 42 primary sources, primarily major narrative chronicles, that were composed during or shortly after the mid-15th century. As primary sources sometimes contain errors or fabrications, in all cases I use modern critical editions with corrections, rather than original manuscripts. To reduce the risk of selection bias, I ensured that all major sovereign polities and conflicts with at least 32 battle deaths between 1443 and 1463, as recorded by Brecke (1999), were covered by at least one primary source. I supplemented the primary sources with over 200 secondary sources and ensured data reliability by only including sieges with at least two separate attestations. The full dataset contains 431 sieges associated with over 50 named conflicts. It constitutes one of the most comprehensive accounts of warfare in Europe during the mid-15th century, with 25 times more sieges documented than Jaques (2006), whose data are commonly used in studies of historical conflict.⁹

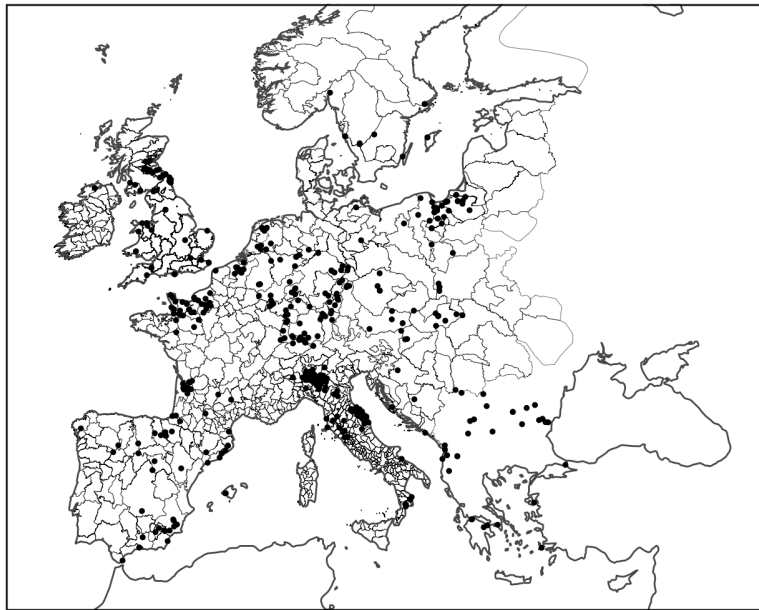


Figure 1: Diocese boundaries and siege locations

Sieges are aggregated by 1450 A.D. Roman Catholic diocese using boundary data from Dorin

⁹Details on primary sources and siege coding rules are contained in Appendix F. Descriptive statistics for the siege data are in Appendix B.

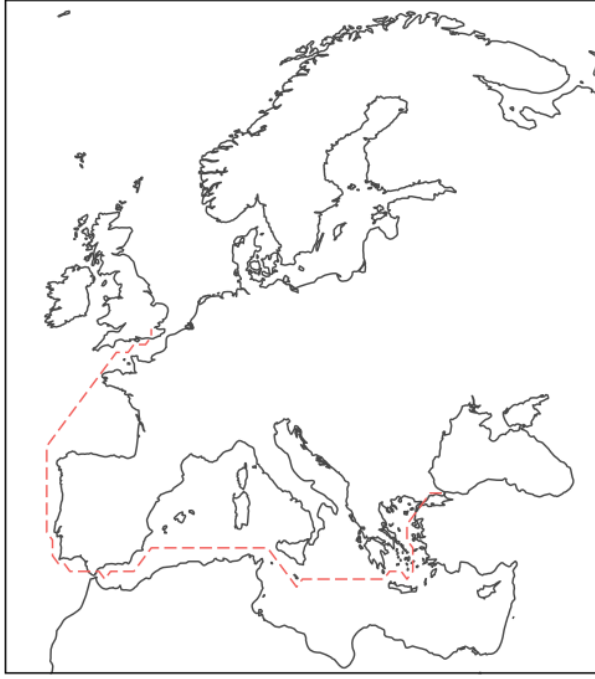
(2021). Figure 1 displays siege locations overlaid on the boundaries for all 696 dioceses in Europe.¹⁰ Aggregating by diocese offers several advantages. First, dioceses were generally centered on cathedral cities, which were often important regional political and economic centers (Dorin, 2021). Consequently, military campaigns often produced clusters of sieges around these cities, making the diocese a natural unit of aggregation. Second, dioceses are the smallest medieval territorial units that had political significance and for which reliable boundary data exist. Medieval dioceses were not purely ecclesiastical institutions. Bishops in charge of dioceses owned land and castles that generated income—there were even sovereign prince-bishops who ruled independent polities and led armies. The capital and prestige associated with control of dioceses made them the objects of political contestation. Temporal rulers seeking income or influence with the Church tried to exert control over the appointment of bishops or even conquer dioceses (Iogna-Prat, 2017; Parsons, 2020). Wars like the Mainz Diocesan Feud (1461-1462) and Ulrich of Celje’s invasion of Slavonia (1445) were fought over control of dioceses (Engel, 2001; Sprenger, 1999). Finally, the portion of Europe with established dioceses in 1450 corresponded to Western Christendom, the audience that would have most closely followed news of the Fall of Constantinople (Bisaha, 2017; Schwoebel, 1967).

4.2 Treatment: transport cost

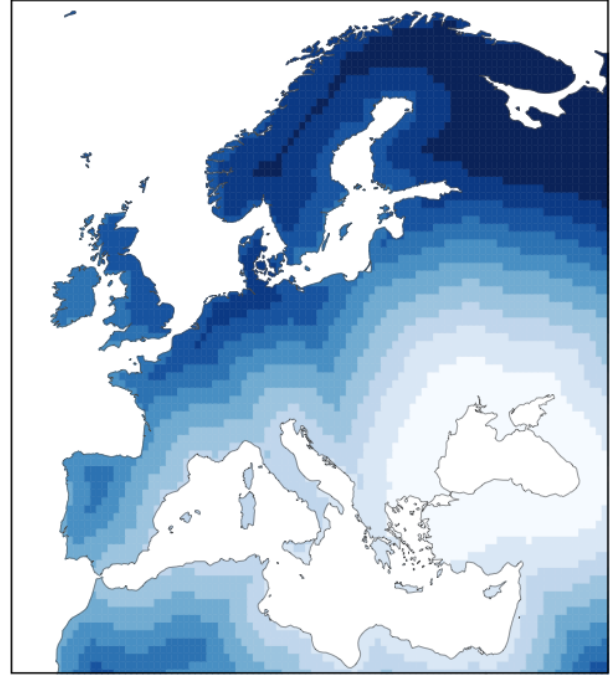
The main treatment variable of the study is the transport cost of a diocese from Constantinople, which I use as a proxy for when locals learned how cannons were used at the Siege of Constantinople. Because of a lack of data on the cost of travel in the 15th century, I estimate the transport cost of all dioceses from Constantinople using a shortest-path approach common in the literature on international trade.¹¹ To do so, I divide Europe into grid cells of roughly $0.5^\circ \times 0.5^\circ$, with cells classified as either sea or land. The cost of traveling between adjacent cells i and j , denoted $c(i, j)$, is a function of the distance in kilometers between the centers of the

¹⁰The dioceses on the Canary Islands, the Faroe Islands, Greenland, and Iceland are removed from the data due to their low populations and limited contact with continental Europe during the period under study. Their exclusion does not affect any results presented subsequently.

¹¹See Flückiger et al. (2024) and Juhász (2018) for recent papers that use a similar methodology.



— Least-cost Path



Transport Cost Decile

1	3	5	7	9
2	4	6	8	10

Figure 2: Least-cost path to London

Figure 3: Transport costs by grid cell

cells, denoted $d(i, j)$, and a per-kilometer cost determined by the terrain of the destination cell, denoted c_j^{mode} . Technological constraints in the Middle Ages made travel by sea much cheaper than travel by land—an analysis of medieval shipping records by Masschaele (1993) reveals that it was roughly eight times more expensive to ship identical goods by land than by sea. I therefore set $c^{land} = 8$ and $c^{sea} = 1$, and define the directed cost of travel between adjacent cells i and j as $c(i, j) = c_j^{mode} \times d(i, j)$. The transportation cost of diocese d from Constantinople is then the cost of the shortest path through adjacent cells between the cells containing d and Constantinople, or

$$\text{transportCost}_d = \min_{p \in \mathcal{P}_d} \sum_{(i,j) \in p} c(i, j)$$

where $\mathcal{P}_d$ is the set of all paths between Constantinople and a diocese d . The variable transportCost_d is computed using Dijkstra's shortest path algorithm (Dijkstra, 2022). An example of such a path

between Constantinople and London is shown in Figure 2, while Figure 3 visualizes transport costs to Constantinople for every grid cell on land.

Proxy validation and alternative measures

The validity of the transportation cost proxy depends on an assumption that dioceses with lower transport costs learned about the use of cannons at Constantinople earlier than dioceses with greater transportation costs. To empirically assess if this is the case, I compare my transport cost estimates with the dates that major cities first learned about the Fall of Constantinople, drawn from primary sources. Note that this initial wave of information communicated few specifics about the course of the siege, mentioning only that the city had fallen and diffusing relatively quickly throughout the continent (Pertusi, 1976; Schwoebel, 1967). Nevertheless, it is the only news that can be reliably dated. The spread of this information is plotted in Figure 4.¹² The figure illustrates a clear gradient in the timing of the arrival of news about Constantinople's capture. The news spread first to the Balkans and Italy, moved into Central Europe afterward, and reached the British Isles and Northern Europe last of all.

Figure 5 plots the time it took news of Constantinople's capture to reach a location by its estimated transport cost to Constantinople. The figure reveals that there is a strong linear relationship between transport costs and the timing of initial news of the Fall of Constantinople. Although more detailed reports about the use of gunpowder artillery at Constantinople would begin to diffuse months later with the movement of refugees and the publication of firsthand accounts, the correlation displayed in Figure 5 provides some evidence that transportation costs accurately track how information spread in the mid-15th century. These results are also consistent with past research on the spread of information in pre-industrial times, which finds that news tended to move along trade routes (Birkett, 2018; Robb, 2007).

¹²Details of the sources used to construct Figure 4 can be found in Appendix A.3.

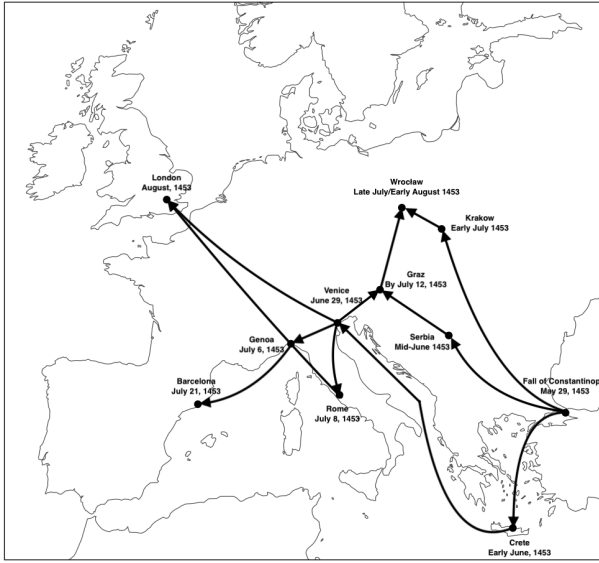


Figure 4: Diffusion of news about Constantinople

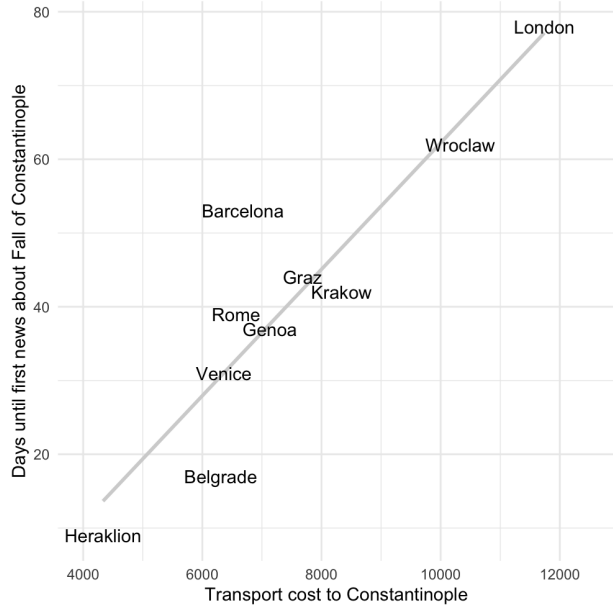


Figure 5: Timing of news vs. transport cost

4.3 Additional variables

For use in robustness checks, I collect diocese-level data on urbanization, number of fortresses, presence on a trade route, conflicts, distance from political border, and sovereign state affiliation. I measure a diocese's level of urbanization by the number of settlements it contained with a population of at least 1,000 in the year 1400, drawn from Buringh (2021). The number of fortresses extant between 1443 and 1463 is sourced from Cappelen and Hariri (2026), which I supplement with a scrape of crowd-sourced records on Wikidata. I draw historical overland trade route data from Shepherd (1911). To measure the number of ongoing wars in a given diocese, I geocode the extent of relevant European wars included in Brecke's 1999 catalogue of historical conflicts with at least 32 battle deaths. Finally, I determine each diocese's political affiliation in 1400 using the historical atlas maintained by Nüssli (2009), which I also use to compute each diocese's distance from a political border.¹³

¹³Descriptive statistics and maps of the control variables can be found in Appendix B.

5 Empirical strategy

5.1 Difference-in-differences

To examine how exposure to information about the use of cannons at the Siege of Constantinople affected the frequency of siege warfare in Europe, I employ a continuous difference-in-differences design implemented using a two-way fixed effects (TWFE) estimator. Specifically, I estimate the following fixed effects Poisson model using quasi-maximum likelihood

$$E[sieges_{dt} | \beta_d, \gamma_t] = \exp(\alpha \text{post1453}_t \times \text{transportCost}_d + \beta_d + \gamma_t)$$

where d indexes dioceses and t indexes calendar years between 1443 and 1463. The main dependent variable, $sieges_{dt}$, is the number of sieges that occurred in diocese d during year t (Hausman, Hall and Griliches, 1984). The variable post1453_t is a dummy that takes the value 1 in the years following the Fall of Constantinople in 1453 and 0 otherwise. The variable transportCost_d is the transport cost of a diocese from Constantinople. The terms β_d and γ_t are diocese and year fixed effects, respectively. The difference-in-differences estimator is the interaction term $\text{post1453}_t \times \text{transportCost}_d$. It takes the value of the transport cost of diocese d after 1453 and zero otherwise. Therefore, the coefficient α captures how a diocese's transport cost from Constantinople affected the frequency of siege warfare in years after the Fall of Constantinople, holding other factors fixed. Intuitively, the difference-in-differences estimator compares the frequency of siege warfare in dioceses at varying distances from Constantinople before and after 1453.

I use a Poisson regression model for ease of interpretation and robustness. In the model, the regression coefficient α can be interpreted as a semi-elasticity, meaning that the transformation $100 \times (e^\alpha - 1)$ yields the expected percentage change in the number of sieges associated with a one-unit increase in transport cost to Constantinople. Attempts to achieve a similar interpretation using a linear model require transformations to address zero-valued outcomes, which can

potentially introduce bias (O’Hara and Kotze, 2010). Moreover, the fixed-effect Poisson model is robust, producing consistent estimates for any mean-variance relationship in the outcome data, including when it does not follow a Poisson distribution (Wooldridge, 1999, 2023). Linear models lack such desirable properties when applied to count data, and have been shown to perform poorly when applied to count and count-like data (Cohn, Liu and Wardlaw, 2022; King, 1988).

A potential issue with implementing a two-way fixed-effects estimator via Poisson regression is the exclusion of dioceses that experienced no sieges during the period under study. When fitting models with (pseudo-)maximum likelihood estimation, panel units that contain only zero observations are dropped when unit fixed effects are included. For these panel units, the fixed effects perfectly explain the lack of observed variation and the zero counts contribute no information to the parameter likelihoods (Silva and Tenreyro, 2010). This means that the effect of proximity to Constantinople on siege warfare will be estimated using dioceses that experienced at least one siege between 1443 and 1463. The exclusion of dioceses that had no sieges is not necessarily problematic. Many dioceses were sparsely populated and had no castles or fortified towns—in these cases the diocese fixed effects do fully explain the lack of sieges. Nevertheless, to account for the possibility that important observations may be left out, I estimate additional Poisson models using correlated random effects, ensuring all observations are used. I also estimate Poisson regressions with polity fixed effects. The additional models can be found in Appendix D.

5.2 Causal identification

Causal identification in a difference-in-differences design implemented with a two-way fixed effects estimator requires the assumption of no anticipation and parallel trends (Roth et al., 2023). In addition, while the common treatment time in my design makes negative weights less of a concern, safely interpreting the DiD estimate as a marginal effect necessitates the assumption of treatment effect homogeneity (Callaway, Goodman-Bacon and Sant’Anna, 2024).¹⁴ I now offer evidence that my empirical design satisfies these assumptions, permitting causal interpretation

¹⁴For TWFE with Poisson regression, the homogeneity assumption is more precisely stated as constant semi-elasticity.

of a potential effect of proximity to Constantinople on siege warfare.

No anticipation mandates that siege warfare did not begin to decline in places near Constantinople prior to the city's capture in 1453. This assumption would be violated if the Ottoman Empire's European neighbors regarded the conquest of the moribund Byzantines as inevitable and anticipated that the Ottomans would continue their conquests westward afterward. Fear of an Ottoman invasion could have induced the polities bordering the Ottoman Empire to cease fighting among themselves and form a balancing coalition to deter invasion, thereby causing a decline in near-Constantinople siege warfare before 1453. However, the historical record indicates this is unlikely to have occurred. While there is a consensus among historians that Europeans saw the Byzantine Empire as doomed, there is also evidence that the timing of the Fall of Constantinople came as a surprise (DeVries, 2017; Pertusi, 1976; Schwoebel, 1967). The 1453 siege was the fifth time the Ottoman Empire had attempted to capture Constantinople, with the city even resisting an eight-year blockade between 1394 and 1402 (Harris, 2012). There was no guarantee that the Ottomans would succeed in 1453; indeed, some of Mehmed II's military advisors advised him to abandon the siege after early setbacks (Runciman, 1965). Moreover, historical accounts suggest that the Fall of Constantinople came as a shock, making it unlikely that Europeans changed their behavior prior to hearing news of the city's fall (Bisaha, 2017).

The assumption of parallel trends requires that, prior to 1453, the frequency of siege warfare in locations both near to and far from Constantinople was trending in the same direction, and that these trends would have continued had the city not fallen in 1453. I assess the plausibility of the no anticipation and parallel trends assumptions by conducting an event study of the effect of transport costs on sieges for each year between 1443 and 1463. The resulting coefficients and 95% confidence intervals are plotted in Figure 6.¹⁵ The figure shows that, prior to 1453, the effect of transport costs to Constantinople on sieges is generally statistically indistinguishable from zero, but increases notably after 1453 before declining after about five years. This pattern provides evidence for parallel trends, a post-1453 effect, and is consistent with a theory of learning. In

¹⁵The specification and coefficients for the event study are reported in Appendix C.1.

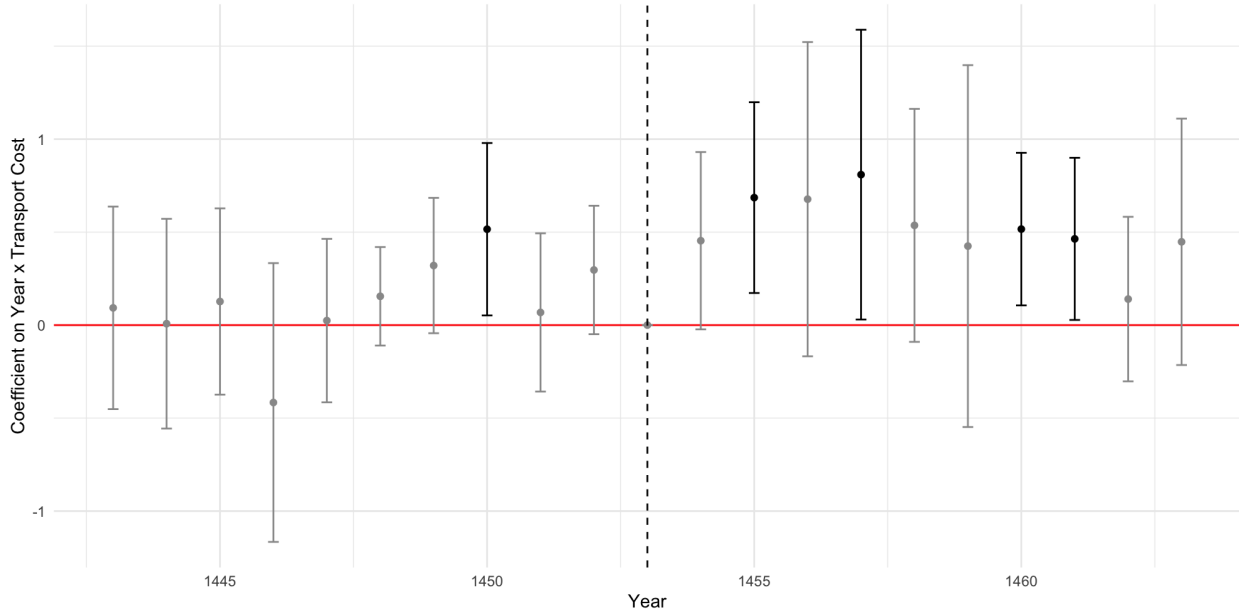


Figure 6: Event study coefficient plot

Appendix C, I provide additional evidence for the parallel trends assumption with placebo tests using alternative treatment years and cities, and alternative event study specifications.

The parallel trends assumption could be violated if there are time-varying unobserved confounders influencing the frequency of siege warfare. Two major sources of confounding are selection and survival biases in the recording of sieges. The quality of primary sources and historiographical attention to particular regions may vary in a way that is correlated with those regions' proximity to Constantinople after 1453. If so, post-1453 changes in siege warfare may be artifacts of data collection rather than effects of a Fall of Constantinople information shock about artillery. Another potential confounder is the local frequency of disputes. Because my outcome variable counts only confirmed sieges, it cannot distinguish between declines in siege warfare due to defenders surrendering in the face of artillery and declines due to fewer other reasons to fight. This could confound the analysis if wars became less common nearer to Constantinople after 1453 for reasons unrelated to information about artillery.

I address these concerns by adding polity-year fixed effects to the baseline TWFE estimator. Polity-year fixed effects absorb time-varying regional shocks that affect the frequency of sieges,

such as the outbreak of wars (including the total number of disputes), natural disasters, and differences in overall siege counts due to systematic over- or under-documentation of conflicts within polities. My specification becomes

$$E[sieges_{dpt} \mid \beta_d, \gamma_t, \delta_{pt}] = \exp(\alpha \text{post1453}_t \times \text{transportCost}_d + \beta_d + \gamma_t + \delta_{pt}) \quad (1)$$

where δ_{pt} is a polity-year fixed effect. There may also be time-varying confounding that is heterogeneous across dioceses within polities, and thus not captured by any of the existing fixed effects. As a robustness check, in Appendix D.5 I estimate additional specifications containing interactions of diocese-level pre-1453 controls with time.

Finally, treatment effect homogeneity means that the influence of proximity to Constantinople on siege warfare should not vary across units. While I cannot definitively prove homogeneity, I provide some suggestive evidence in its favor. First, in Appendix C.7, I interact Post-1453 $\times$ Transport cost and diocese-level controls. In all cases, the triple interaction term is statistically insignificant and the coefficient on Post-1453 $\times$ Transport cost remains similar in magnitude to the estimate in the main analysis, suggesting the effect size does not vary with some diocese-level characteristics. Second, Appendix D.10 displays results of leave-one-out analyses for the six-year and ten-year windows. Excluding individual polities does not have a large effect on the main coefficient, indicating that the treatment effect does not vary across polities. Finally, I relax the functional form assumption by estimating OLS and logistic regression models in Appendices D.6 and D.7, which yield directionally similar results to those in the main analysis.

6 Results

6.1 Main results

I first present two figures illustrating that siege warfare became less frequent near Constantinople after 1453. Figures 7 and 8 map the locations of sieges before and after the Fall of Constantinople

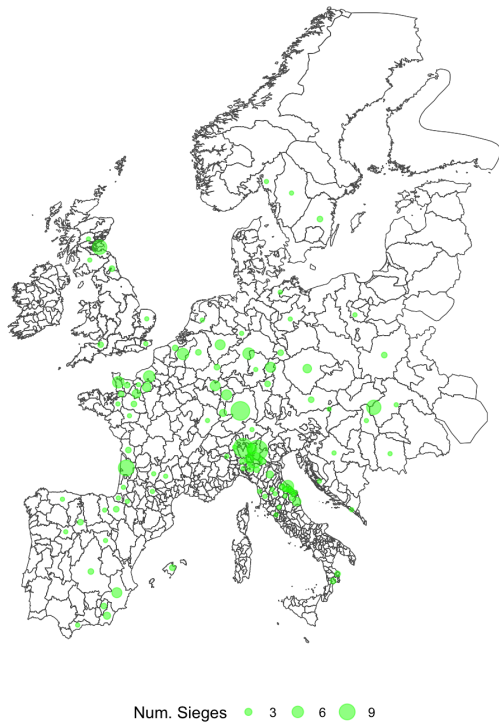


Figure 7: Siege locations before 1453

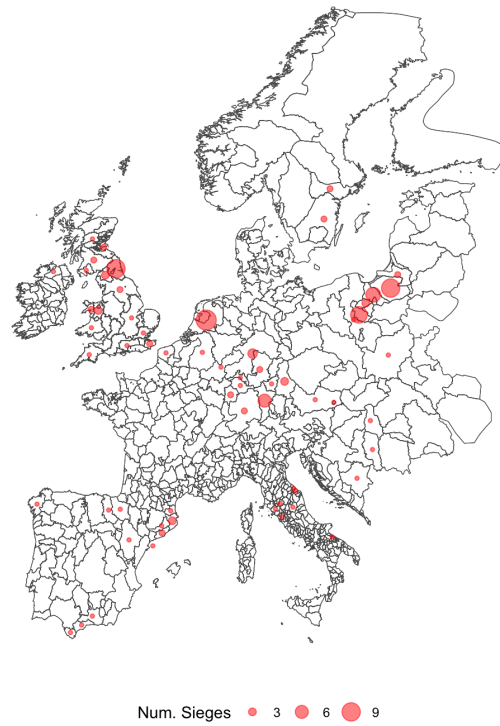


Figure 8: Siege locations after 1453

in 1453, aggregated by decade. The maps reveal a geographic shift in the concentration of siege warfare from the Mediterranean Basin and Central Europe, the places with the lowest transportation costs to Constantinople, to the northern and eastern fringes of the continent. In particular, there is little change in the incidence of siege warfare in Great Britain and the Low Countries, but declines in Italy and parts of Central Europe.

A similar pattern emerges from plotting the relative frequency of sieges by transportation costs. Figure 9 displays the empirical densities of sieges for ten-year periods before/after the Fall of Constantinople. Before 1453, sieges were nearly uniformly distributed by transport cost to Constantinople, though not perfectly so as population and other factors that influence siege frequency are not uniformly distributed. This suggests that before 1453, proximity to Constantinople had little to no relationship with the incidence of siege warfare. After the capture of Constantinople by the Ottomans, sieges became more concentrated in the latter half of the distribution, indicating that sieges became relatively less common in dioceses nearer to Constantinople. Together,

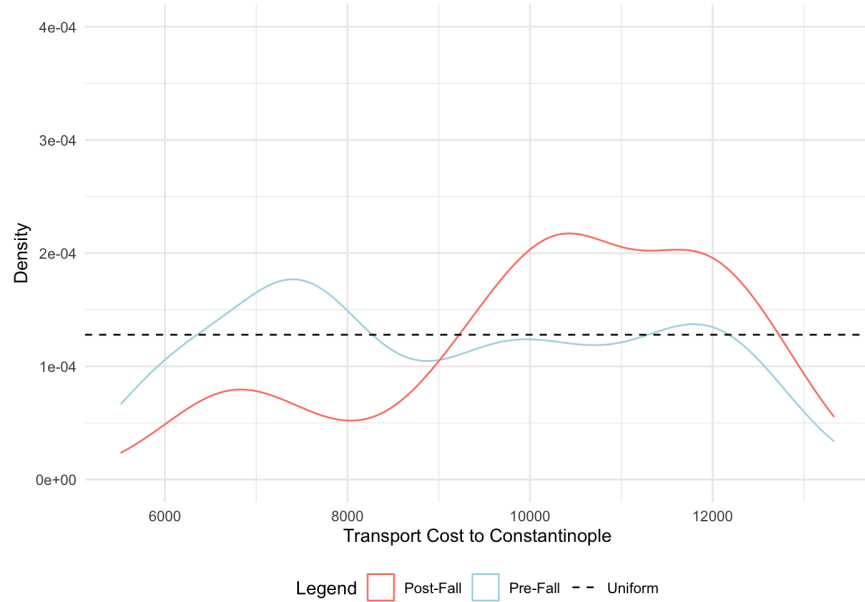


Figure 9: Empirical siege density before/after 1453 (10-year pre-/post)

the figures offer suggestive visual evidence that the geography of siege warfare shifted after 1453 in a way that is consistent with exposure to information about the Fall of Constantinople after 1453.

I now use my difference-in-differences identification strategy to test whether the descriptive patterns observed in the plots can be ascribed to an information shock about the effectiveness of artillery from the Fall of Constantinople. Figures 10 and 11 report the DiD coefficients for the baseline TWFE and polity-year fixed effects estimators, respectively. For both estimators, I perform the DiD analysis on nine window sizes around the Fall of Constantinople, ranging from four years (1451-1455) to twenty years (1443 to 1463). Vertical bars represent 95% confidence intervals computed using robust standard errors clustered by diocese.

For nearly all specifications and window sizes, the coefficient on $\text{Post-1453} \times \text{Transport Cost}$ is positive and statistically significant, indicating that locations near Constantinople experienced fewer sieges in the decade following the capture of the city. Using a four-year window (1451-1455) and TWFE estimator, a 1,000-unit decrease in transport costs (equivalent to moving from the 50th percentile to the 25th percentile, or between London and Prague) is associated with a

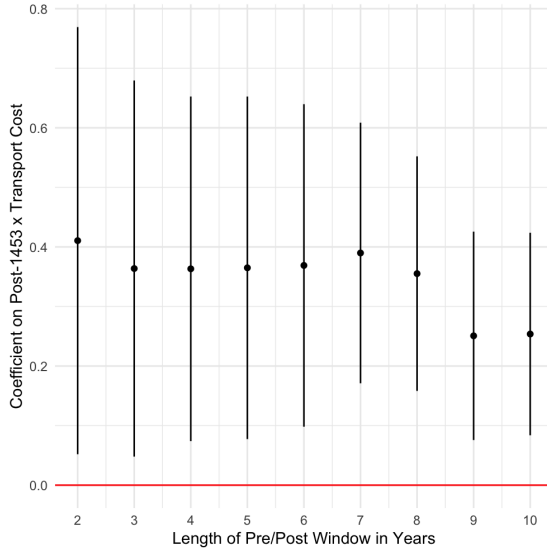


Figure 10: Baseline TWFE estimates

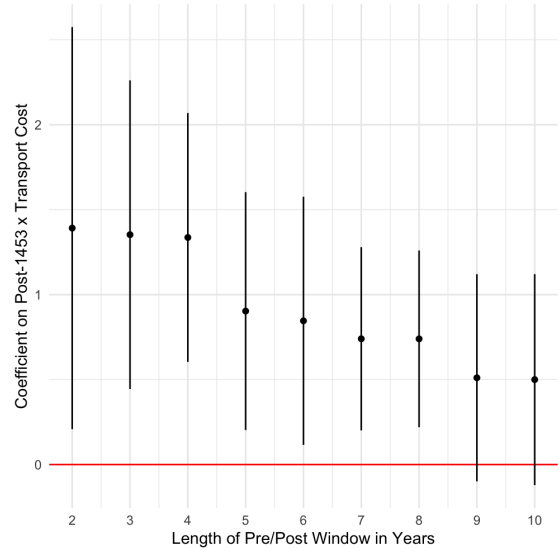


Figure 11: TWFE with polity-year FE estimates

33.6% decrease in sieges per diocese-year. This corresponds to 0.15 fewer sieges per diocese-year on average in the retained sample of dioceses. The magnitude of the coefficients increases when polity-year fixed effects are included, likely due to the absorption of time-varying shocks that also influence siege frequency. Overall, the difference-in-differences estimates support the hypothesis that fewer sieges took place in dioceses with earlier exposure to information about the use of cannons at Constantinople.

The empirical analysis also reveals that the effect of transport costs to Constantinople on siege warfare decreases near-monotonically as the window size expands from four years to twenty. Proximity to Constantinople has a larger estimated effect when comparing the years immediately before and after the capture of the city: the estimated percent decline in sieges associated with a 1,000-unit decrease in transport cost falls from 33.6% in the four-year window to 22.1% in the twenty-year window. Declining effect sizes in larger windows provide further evidence for the hypothesis that the decline in siege warfare is due to Europeans learning about the effectiveness of cannons from Constantinople, rather than some other variable correlated with proximity to Constantinople. Specific information about the Siege of Constantinople would likely have diffused throughout the continent within two to three years of 1453. After this point, both near and far dioceses would have equal information about the utility of cannons in siege warfare. With

full information diffusion, there should no longer be a difference in siege frequency by proximity to Constantinople when controlling for all other factors. Nevertheless, because the coefficient on $\text{Post-1453} \times \text{Transport cost}$ averages large declines in early years with later null effects, it may remain statistically significant in larger windows.

A caveat to the results is that the estimates capture the post-1453 effect of proximity to Constantinople on the raw count of sieges in a diocese-year. The coefficient may therefore conflate declines in sieges due to negotiated surrenders with fewer siege attempts. A threat to identification can arise if there are time-varying confounders influencing local dispute frequency. While the inclusion of polity-year fixed effects captures some of these potential confounders, I also systematically address possible alternative explanations in Section 7.

6.2 Robustness checks

Alternative measures of proximity to Constantinople

The results may be sensitive to the choice of proxy for exposure to information about the use of cannons at Constantinople. To test whether this is the case, I conduct the analysis again using two alternative measures of proximity: the great-circle distance and travel time. Details of variable construction and the empirical results can be found in Appendices D.1 and D.2. Both alternative measures yield conclusions similar to those obtained for transport costs.

Alternative functional forms

The results of the analysis may be sensitive to the use of a specific modeling approach. To account for this, I show that the results are robust to changes in functional form. In Appendix D.6, I apply a logit model to a binary indicator of whether a siege occurred. In Appendix D.7, I redo the main analysis using a linear model with a logged outcome variable. For both alternative functional forms, conclusions regarding the direction of the effect of information exposure on siege warfare are unchanged.

Outlier polities

Another potential concern is that the statistical findings could be driven by shifts in patterns of conflict within a single outlier polity. To examine whether this could be the case, I perform a leave-one-out analysis by excluding each polity one by one and estimating the full specifications again. The results, reported in Appendix D.10, show that no single polity drives the effect.

Standard error adjustments for spatial correlation

Medieval territorial conquest required capturing all castles and fortified towns to guarantee control of land, naturally producing clusters of sieges. As a result, sieges located in nearby dioceses are unlikely to be independent observations. Treating them as such may produce deflated estimates of standard errors and unwarranted confidence in the results of the empirical analysis. I take three approaches to address spatial correlation. First, I use the standard error adjustment proposed by Conley (1999), using various distance cutoffs selected by estimating the geographic extent of historic conflicts. Second, I cluster standard errors by sovereign state, using both 1400 A.D. and present-day boundaries. Finally, to account for both autocorrelation and cross-sectional spatial shocks, such as a strategic territory changing hands multiple times, I also compute Driscoll-Kraay standard errors. Standard errors obtained from these methods are reported in Appendix D.4. Adjustments for spatial correlation produce larger standard errors but do not change conclusions about the statistical significance of the coefficient of interest.

Tipping-point analysis

A challenge of collecting new historical data is the possibility that some data points are omitted due to variation in the availability of sources (Cirone, 2024). Because there has been no previous attempt to exhaustively document all sieges in the mid-15th century, it is difficult to assess how many sieges are missing and whether omitted sieges systematically differ from observed sieges in ways that could bias the empirical results. I quantify the risk of bias from unobserved sieges by conducting adversarial tipping-point simulations (Liu, Zhou and Sims, 2025). The simulations

estimate how many new sieges, drawn from a distribution that strongly contradicts my hypothesis, would be needed to overturn the results. The simulation process and results are reported in Appendix D.12. The simulations show that the number of contra-hypothesis sieges needed for the main coefficient to lose significance in the six-year window would be between one-sixth and one-third of the size of the current data, depending on the severity of the adversarial simulation. Therefore, a small number of missing sieges is unlikely to overturn the results.

7 Alternative explanations

The Fall of Constantinople was not the only event with implications for siege warfare in the 1450s. Another threat to identification is the existence of simultaneous unobserved confounders that are correlated with proximity to Constantinople after 1453. In this section, I examine two simultaneous shocks: the rise of the Ottoman Empire and the nearly concurrent ends of the War of Milanese Succession and the Hundred Years' War in 1453/1454. These events are potentially problematic because my outcome variable is the raw count of sieges, which cannot distinguish whether declines in siege warfare were due to bargaining settlements or fewer reasons to fight.

7.1 The rise of the Ottoman Empire

7.1.1 Formation of a balancing coalition

The Fall of Constantinople marked the end of the Byzantine Empire, a buffer state between the Catholic West and the Islamic East. Polities located near the Ottoman Empire (and thus Constantinople) may have seen the Fall of Constantinople as heralding a threat to their own security, as the leaders of the Ottoman Empire openly professed a desire to conquer Europe (DeVries, 2017). Under a balance of power logic, the European polities close to Constantinople may have ceased fighting among themselves to form a balancing coalition (Waltz, 1979). More insulated polities in Northern Europe may have been less inclined to join such a coalition. Consequently, another channel through which proximity to Constantinople could influence siege warfare after 1453 is

through a decline in warfare among states near the Ottoman Empire. While the discussion of the no anticipation assumption in Section 5.2 rules out balancing before 1453, it is still possible that states balanced after 1453.

There is little historical evidence that balancing occurred, though some calls for European unity did occur in the aftermath of the Fall of Constantinople, including in regions threatened by Ottoman expansion, such as Italy and the Habsburg domains in Central Europe. Three successive popes—Nicholas V (r. 1447-55), Callixtus III (r. 1455-58), and Pius II (r. 1458-64)—made organizing a crusade to retake Constantinople an objective of their papacies (Pastor, 1906). In addition, Holy Roman Emperor Frederick III held several “Turkish Diets,” in an attempt to corral support among the German princes for a crusade (DeVries, 2017). Yet appetite for a crusade, or even the appearance of unity, was minimal. The inclination of many leaders was to exploit the attempts to organize a crusade for personal benefit. For instance, negotiations in Italy collapsed in 1455 after Alfonso V, King of Naples, attacked Genoa using a fleet of galleys purchased by the Papacy and left in Alfonso’s care for the purposes of a crusade (Schwoebel, 1967). This pattern of disunity continued throughout the decade. The frustration of several attempts at organizing a crusade even led Pope Pius II to exclaim:

I cannot persuade myself that there is anything good in the prospect [of a crusade].
Who will make the English love the French? Who will unite the Genoese and the
Aragonese? Who will reconcile the Germans with the Hungarians and Bohemians?
(Cipolla, 1965, 17)

Pius attributed the reluctance to crusade against the Turks to the fact that Europeans feared each other more than they did the Turks, lamenting: “no king could be found who did not stand in terror of his neighbor and fear to leave his own house empty” (DeVries, 2017, 545).

Attempts at unity were no more successful in the Holy Roman Empire. The German princes were more interested in extracting concessions from Emperor Frederick III or bickering among themselves than constructing a plan to take back Constantinople (Schwoebel, 1967). One chronicler recounted that the princes “had too many quarrels among themselves on their hands to want

another with the Turks” (Schwoebel, 1967, 33). Even rulers who committed to a crusade were waylaid by infighting. For instance, Philip the Good, the Duke of Burgundy, pledged himself to the crusading effort only to reverse course a year later due to the risk of war in his own domain (Davies, 1851).

To supplement this qualitative evidence, I conduct a placebo test to quantitatively assess the hypothesis that polities near Constantinople ceased fighting to balance against the Ottoman Empire. The results are reported in Appendix D.13. I find no evidence that the rate of major wars (as coded by Brecke (1999)) varied with transport cost to Constantinople after 1453. This indicates that it is unlikely that proximity to the expanding Ottoman Empire induced polities to cease fighting each other and form a balancing coalition. Moreover, together with the main analysis, these results imply that while wars did not become less likely near Constantinople, the strategic significance of sieges within conflicts diminished.

While the placebo test suggests that no balancing occurred, it may nevertheless seem at odds with my theory. If better information about the combat effectiveness of cannons resolves uncertainty about military capabilities and makes peaceful settlements more likely, why do we observe fewer sieges but not fewer wars? The explanation has to do with the relative importance of cannons for each type of conflict. While both sieges and wars can result from bargaining failure, sieges are micro-conflicts where artillery has a large effect on which side wins, while wars are macro-conflicts whose outcomes depend on a wide range of factors. In large conflicts, military planners can employ countermeasures to minimize the advantage of artillery in ways that are not possible in sieges. Historians such as Rogers (2018) have documented that as the primacy of gunpowder artillery over fortifications became evident, pitched battles increased in importance. By the mid-15th century, gunpowder artillery could fire projectiles at velocities capable of piercing curtain walls, but issues of low mobility, inaccuracy, and slow rate of fire limited applications to fixed targets and situations where the artillery was not threatened by enemy forces. As a result, gunpowder artillery offered a major advantage to attackers in sieges but not yet in battles. The availability of such countermeasures is one of many additional factors that influence decisions to

go to war. Thus, sieges present a special case of conflict where technology was so instrumental to victory that the effect of information on conflict escalation is detectable.

7.1.2 Other impacts of the rise of the Ottoman Empire

While there is little evidence for the formation of a European balancing coalition, the conquest of Constantinople may also have affected the geography of siege warfare by orienting Ottoman expansion toward Europe. With Constantinople taken, the Ottomans proceeded to conquer Serbia in 1459, the Greek Byzantine remnant in 1460, and Bosnia in 1463 (Babinger, 1978). But except for portions of Bosnia and Serbia, these areas are excluded from the empirical analysis because they lacked Catholic dioceses. The omission of the Balkans may bias the empirical analysis in favor of my hypothesis by excluding near-Constantinople sieges that were part of post-1453 Ottoman conquests.

To address the potential bias caused by the omission of the Balkan conflicts, I transform the dataset to use spatial grid cells with a resolution of about $0.5^\circ \times 0.5^\circ$ as panel units rather than dioceses. Doing so brings in areas outside of Western Europe and adds 34 more sieges from the Balkans to the data. These new sieges are all associated with conflicts between the Ottomans and Balkan polities, such as the Albanian-Turkish Wars, the Crusade of Varna, and the Ottoman conquest of the Byzantine remnant in the Morea. Appendix D.11 reports the results of performing the difference-in-differences design on the grid cell dataset. The obtained coefficients are similar in size and statistical significance for all window sizes, suggesting that the Balkan sieges followed the same pattern as those in Western Europe and that the results are robust to alternative forms of spatial aggregation.

Besides encouraging an invasion of the Balkans or European balancing, there are unlikely to be other consequences of the rise of the Ottoman Empire that could confound a post-1453 relationship between proximity to Constantinople and siege warfare, especially in the short term. Historical studies of European reactions to the Fall of Constantinople suggest that the most common political reaction was a continuation of the status quo (Bisaha, 2017; DeVries, 2017; Schwoebel,

1967). Most European polities already had hostile relationships with the Ottoman Empire and limited contact with the Byzantines prior to 1453. A few polities, like Venice and Genoa, lost trading privileges in Constantinople with the destruction of the Byzantine Empire. Nevertheless, they rapidly adapted and signed treaties with the Ottomans to resume their commercial activity (Schwoebel, 1967).

7.2 The end of the Hundred Years' War and the Peace of Lodi

Two other events with significance for the geography of siege warfare took place almost contemporaneously with the Fall of Constantinople: the end of the Hundred Years' War in October 1453 and the signing of the Peace of Lodi in April 1454. The end of the Hundred Years' War led to peace in France, but conflict in England. England's defeat in the Hundred Years' War generated domestic political turmoil, which in 1455 crescendoed into the series of civil wars today known as the Wars of the Roses. Similarly, the ratification of the Peace of Lodi by Florence, Milan, Naples, and Venice brought an end to thirty years of fighting in Northern Italy (Urquhart, 1852). The peace agreement was preceded by intense fighting in the War of Milanese Succession but ushered in a period of relative peace in Italy.

The consequences of both events changed patterns of conflict across Europe in ways that made conflict relatively less frequent near Constantinople after 1453. The frequency of sieges fell after 1453 in France and Italy, but grew in Great Britain. As France, Great Britain, and Italy contained some of the largest polities during this period, this raises concerns that the results of the main analyses are driven by the Peace of Lodi and the Hundred Years' War rather than learning about artillery use. To account for this issue, I estimate Equation 1 on subsets of the data that exclude polities in France, Great Britain, and Italy. Figure 12 displays the coefficients obtained using six-year (1450-1456), ten-year (1448-1458), and twenty-year (1443-1463) windows when excluding each region individually and collectively. The figure shows that exclusion of these polities does not have a large effect on the magnitude or precision of the coefficients—perhaps indicating the polity-year fixed effects are effective at capturing regional time-varying shocks. Moreover, the

effect size is again decreasing with window size, consistent with the learning hypothesis. The empirical results suggest that it is unlikely that simultaneous events drove a post-1453 decline in siege warfare near Constantinople.

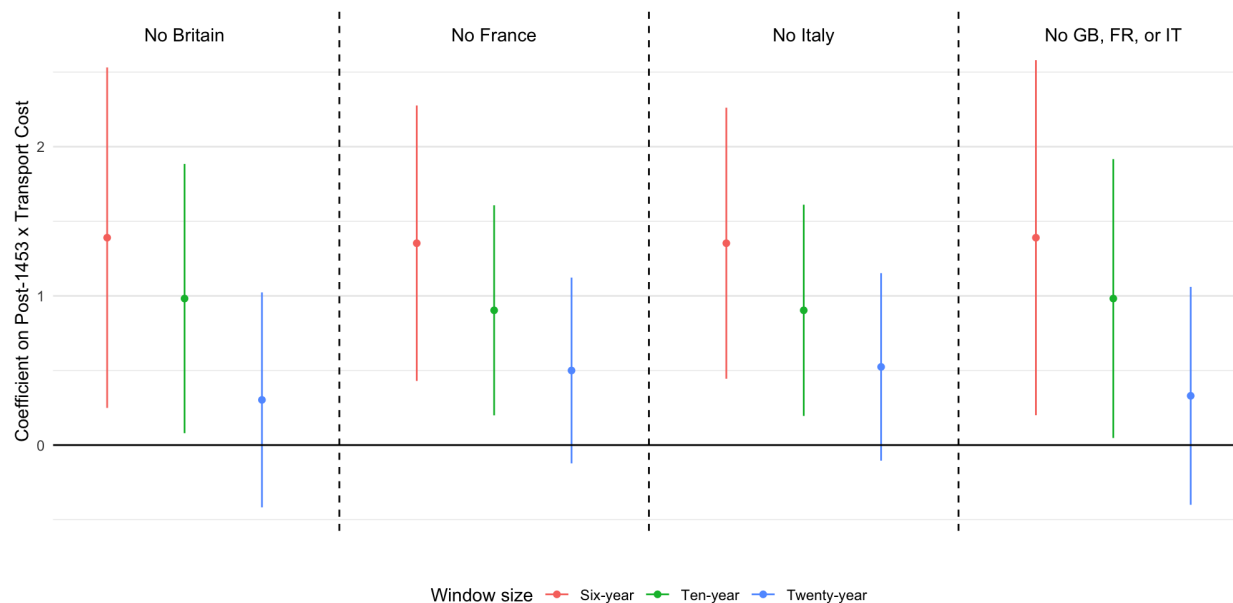


Figure 12: Results when excluding polities involved in simultaneous events

8 Conclusion

I posit that technological change can increase the frequency of international conflict by introducing uncertainty about the balance of power. This uncertainty arises from the fact that states must independently develop doctrine to apply the technology and assess its effect on military capabilities. Such uncertainty can be mitigated if states observe the new technology being used in combat, which provides a common doctrinal reference point to judge the efficacy of the new technology. I test this theory by examining how the Fall of Constantinople in 1453 influenced the frequency of siege warfare across Europe. The Siege of Constantinople featured Ottoman use of gunpowder artillery to great effect against the most sophisticated fortifications in Europe. Anecdotal evidence indicates that Europeans quickly adopted these new tactics and the defenders of castles and settlements realized that existing fortifications could not withstand artillery fire,

leading them to surrender more often when faced with an adversary equipped with cannons. A difference-in-differences approach confirms that locations that learned about the Fall of Constantinople earlier, as measured by transportation cost, experienced fewer sieges after the city's capture. I rule out important alternative explanations for this relationship, strengthening the conclusion that the decline in siege warfare was due to better information about the effectiveness of gunpowder artillery.

These findings suggest that perceptions of how new technologies affect military capabilities are important factors in conflict escalation. A focus on beliefs and doctrine distinguishes my theory from prior research, which has sought to link particular attributes of technologies (i.e., creating offensive advantage) to conflict (Glaser and Kaufmann, 1998; Levy, 1984). Yet the empirical results of this paper indicate that how new technologies impact capabilities contributes less to the outbreak of conflict than the degree to which states agree on what that impact is. As in the case of artillery, an offense-enhancing innovation need not generate conflict if states acknowledge it as such and adjust their bargaining behavior accordingly. Of course, nothing prevents states from seeking countermeasures to the new technological reality. Indeed, after the Fall of Constantinople, rulers modified their military strategies to avoid sieges and invested in new cannon-resistant fortifications (Mangini and Petroff, 2022; Rogers, 2018). Therefore, the relationship between technological change and conflict may be cyclical: unproven technologies developed to counter existing equipment may blur perceptions of the balance of power.

In addition, the new theory and evidence presented by this paper can contribute to ongoing debates about the impact of emerging technologies on international politics. While the empirical test conducted focuses on the historic case of gunpowder artillery, no characteristic of medieval warfare is essential to the theory—the slow pace of information diffusion and high salience of artillery to siege outcomes merely aid a causal test. But the causal logic of the theory and each individual component—variation in initial doctrine, bargaining failure, battlefield demonstrations, and convergence—are just as salient to modern interstate conflict. The fast pace of information diffusion in the present implies that battlefield demonstrations are communicated much more

rapidly than in the past, but convergence should still occur. Indeed, recent reporting indicates that the U.S. Army's drone program is imitating successful innovations from Ukraine, adapting equipment like first-person-view drones and command-and-control links as well as training procedures (Curtatone and Nagl, 2026; Myers, 2026). Future research could examine whether learning from drone use in Ukraine affected decisions to use force in other settings.

Modern interstate war also has aspects that present opportunities to clarify the scope and empirically test the logic of the theory. First, some emerging technologies, such as artificial intelligence (AI), can be employed in a larger variety of ways than older technologies like cannons. Greater versatility may exacerbate initial divergence over doctrine, as wider applicability may encourage militaries to envision a wider range of operational use cases for the technology in combat. AI and other versatile technologies may therefore present greater risks for disagreements over the balance of power and thus bargaining failure. Moreover, for complex and multifaceted technologies, a single battlefield demonstration may not be sufficient to achieve convergence in doctrine. Finally, modern militaries may have a greater interest in and capabilities for concealing technological innovation to retain a potential combat advantage (Kreps, 2025; Slantchev, 2010). Incentives to hide strength can exacerbate diverging expectations about the balance of power and limit postwar learning. Modern conflict therefore offers at least two additional variables that complicate the predictions of my theory. Future research in this area would elaborate the mechanisms of the theory while clearly delimiting its scope.

Finally, the empirical design of this paper can serve as a model for other studies seeking to test predictions of international relations theory. There are other historical episodes similar to the Fall of Constantinople with characteristics that make them natural experiments for theories which are difficult to assess using modern data. But identifying and exploiting such episodes has been relatively rare in the field of international security. While caution must be taken to ensure that historical contexts are analogous to present-day events, scholars seeking to causally test general theories of international politics can use the past as a fruitful source of data (Charnysh, Finkel and Gehlbach, 2023; Torreblanca et al., 2025).

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A Qualitative appendix

A.1 Examples of sources of technological uncertainty

Technical difficulty of assessing new technologies This dynamic is visible in the German nuclear program during the Second World War. The Nazis’ repressive and antisemitic policies against academics created a shortage of scientific talent. This contributed to a false scientific conclusion that graphite was infeasible as a moderator for fission, leading the Germans to focus research on reactors that use expensive and hard-to-obtain heavy water instead (Bethe, 2000; Popp and de Klerk, 2023). By 1942, setbacks convinced German scientists and military leaders that nuclear weapons could not be produced in time to influence the course of the war and returned the nuclear program to civilian control (Bethe, 2000; Popp and de Klerk, 2023).

Different approaches to doctrine An example is the German development of mechanized combined arms warfare and its application in the Battle of France. While the French military establishment appears to have been aware of changes in German doctrine on the use of armor, France’s own doctrine subordinated tanks to the infantry as support units, and as a result perceived them to be far less mobile than the independent armored units Germany would employ during its invasion (Kier, 2017). Moreover, French military officers adopted a defensive mindset following the experience of the First World War, which influenced their view of how technology would be employed in the next war. This is exemplified by a contemporary French tank officer manual: “At the present time, the anti-tank weapon confronts the tank as, during the last war, the machine gun confronted the infantry” (Ministry of War of the French Third Republic, 1937). This attitude led to a belief that German armored units would be slow enough that, together with the Maginot Line, the French army would have enough time to mobilize and repulse a German incursion (Doughty, 1974).

Competition between bureaucratic actors In the interwar period, cavalry officers in the Soviet Union, United Kingdom, and United States organized to lobby against the replacement of roles traditionally performed by horse cavalry by mechanized transport, arguing that tanks and trucks were less versatile than horses. (Katzenbach, 1958; Searle, 2023). Around the same time, there was competition for funding between fighter and bomber aircraft programs within the nascent U.S. Army Air Corps. A so-called “Bomber Mafia” embarked on a campaign to demonstrate the superiority of bombers to fighters, even conducting rigged tests using outdated fighter models to “prove” that bombers could outrace fighters, thereby obviating the need for resources to be committed to fighter interceptor or escort programs (Boyne, 2003; Wagner, 2012). These efforts successfully influenced U.S. air power doctrine, with Chief of Army Air Corps, Henry “Hap” Arnold stating “[fighters] rarely intercept modern bombers except accidentally. Such being the case...it is doubtful whether such operations justify their existence” (Tate, 1998). This in turn provided the justification for the ill-conceived initial Allied strategic bombing campaign against Germany in the Second World War, where bombing runs were planned without fighter escorts under the premise that “the bomber will always get through,” resulting in heavy losses.

A.2 Mentions of the Fall of Constantinople in primary sources

Table A.1 lists mentions of the Fall of Constantinople in the primary sources used to construct the siege dataset. Descriptions of the primary sources can be found in Appendix F.1. Of the 42 primary sources used to construct the siege dataset, 17 (40%) mention the Fall of Constantinople. Every major region of Europe, save Scandinavia, features at least one narrative chronicle that describes the Fall of Constantinople, and several sources contain extended descriptions of how the city was taken. In contrast, outside the English and French chronicles, the Hundred Years' War receives only a mention in Piccolomini, who notes in *Europe* (2013, 204-207) the capture of Bordeaux but does not make any reference to gunpowder or the fact that France reclaimed Normandy and Gascony.

Source	Pages	Location
<i>Annals of Jan Długosz</i>	pg. 516	Poland
<i>Augsburg Chronicle</i>	pg. 325	Bavaria/Holy Roman Empire
<i>Chronica de El-Rei D. Affonso V</i>	Vol. 2, pg. 133	Portugal
<i>Chronicle of the Hungarians</i>	pgs. 172-173	Hungary
<i>Chronique de Charles VII, Roi de France</i>	Vol. 3, pgs. 19-35	France
<i>Chronique de Mathieu D'Escouchy</i>	Vol. 2, pgs. 50-58	Burgundy and France
<i>Cronaca di Anonimo Veronese</i>	pgs. 50-51	Venice/Italy
<i>Cronaca di Ser Guerriero da Gubbio</i>	pg. 64	Urbino/Italy
<i>Cronica de Enrique IV</i>	pg. 143	Castile
<i>Dietaris de la Generalitat de Catalunya</i>	pg. 120	Catalonia
<i>Die Geschichte Kaiser Fredrichs III</i>	Vol. 2 pg. 294	Austria/Holy Roman Empire
<i>Koelhoff's Chronicle</i>	pg. 796	Westphalia/Holy Roman Empire
<i>La Sforziada</i>	pg. 407	Milan/Italy
<i>Piccolomini's Europe</i>	pgs. 93-100	Europe
<i>Speierische Chronik</i>	pg. 392	Rhineland/Holy Roman Empire
<i>Storia di Milano</i>	Vol. 3 pg. 205	Milan/Italy
<i>Vitellius Chronicle of London</i>	pg. 164	England

Table A.1: Mentions of Constantinople in primary sources

Figures A.1 and A.2 contain two translated passages from Aeneas Silvius Piccolomini's *Europe* and Janos Thuróczy's *Chronicle of the Hungarians* which emphasize both the strength of the city's walls and the innovative military tactics used by the Turks. Together the primary sources demonstrate that contemporaries viewed the Fall of Constantinople as a singular event, and were aware of the military implications.

Meanwhile, Mehmed mustered forces from all quarters and attacked the imperial city by land and sea with impressive armaments and formidable vigor. He excavated tunnels and hidden ditches of great depth; raised a broad ramp; rapidly erected a pontoon bridge, extending for two miles of the city's length, where the sea laps the walls on the side facing the town of Pera; built wooden siege towers of such a height that they topped the city walls, though these were very high; and installed a variety of siege-engines and artillery...The main walls of [Constantinople] were famous throughout the world for their height and thickness... the Turks bore down more fiercely and, since a part of the wall had been shattered by bronze cannon and filled up most of the ditch, they climbed over the debris, scaled the outer wall, and dislodged the Greeks.

Figure A.1: Piccolomini, *Europe*, Brown's (2013) translation, 94-97

And after collecting countless troops from every source and setting in motion all the instruments of war in his kingdom, with a blockade by land and sea [Mehmed II] violently and arrogantly attacked that most celebrated city of Constantinople, formerly called Byzantium, child of the ancient emperors and mistress of the world. And with the unremitting attacks of his arms he so devastated the sad city, which was begging for reinforcements from Christian kings and yet was abandoned by everyone to disaster and destruction, that even women were compelled to come to its defense. But what good was it for the Greeks to raise a mismatched hand against so many countless foes? Devising a new kind of assault, the sultan had subterranean tunnels and trenches excavated, which he used for an ambushade; and when the sea had been spanned both with chains and by a bridge, and wooden towers had been erected to such a height that they were taller than the walls of the city, which were exceedingly tall, and from their tops the Turks were with thick showers of arrows striking everything that was being mobilized in the city, he caused, with the additional use of an assemblage of siege-engines and ballistas of many kinds and of unforeseen size, the city's walls to be reduced to debris.

Figure A.2: Thuróczy, *Chronicle of the Hungarians*, Mantello's (1991) translation, 172

A.3 Diffusion of initial news about the Fall of Constantinople

Table A.2 describes the sources used in the construction of Figure 4. In this first wave, few specifics about what occurred during the siege were transmitted. The initial wave of information traveled quickly by medieval standards, diffusing throughout the continent in roughly six months, spread primarily by letters. An excerpt from a translated letter sent by Aeneas Silvius Piccolomini to Pope Nicholas V on July 12, 1453 is provided in Figure A.3.

Location	Date	Source
Crete/Candia	Early June 1453	A Greek refugee ship reached Crete in early June according to Schwoebel (1967, 1), citing a primary source written by a monk at the monastery of Agarathos.
Belgrade/Serbia	Mid-June 1453	News of the fall of the city reached Despot Durađ Branković overland sometime in June (Pertusi, 1976, xxvii; Schwoebel, 1967, 3).
Venice	June 29, 1453	Three ships sailing from Constantinople reached Venice on June 29 (Schwoebel, 1967, 1; Pastor, 1906, 271-272).
Genoa	July 6, 1453	A letter from Venice about the Fall of Constantinople reached Genoa on July 6 (Schwoebel, 1967, 2).
Rome	July 8, 1453	A letter sent from Venice to Pope Nicholas V in Rome arrived on July 8 (Schwoebel, 1967, 2; Pastor, 1906, 272).
Krakow	Early July 1453	The Annals of Jan Długosz state that news of the Fall of Constantinople reached the court of Casimir IV at Krakow via Moldavia in "early July" (Michael, 1997, 516).
Graz	By July 12, 1453	A letter from Aeneas Silvius Piccolomini to Pope Nicholas V dated July 12 relates recent news of the city's fall reaching Graz from Rascia (Serbia) (Von Cotta-Schönberg, 2025, 235).
Barcelona	July 21, 1453	The Dietari de la Generalitat de Catalunya record a messenger arriving from Genoa on July 21, bringing news of Constantinople's fall (Generalitat de Catalunya, 1994, 120).
Wroclaw	By August 1453	Estimated from the movements and correspondence of Saint John Capistran, then an itinerant preacher in Poland (Cygielman, 1987; Fitzgerald, 1911; Schwoebel, 1967).
London	August 1453	Estimated from the reference to news of the city's fall provided in the Vitellius Chronicle, which puts it at the end of the mayoralty of Godfrey Fielding and Spufford's (2003, 27) approximation using historical letters that about 30 days were needed for a letter to travel from Northern Italy to London.

Table A.2: Arrival of initial news about Fall of Constantinople with sources

But what about the horrible news from Constantinople? My hand trembles as I write this; the mind shudders, and indignation bids me to speak while pain prevents it. Alas, poor Christianity! We are ashamed to live and should happily have died before this catastrophe! While Germany, France and most of Spain were safe, we - oh, shame! - allowed the illustrious City of Constantinople to be captured by the effeminate Turks. I believe Your Piety has already heard it: I leave Germany and pass on to Greece. With great forces, the Turkish ruler recently besieged Constantinople from land and sea and conquered it, having brought in machines of war and stormed it three times. He killed the whole population, tortured the priests to death, and spared neither sex nor age. They say that more than 40,000 people were killed. These events have been reported by people coming from Rascia. They also tell that the Paleologue emperor was beheaded and that his son took refuge in Pera, now under siege. I shall not describe each of the events since others have undoubtedly informed Your Holiness about them better and more correctly than I can. It is enough to mourn, with Your Piety, such a great catastrophe for Christianity.

In this unhappy year, the City of Constantinople, which had lasted for 1,100 years and never come into the power of the infidels, was sacked by the foul Turks. Rome, too, is reported to have been sacked by the Goth King Alaric, but he ordered that the temples of the saints should be spared. Who doubts that the Turks will attack God's churches ferociously? I grieve that Hagia Sophia, the most famous church in all the world, has been ruined or polluted. I grieve that saints' basilicas without number, built with wondrous skill, should suffer ruin or desecration by Mehmed. What shall I say about the countless books not yet discovered by the Latins? Alas, how many names of great men will disappear! This is Homer's second death; this is Plato's second demise. Where shall we now search for the genius of philosophers and poets? The fountain of the Muses flows no longer. Hopefully, we have enough intellect left to adequately deplore this catastrophe. Holy Father, these events lie heavily on my mind (Von Cotta-Schönberg, 2025, 235).

Figure A.3: Letter from Aeneas Silvius Piccolomini to Pope Nicholas V, July 12, 1453, translated by Von Cotta-Schönberg (2025)

B Data appendix

B.1 Sieges by war

Conflict	Location	Duration	Number of Sieges
Albanian-Turkish Wars	Southwestern Balkans	1432-1479	5
Albanian-Venetian Wars	Albania	1447-1448	3
Anglo-Scottish Border Wars	Northern England	1440-1460	4
Armagnac Revolt	France	1440s	4
Austrian-Hungarian Border Conflicts	Eastern Austria	15th century	1
Bavarian War / Princes' War	Bavaria	1459-1463	17
Bonville-Courtenay Feud	Devon	1449-1455	3
Byzantine-Ottoman Wars	Anatolia and Greece	1299-1462	8
Castile-Granada Conflicts	Granada	718-1492	11
Castilian Civil War	Castile	1437-1451	7
Catalan Civil War	Catalonia	1462-1472	8
Cleves-Cologne Conflict	Rhineland	1443	1
Colville-Douglas Feud	Scotland	1449	1
Crichton Rebellion	Scotland	1454	1
Crichton-Douglas Feud	Scotland	1443-1445	5
Crusade of Varna	Bulgaria and Serbia	1444	8
Czech Civil War	Bohemia	1450-1451	4
Czech-Polish Conflict of 1453	Bohemia and Poland	1453	2
Dano-Swedish War	Norway and Sweden	1448-1456	7
Donia War	Frisia	1458-1463	3
Douglas Rebellion	Scotland	1453-1455	4
Dunbar Revolt	Scotland	1446	1
Dundas Revolt	Scotland	1449	1
Fajardo Feud / Murcian Civil War	Murcia	1448-1450	3
First Margrave War	Franconia	1449-1450	6
Hohenberg-Wittelsbach Feud	Saxony	1447	1
Hundred Years' War	France	1337-1453	43
Hunyadi-Jiskra Conflict	Bohemian-Hungarian Border	1447-1452	8
Jack Cade's Rebellion	Southeast England	1450	1
Mainz Diocesan Feud	Mainz	1461-1462	2
Metz Campaign	Northeast France	1444	4
Milanese Succession War	Lombardy	1447-1454	33
Navarrese Civil War	Navarra	1451-1455	6
Old Zurich War	Switzerland	1440-1446	11
Palma Revolt	Mallorca	1450-1452	2
Recovery of Luxembourg	Luxembourg	1443	3
Revolt Against Arnold van Egmond	Netherlands	1459	1
Revolt Against Christian I	Sweden	1463	1
Revolt Against Frederick III	Austria	1452	2
Revolt of Antonio Centelles	Calabria	1444-1445	4
Revolt of Ghent	Flanders	1449-1453	9
Revolt of Hans von Rechberg	Austria	1452	2
Saxon Fratricidal War	Saxony	1446-1451	6
Silesia Wars	Silesia	1443	1
Soest Feud	Westphalia	1444-1449	5
Thirteen Years' War	Pomerania and Prussia	1454-1466	29
Tir Connaill Civil War	County Donegal	1454	1
Turkish-Hungarian Wars	Hungary and Romania	1366-1526	10
Ulrich of Celje's Conquest of Slavonia	Croatia	1445	2
Uprising of Evrard de la Marck	Wallonia	1445	2
Utrecht War	Utrecht	1455-1457	5
Waldenfels Feud	Upper Franconia	1441-1446	2
War of Deposition against Karl Knutsson 1457	Sweden	1457	2
War of Neapolitan Succession	Southern Italy	1458-1462	4
Wars in Lombardy	Lombardy	1423-1454	55
Wars of the Roses	England and Wales	1455-1487	26
Wasselonne War	Alsace	1446-1448	2
Unknown/Minor			28

Table A.3: List of wars, 1443-1463

B.2 Dioceses and sieges by polity

Polity	Num. Dioceses	Num. Sieges
Bishopric of Ösel-Wiek	1	0
Counties of Hainaut and Holland	1	0
County of Corsica	2	0
County of Holstein	1	0
County of Mantua	1	0
County of Provence	16	3
County of Savoy	13	1
County of Savana	2	1
Crown of Aragon	15	11
Crown of Bohemia	4	4
Crown of Castile	28	16
Duchy of Bar	1	0
Duchy of Brabant	1	0
Duchy of Brittany	9	1
Duchy of Brunswick-Lüneburg	1	0
Duchy of Finland	1	0
Duchy of Lorraine	1	0
Duchy of Lower Bavaria in Straubing	1	0
Duchy of Mazovia	1	1
Duchy of Mecklenburg	2	1
Duchy of Milan	29	53
Duchy of Split under Hrvoje Vukčić	3	1
Duchy of Spoleto	6	0
Duchy of Upper Bavaria-Munich	2	3
Duchy of Zachlunia	3	0
Earldom of Desmond	4	0
Earldom of Orkney	1	0
Earldom of Ormonde	5	0
Earldom of Ulster	3	0
Electorate of Brandenburg	3	1
Garay Lordship	10	1
Giudicato of Arborea	18	0
Habsburg Dominions	13	12
House of Este	5	1
Kalmar Union between the Kingdoms of Denmark, Sweden and Norway	19	4
Kingdom of Bosnia	2	1
Kingdom of Breifne	3	0
Kingdom of England	28	40
Kingdom of France	81	30
Kingdom of Granada	5	7
Kingdom of Hungary	9	11
Kingdom of Leinster	1	0
Kingdom of Naples	142	4
Kingdom of Navarre	2	3
Kingdom of Portugal	10	0
Kingdom of Scotland	8	11
Kingdom of Sicily	11	0
Kingdom of Thomond	4	0
Kingdom of Tir Conaill	1	0
Kingdom of Tir Eógain	3	0
Landgraviate of Thuringia and Margravate of Meissen	1	3
Lordship of Bologna	2	2
Lordship of Caithness	1	0
Lordship of Cilli	1	1
Lordship of Connacht in Mayo	3	0
Lordship of Connacht in Sligo Under the O Conor Sligo	1	0
Lordship of Cortona	1	0
Lordship of Forli	1	0
Lordship of Padua	1	0
Lordship of Perugia	1	0
Lordship of Rimini	1	0
Lordship of the Isles	3	0
Lordship of Travunia	1	0
Lordship of Zeta and The Sea	4	0
Lordships of Romagna	5	0
Lordships of the House of Burgundy	7	10
Mac Carthy Mor Lordship	1	0
March of Ancona	10	25
Margravate of Montferrat	2	0
Margravate of Moravia	1	0
Margravate of Savona	1	0
Monastic State of the Teutonic Knights	9	28
Montefeltro Lordship	6	3
Principality of Moldavia	1	0
Principality of Wales	1	2
Republic of Florence	4	3
Republic of Genoa	9	0
Republic of Venice	12	4
Small States of the Holy Roman Empire	28	50
State of the Church	30	3
United Kingdom of Poland and Lithuania	10	2

Table A.4: List of polities, 1400 A.D.

B.3 Dioceses with most sieges

Diocese	Num. Sieges	Diocese	Num. Sieges
Brescia	12	Utrecht	10
Milano	11	Durham	9
Konstanz	11	Warmia (Ermland)	9
Cremona	9	Gniezno	8
Bordeaux	9	Pomesania	7
St. Andrews	8	Augsburg	6
Esztergom	8	Mainz	4
Pesaro	6	Carlisle	3
Cambrai	6	St. Asaph	3
Rouen	6	Regensburg	3

(a) Before FoC

(b) After FoC

Table A.5: Dioceses with most sieges

B.4 Descriptive statistics

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
Num. sieges	14616	0.027	0.24	0	0	0	9
Num. towns	14616	2.6	4.9	0	0	3	45
Num. castles	14616	5.6	17	0	0	5	222
On trade route	14616	0.31	0.46	0	0	1	1
Distance from border (km)	14595	166	181	0	44	235	1011
Transport cost	14616	8106	2365	3475	6133	10346	14318
Distance (km)	14616	1809	647	731	1261	2282	3204
Travel time (days)	14616	30	8	12	24	36	52

Table A.6: Descriptive statistics

B.5 Covariate maps

This section contains maps depicting the distribution of relevant spatial variables.

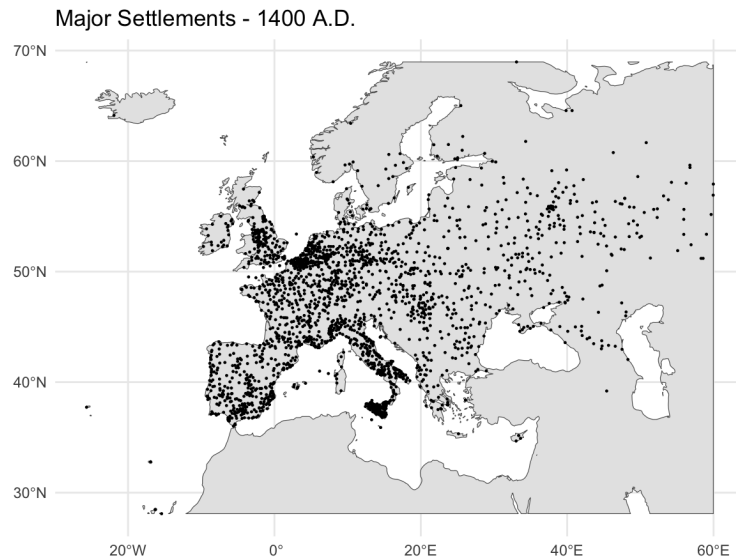


Figure A.4: Settlements with 1,000 or more inhabitants, 1400 A.D.

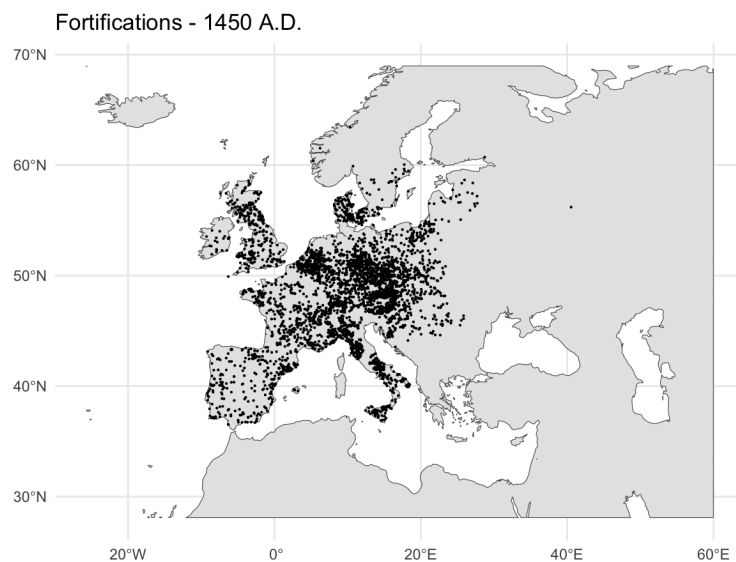


Figure A.5: Non-settlement fortifications, 1450 A.D.

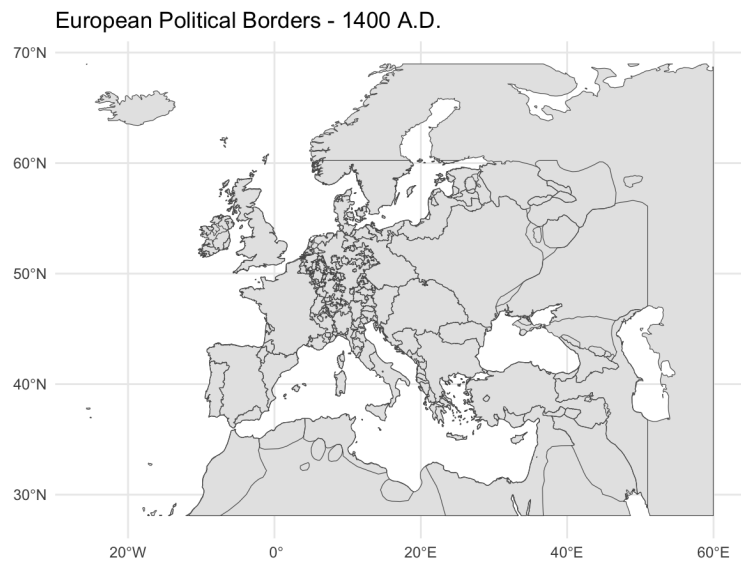


Figure A.6: Political boundaries, 1400 A.D.



Figure A.7: Medieval overland trade routes, from Shepherd (1911)

C Identification assumptions

C.1 Event study specification and results

To conduct the event study reported in Section 5.2, I estimate the following specification via Poisson pseudo-maximum likelihood

$$E[\text{sieges}_{dt} \mid \beta_d, \gamma_t] = \exp \left(\sum_{\tau=1443}^{1463} \alpha_{\tau} \mathbf{1}_{\tau=t} \text{transportCost}_d + \beta_d + \gamma_t \right)$$

where d indexes dioceses and t indexes years. The term β_d is a diocese fixed effect and γ_t is a year fixed effect. The variable sieges_{dt} is the number of sieges occurring in diocese d in year t . The variable transportCost_d is the transport cost of diocese d from Constantinople. The transport cost is interacted with the function $\mathbf{1}_{\tau=t}$, which is an indicator function that takes value 1 when the year is equal to t and 0 otherwise. The coefficients of interest are the α_t terms, which represent the effect of a one-unit increase in transport cost during year t . I cluster standard errors at the level of the polity to account for spatial correlation in sieges. The results are in Table A.7.

	Num. sieges
	Model 1
1443 × Transport Cost	0.09 (0.28)
1444 × Transport Cost	0.01 (0.29)
1445 × Transport Cost	0.13 (0.26)
1446 × Transport Cost	−0.42 (0.38)
1447 × Transport Cost	0.02 (0.22)
1448 × Transport Cost	0.15 (0.14)
1449 × Transport Cost	0.32* (0.19)
1450 × Transport Cost	0.52** (0.24)
1451 × Transport Cost	0.07 (0.22)
1452 × Transport Cost	0.30* (0.18)
1454 × Transport Cost	0.45* (0.24)
1455 × Transport Cost	0.69*** (0.26)
1456 × Transport Cost	0.68 (0.43)
1457 × Transport Cost	0.81** (0.40)
1458 × Transport Cost	0.54* (0.32)
1459 × Transport Cost	0.42 (0.50)
1460 × Transport Cost	0.52** (0.21)
1461 × Transport Cost	0.46** (0.22)
1462 × Transport Cost	0.14 (0.23)
1463 × Transport Cost	0.45 (0.34)
DV Mean:	0.13
Num. clusters:	39
N	3024

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by polity. Unit of analysis is diocese-year.

Table A.7: Event study of exposure to Constantinople on sieges (transport costs)

C.2 Event study minimum detectable effects

I compute the minimum detectable effect for each event study coefficient with a significance level of 0.05 and a power level of 0.80, displayed in the left column of Table A.8. The right column reports the corresponding absolute effect on the frequency of sieges per diocese-year for a one standard deviation decrease in transport cost.

Year	Min. Detectable Coef.	Δ Siege Freq. ($\downarrow$ 1 SD)
1443	0.778	-0.110
1444	0.806	-0.112
1445	0.716	-0.107
1446	1.071	-0.121
1447	0.628	-0.102
1448	0.379	-0.078
1449	0.520	-0.093
1450	0.663	-0.104
1451	0.608	-0.100
1452	0.494	-0.090
1454	0.681	-0.105
1455	0.733	-0.108
1456	1.208	-0.124
1457	1.114	-0.122
1458	0.895	-0.115
1459	1.391	-0.126
1460	0.586	-0.098
1461	0.623	-0.101
1462	0.632	-0.102
1463	0.947	-0.117

Table A.8: Event study minimum detectable effects

C.3 Event study with two-year periods

I conduct another event study of the effect of transportation costs to Constantinople on siege frequency where sieges are aggregated at the diocese-two-year-level. Given the sparsity of sieges, binning periods together provides greater statistical power to detect medium-term trends but is less sensitive to yearly swings. I estimate the following specification via Poisson pseudo-maximum likelihood

$$E[\text{sieges}_{dt} \mid \beta_d, \gamma_t] = \exp \left(\sum_{\tau=0}^9 \alpha_{1444+2\tau} \mathbf{1}_{1444+2\tau=t} \text{transportCost}_d + \beta_d + \gamma_t \right)$$

where d indexes dioceses and t indexes two-year periods. The term β_d is a diocese fixed effect and γ_t is a two-year-period fixed effect. The variable sieges_{dt} is the number of sieges occurring in diocese d in two-year period t . The variable transportCost_d is the transport cost of diocese d from Constantinople. The transport cost is interacted with the function $\mathbf{1}_{\tau=t}$, which is an indicator function that takes value 1 when the two-year period is equal to t and 0 otherwise. The coefficients of interest are the α_t terms, which represent the effect of a one-unit increase in transport cost during the two-year period t . The omitted period is 1452-1453. The year 1443 is dropped due to a lack of data for 1442. The resulting coefficients are plotted in Figure A.8, with vertical bars representing 95% confidence intervals computed using robust standard errors clustered at the polity level. The coefficients and standard errors are also reported in Table A.9.

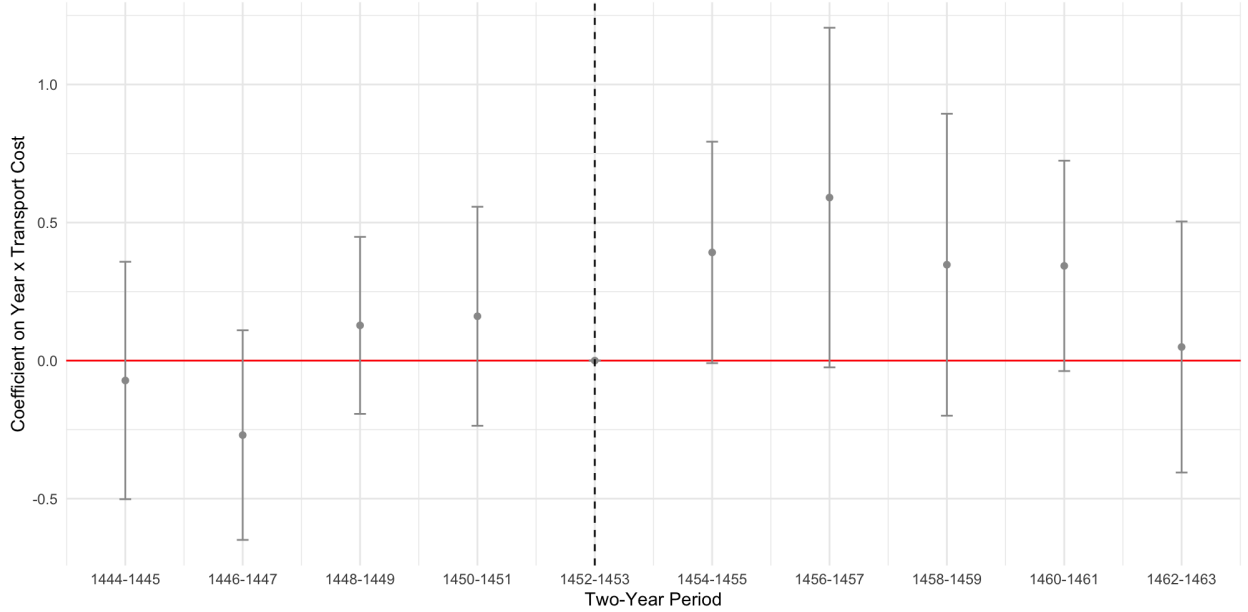


Figure A.8: Event study plot, two-year periods

The estimates, while noisy, show a lack of correlation between transportation cost and siege frequency before 1453, but a positive correlation after 1453, especially in 1454-1457. After 1458, the correlation disappears. This provides evidence for both a lack of pre-trends and a short-term post-1453 effect.

	Num. sieges
	Model 1
1444 $\times$ Transport Cost	−0.07 (0.22)
1446 $\times$ Transport Cost	−0.27 (0.19)
1448 $\times$ Transport Cost	0.13 (0.16)
1450 $\times$ Transport Cost	0.16 (0.20)
1454 $\times$ Transport Cost	0.39* (0.20)
1456 $\times$ Transport Cost	0.59* (0.31)
1458 $\times$ Transport Cost	0.35 (0.28)
1460 $\times$ Transport Cost	0.34* (0.19)
1462 $\times$ Transport Cost	0.05 (0.23)
DV Mean:	0.27
Num. clusters:	39
N	1400

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by polity. Unit of analysis is diocese-two-year.

Table A.9: Event study of exposure to Constantinople on sieges (transport costs, two-year periods)

C.4 Alternative treatment location placebos

In this section, I further assess the parallel trends assumption by conducting a series of placebo tests that examine whether proximity to other besieged cities also influenced the frequency of siege warfare. I select five other politically significant sieges that occurred between 1443 and 1463, and test whether proximity to these sieges had an effect on siege warfare in a three-year pre/post window. The sieges I use for the placebos are:

1. Belgrade (July 2, 1456 to July 22, 1456): Hungarian and Serbian defenders successfully defended Belgrade from an Ottoman siege, temporarily halting the Ottomans' westward advance. The defense of Belgrade was hailed by contemporaries as one of the most important Christian victories against the Ottomans.
2. Bordeaux (August 13, 1453 to October 19, 1453): The final engagement in the Hundred Years' War. France conquered the final English stronghold in Gascony, recapturing all continental territory held by England save for Calais.
3. Caen (June 5, 1450 to July 1, 1450): One of the most important English strongholds in Normandy, its capture by the French in July 1450 led to the collapse of English forces in the region, precipitating the French conquest of Normandy by August.
4. Milan (mid-September 1449 to February 25, 1450): After a five-month siege, the condottiero Francesco Sforza captured Milan, his former employer, becoming Duke of Milan shortly thereafter.
5. Wiener Neustadt (August 29, 1452 to September 4, 1452): Holy Roman Emperor Frederick III was besieged by the Austrian Estates in his main residence at Wiener Neustadt. The estates forced Frederick to relinquish guardianship of twelve-year-old Ladislaus the Posthumous, who held the titles of Duke of Austria, King of Croatia, and King of Hungary.

Table A.10 reports the correlation coefficients between transport costs for the various placebo sieges as well as Constantinople. Figures A.9 through A.13 plot transport costs for each of the alternative treatment sieges.

	Belgrade	Bordeaux	Caen	Constantinople	Milan	Wiener Neustadt
Belgrade	1.00	-0.53	-0.88	0.94	0.71	0.87
Bordeaux	-0.53	1.00	0.81	-0.37	-0.05	-0.74
Caen	-0.88	0.81	1.00	-0.79	-0.49	-0.89
Constantinople	0.94	-0.37	-0.79	1.00	0.75	0.73
Milan	0.71	-0.05	-0.49	0.75	1.00	0.59
Wiener Neustadt	0.87	-0.74	-0.89	0.73	0.59	1.00

Table A.10: Correlation between transport costs

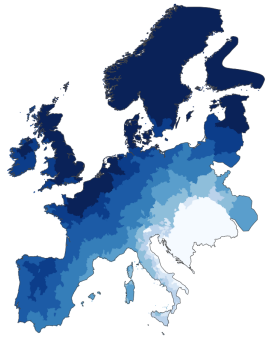


Figure A.9: Transport cost to Belgrade

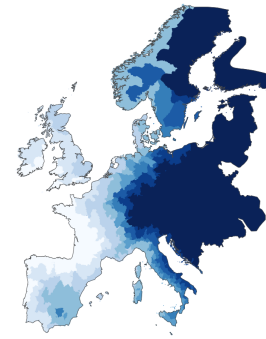


Figure A.10: Transport cost to Bordeaux

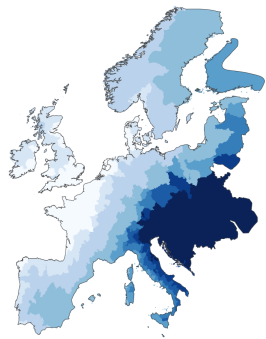


Figure A.11: Transport cost to Caen

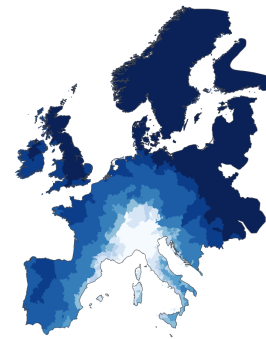


Figure A.12: Transport cost to Milan

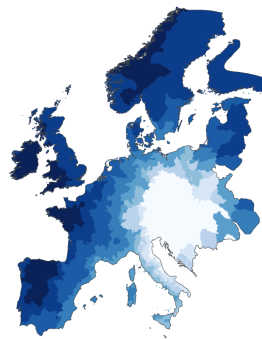


Figure A.13: Transport cost to Wiener Neustadt

Table A.11 displays the coefficients obtained from the placebo tests. For no placebo treatment siege does transport cost have a significant effect on siege warfare, in specifications using both two-way fixed effects and polity-year fixed effects. The results of the tests support the parallel-trends assumption and show that Bordeaux/Caen are unlikely to have been alternative demonstration points for the effectiveness of cannons.

	Belgrade (1456 $\pm$ 3)		Bordeaux (1453 $\pm$ 3)		Caen (1450 $\pm$ 3)		Milan (1450 $\pm$ 3)		Wiener Neustadt (1452 $\pm$ 3)	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Siege Date $\times$ Transport cost (1000s)	0.19 (0.14)	0.17 (0.44)	0.10 (0.13)	-0.59 (0.52)	0.13 (0.11)	0.65 (0.81)	0.04 (0.12)	-0.07 (0.30)	-0.01 (0.10)	0.45 (0.40)
DV Mean:	0.28	0.63	0.31	0.70	0.37	0.61	0.37	0.52	0.33	0.68
Polity $\times$ Year FEs	-	✓	-	✓	-	✓	-	✓	-	✓
Num. clusters:	40	24	57	41	71	55	71	81	71	55
N	280	83	399	142	497	251	497	419	497	206

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.11: Siege location placebos

C.5 Alternative treatment year placebo tests

The event study displayed in Figure 6 shows that there is a positive correlation between transportation cost and siege frequency in some years prior to 1453. To further test for the existence of pre-trends, I conduct a series of placebo tests using each of the years between 1448 and 1451 as alternative treatment dates.

Tables A.12 and A.13 report coefficients and standard errors obtained from performing the DiD analysis on alternate treatment years for the six-year (± 3 years around treatment year) and ten-year (± 5 years) windows. I include coefficients obtained from both the base TWFE analysis and including polity $\times$ year fixed effects. All estimates, save for the one obtained from TWFE on the six-year window around 1448, are not statistically significant at the 5% or 10% level. Moreover, the result for 1448 is not robust to inclusion of polity $\times$ year fixed effects or alternative standard error computations, unlike the 1453 estimates.

	1448		1449		1450		1451	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Treat. Year $\times$ Transport Cost (1000s)	0.28** (0.12)	-0.29 (0.32)	0.16 (0.12)	-0.23 (0.31)	-0.15 (0.15)	-0.71 (0.50)	-0.08 (0.14)	-0.59 (0.66)
DV Mean:	0.31	0.61	0.33	0.60	0.37	0.61	0.35	0.64
Polity $\times$ Year FEs	-	✓	-	✓	-	✓	-	✓
Num. clusters:	76	62	73	61	71	55	73	51
N	532	241	511	249	497	251	511	216

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.12: Alternative treatment date placebos (six-year window)

	1448		1449		1450		1451	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Treat. Year $\times$ Transport Cost (1000s)	0.08 (0.10)	-0.11 (0.24)	0.11 (0.10)	0.04 (0.26)	0.04 (0.11)	-0.24 (0.44)	0.11 (0.12)	-0.59 (0.66)
DV Mean:	0.23	0.50	0.23	0.51	0.23	0.52	0.23	0.52
Polity $\times$ Year FEs	-	✓	-	✓	-	✓	-	✓
Num. clusters:	105	88	105	83	99	79	90	75
N	1155	477	1155	442	1089	412	990	397

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.13: Alternative treatment date placebos (ten-year window)

The results suggest that the pre-1453 correlations are driven by other idiosyncratic events, such as the near-simultaneous resumption of the Hundred Years' War in 1449 and the truce in Milanese War of Succession for the coronation of Frederick III in Rome, and not a persistent pre-1453 trend.

C.6 Parallel trends plot

In Figure A.14 I plot the mean number of sieges taking place in dioceses with greater- or lower-than average transport costs to Constantinople. Sieges are aggregated at five-year intervals to account for the low number of sieges occurring in individual years, and are from dioceses that experienced at least one siege during the period under study. The figure shows that the groups experienced similar numbers of sieges before 1453 but diverged after 1453, with dioceses with higher than average transport costs to Constantinople experiencing more sieges.

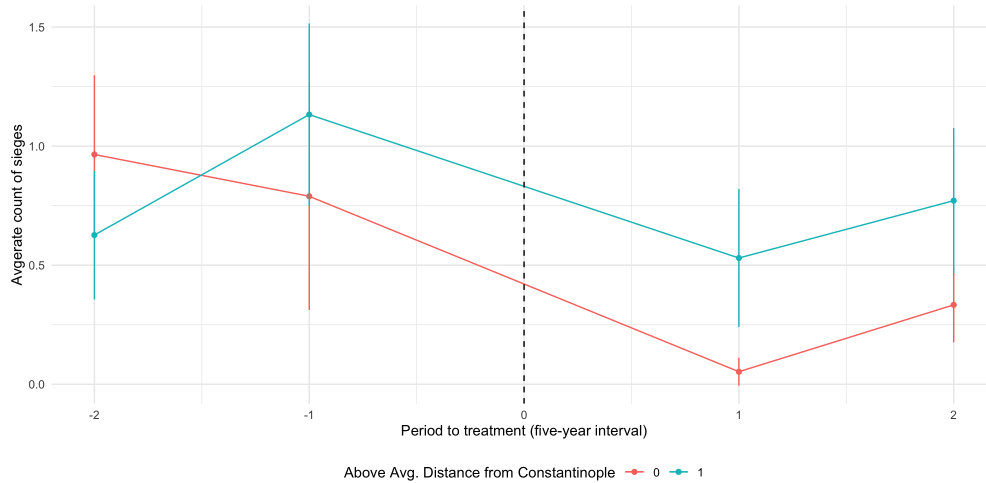


Figure A.14: Trends in sieges

While it may appear that prior to 1453, the frequency of sieges trended upwards in above-average proximity dioceses and downwards in below-average proximity dioceses, it is important to note the large confidence intervals. Indeed, a difference-in-means test for each period, shown in Table A.14, reveals that the average siege frequencies by proximity to Constantinople are not statistically distinguishable in the periods before 1453, but they are post-1453. These comparisons of raw siege counts do not prove parallel trends on their own but they do provide some suggestive evidence that the unconditional frequency of siege warfare did not vary significantly by proximity to Constantinople before 1453. In addition, the post-1453 decline in sieges is consistent with the theory that better information about the effectiveness of cannons when employed correctly led to fewer sieges. The sieges are aggregated by wide enough time intervals that information about the Fall of Constantinople would have fully diffused throughout the continent, leading to a secular decrease in sieges.

Period	Difference in Mean (Above-Below)	Standard Error	Z-score	P-value
1444-1448	-0.338	0.219	-1.548	0.122
1449-1453	0.343	0.312	1.099	0.272
1454-1458	0.477	0.151	3.167	0.002
1459-1463	0.438	0.175	2.500	0.012

Table A.14: Difference in means tests by period

C.7 Treatment effect homogeneity interactions

To assess whether the assumption of treatment effect homogeneity/constant semi-elasticity is plausible, I interact the DiD coefficient with a variety of diocese-level controls. If the triple interaction is statistically significant or the magnitude of the DiD coefficient changes, this could suggest the treatment effect varies over the relevant coefficient. I estimate

$$E[sieges_{dpt} | \beta_d, \gamma_t, \delta_{pt}, X_d] = \exp(\alpha \text{post1453}_t \times \text{transportCost}_d \times X_i + \text{post1453}_t \times \text{transportCost}_d + \beta_d + \gamma_t)$$

where X_i is a pre-1453 diocese-level control and the rest of the terms are defined analogously as in the main analysis. I use four control variables to assess heterogeneity: the number of settlements in a diocese with a population of at least 1,000, drawn from Buringh (2021); the number of castles/fortresses extant in 1450, drawn from Cappelen and Hariri (2026), with gaps filled in the Baltic states, Ireland, Slovenia, and Switzerland using a scrape of Wikidata; an indicator variable for whether a diocese is on an overland trade route, georeferenced from a historical atlas by Shepherd (1911), and distance of a diocese from a political border, computed using the historical atlas by Nüssli (2009). I also estimate models with polity-year fixed effects. The results are shown in Table A.15. Columns 1 and 5 report the estimates with no interactions. All models use three-year windows.

	TWFE					Polity-year FEs				
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Post-1453 $\times$ Transport cost (1000s)	0.36** (0.16)	0.37** (0.16)	0.40** (0.16)	0.46*** (0.17)	0.36** (0.16)	1.35*** (0.46)	1.20** (0.49)	1.45*** (0.39)	1.45** (0.60)	1.13* (0.59)
Post-1453 $\times$ Transport cost $\times$ Border dist. (km)		-0.00 (0.00)					-0.00 (0.00)			
Post-1453 $\times$ Transport cost $\times$ Num. castles			-0.00 (0.00)					-0.00 (0.00)		
Post-1453 $\times$ Transport cost $\times$ Num. settlements				-0.00 (0.00)					-0.00 (0.01)	
Post-1453 $\times$ Transport cost $\times$ On trade route					-0.00 (0.07)					-0.09 (0.11)
DV Mean:	0.31	0.31	0.31	0.31	0.31	0.70	0.70	0.70	0.70	0.70
Polity $\times$ Year FEs	-	-	-	-	-	✓	✓	✓	✓	✓
Num. clusters:	57	57	57	57	57	41	41	41	41	41
N	399	399	399	399	399	142	142	142	142	142

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.15: Interaction term tests for heterogeneity

For all models, the interaction terms are near 0 in magnitude and statistically insignificant. In addition, the magnitude of the coefficients on Post-1453 $\times$ Transport cost do not shift substantially when interaction terms are included. While these tests are limited in that they impose a specific type of linear heterogeneity, the null results provide some suggestive evidence in support of treatment effect homogeneity.

D Robustness checks

D.1 Distance alternative treatment

In this section, I test whether the post-1453 siege decline in dioceses near Constantinople holds under an alternative proximity measure: the great-circle distance. I define the great-circle distance of a diocese from Constantinople as the length in kilometers of the shortest arc over the Earth’s surface linking the centroid of the diocese to Constantinople. This measure of information exposure explicitly ignores all geographic and social factors save distance. The mean diocese is located 1,809 kilometers from Constantinople. The nearest diocese is Halych (in southwestern Ukraine), located 731 kilometers away, while the farthest diocese is Lisbon, located 3,203 kilometers away.

I conduct the main analysis again, using great circle distance as the treatment. I estimate the following via Poisson pseudo maximum likelihood

$$E[sieges_{dpt} \mid \beta_d, \gamma_t, \delta_{pt}] = \exp(\alpha \text{postFall}_t \times \text{distance}_d + \beta_d + \gamma_t + \delta_{pt})$$

where all coefficients and variables are defined as in Equation 1.

The coefficients on $\text{postFall}_t \times \text{distance}_d$ for all window sizes are plotted in Figure A.15, with the vertical bars representing 95% confidence intervals, computed using robust standard errors clustered by diocese. The coefficients are also reported in tabular form in Table A.16.

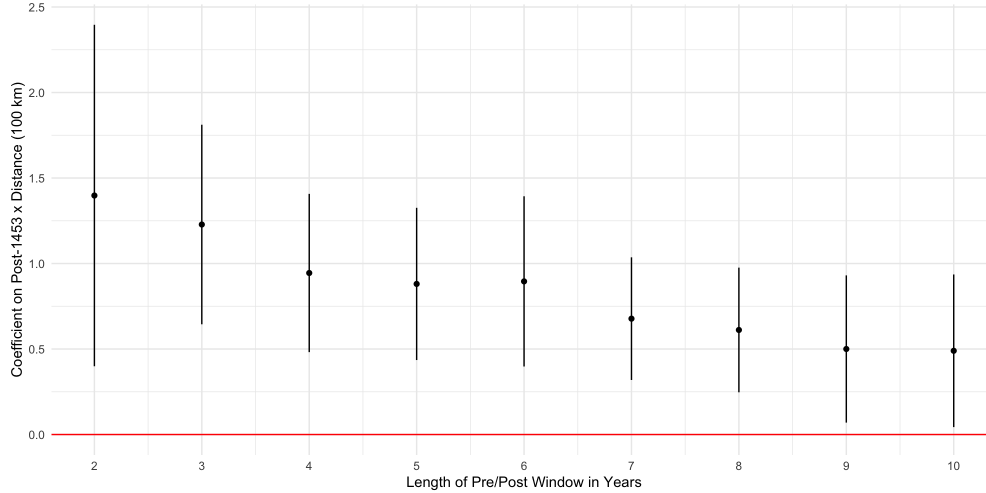


Figure A.15: Effect of distance (100 km) to Constantinople on sieges, all window sizes

As in the main analysis, the coefficients on $\text{postFall}_t \times \text{distance}_d$ are positive and statistically significant. Moreover, we again observe a smaller effect size as the window length increases, consistent with a theory of learning.

	Num. sieges								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 $\times$ Distance (100 km)	1.40*** (0.51)	1.23*** (0.30)	0.94*** (0.24)	0.88*** (0.23)	0.90*** (0.25)	0.68*** (0.18)	0.61*** (0.19)	0.50** (0.22)	0.49** (0.23)
DV Mean:	1.00	0.70	0.59	0.51	0.48	0.42	0.38	0.36	0.36
Polity $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	27	41	57	63	72	88	102	119	125
N	71	142	264	350	423	575	738	932	1019

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.16: Effect of distance (100 km) to Constantinople on sieges, all window sizes

D.2 Travel time alternative treatment

Construction of variable

The other variable I use to operationalize exposure to Constantinople is an estimate of the time it would take to travel from Constantinople using the fastest route available given the modes of transport available at the time. I construct this variable computationally, similar to the procedure for transport costs. As before, I divide Europe into a grid of equally sized square cells. I define $d(i, j)$ to be the great-circle distance, in kilometers, between the centroids of adjacent grid cells i and j and $t(i, j)$ to be the time in hours to travel between cells i and j .

The value of $t(i, j)$ depends on the means of transport used to traverse the cells. I assume that travelers have two options: sailing and walking. To approximate speed of travel for sailing during the 15th century, I use the average speed of 5 knots (9.26 kilometers per hour) offered by Casson (1951), who determined this number by analyzing shipping manifests from antiquity and the Middle Ages.

Computing travel time over land is more involved. To calculate walking speed, I use Tobler's hiking function, which provides an estimate of walking speed given distance and grade, fitted to empirical data on hiking speed collected during the 20th century (Tobler, 1993). Tobler's hiking function for the walking speed between grid cells i and j is

$$v(i, j) = 6e^{-3.5 \left| \frac{h(j) - h(i)}{d(i, j)} + 0.05 \right|}$$

where $h(j) - h(i)$ is the difference in elevation between the centroids of cells j and i , respectively.

Therefore, the travel time $t(i, j)$ needed to move between adjacent grid cells i and j is given by

$$t(i, j) = \mathbf{1}\{i \text{ or } j \text{ on sea}\} \frac{d(i, j)}{9.26} + \mathbf{1}\{i \text{ and } j \text{ on land}\} \frac{d(i, j)}{v(i, j)}$$

If we let $\mathcal{P}_d$ be the set of all paths between Constantinople and a diocese d , then the travel time from Constantinople to that diocese is given by the cost of the shortest path $p \in \mathcal{P}_d$, or

$$\text{travelTime}_d = \min_{p \in \mathcal{P}_d} \sum_{(i, j) \in p} t(i, j)$$

The variable travelTime_d is therefore the minimum time in hours it would take a traveler to move from Constantinople to diocese d , under the speeds estimated above and constant movement. Note that this is certainly an underestimate due to the assumption of uninterrupted travel. Using this measure, the nearest diocese to Constantinople is again Halych, a journey to which would require 281 hours (11.7 days) of uninterrupted travel. The farthest diocese is Moray (in northern Scotland), which requires 1,246 hours of travel (51.9 days). The mean travel time is 721 hours (30 days).

Empirical results

I conduct the main analysis again, using travel time to Constantinople as the treatment. I estimate the following via Poisson pseudo maximum likelihood

$$E[sieges_{dpt} \mid \beta_d, \gamma_t, \delta_{pt}] = \exp(\alpha \text{postFall}_t \times \text{travelTime}_d + \beta_d + \gamma_t + \delta_{pt})$$

where all variables are defined identically to Equation 1.

The coefficients on $\text{postFall}_t \times \text{travelTime}_d$ for all window sizes are plotted in Figure A.16, with the vertical bars representing 95% confidence intervals, computed using robust standard errors clustered by diocese. The coefficients are also reported in tabular form in Table A.17.

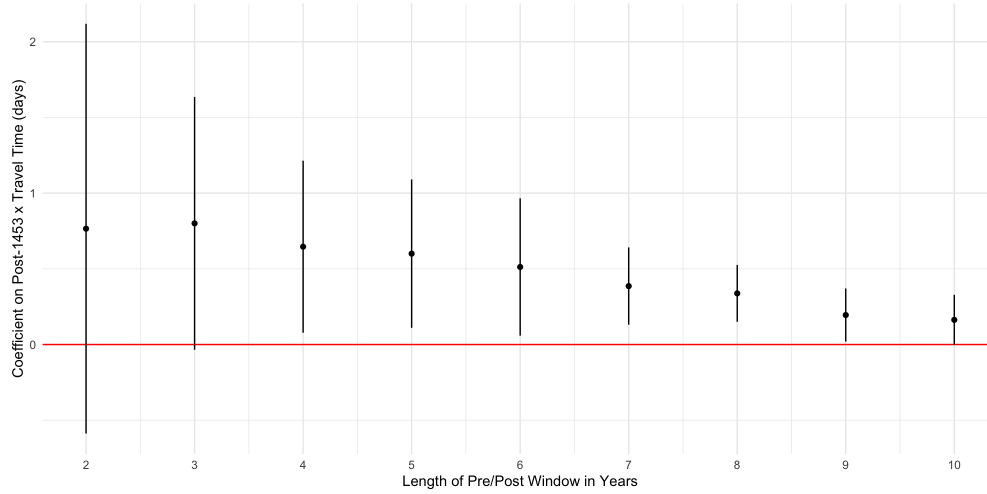


Figure A.16: Effect of travel time (days) to Constantinople on sieges, all window sizes

As in the main analysis, the coefficients on $\text{postFall}_t \times \text{travelTime}_d$ are positive, though somewhat more noisily estimated for smaller window sizes. We again observe a smaller effect size as the window length increases, consistent with a theory of learning.

	Num. sieges								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 × Travel time (days)	0.77 (0.69)	0.80* (0.43)	0.65** (0.29)	0.60** (0.25)	0.51** (0.23)	0.39*** (0.13)	0.34*** (0.10)	0.19** (0.09)	0.16* (0.08)
DV Mean:	1.00	0.70	0.59	0.51	0.48	0.42	0.38	0.36	0.36
Polity × Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	27	41	57	63	72	88	102	119	125
N	71	142	264	350	423	575	738	932	1019

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.17: Effect of proximity to Constantinople on sieges (travel time), all window sizes

D.3 Comparison of proximity measures

Table A.18 displays the correlation between the various measures of exposure to Constantinople. All are positively correlated with each other.

	Transport Cost	Great Circle Distance	Travel Time
Transport Cost	1.00	0.75	0.72
Great Circle Distance	0.75	1.00	0.79
Travel Time	0.72	0.79	1.00

Table A.18: Correlation between proximity measures

Figures A.17 through A.19 depict each diocese's exposure to Constantinople based on the three accessibility measures. The distance measure marks the dioceses at the fringe of the continent as farthest from Constantinople (i.e. the arc from Lisbon to Norway), disregarding topography and travel times. The transport cost measure marks Northern Europe as the farthest from Constantinople, and takes into account the fact that coastal dioceses are generally more accessible than inland ones. Travel time extends this by directly incorporating topography.

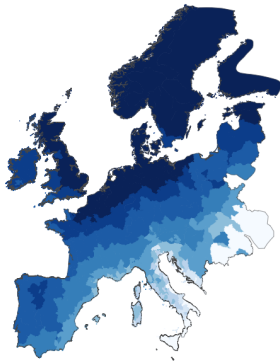


Figure A.17: Transport cost

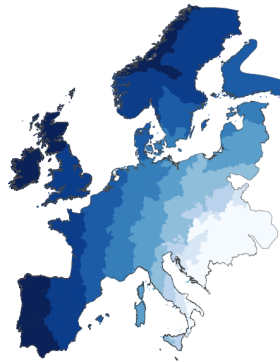


Figure A.18: Distance

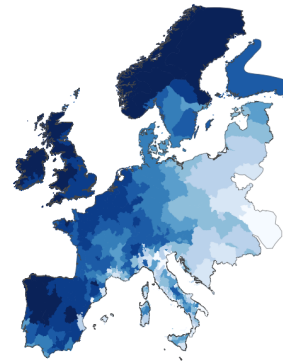


Figure A.19: Travel time

D.4 Standard error adjustments for spatial and temporal correlations

In this section, I report 10 standard error adjustments for the six- and ten-year window models. Sieges were often part of larger conflicts—their occurrence could be correlated with nearby sieges in the same conflict or with future sieges in the same location as strategic fortifications often changed hands multiple times in a conflict. Treating sieges as independent observations could lead to misleadingly small standard errors. I adjust for spatial correlation using Conley standard errors, as well as by clustering standard errors by modern and medieval state as well as by state-year. I also compute Driscoll-Kraay standard errors to address autocorrelation with cross-sectional shocks not captured by the diocese clustering used in the main analysis.

Conley standard errors treat all units within a certain distance from each other as potentially correlated, but this distance must be provided by the researcher. I determine distance cutoffs using two approaches. First, for all wars involving more than a single siege, I compute the radius of the minimum bounding circle that contains all sieges associated with the conflict. This provides a rough approximation of the extent of a war, and therefore how far away potentially related sieges are. Second, since most conflicts during this time took place within polities, I also compute the radius of the minimum bounding circle containing each polity. Figures A.20 and A.21 plot example circles and Table A.20 reports summary statistics for the radii. The circles appear somewhat distorted due to the flattened map projection and Europe’s latitude.

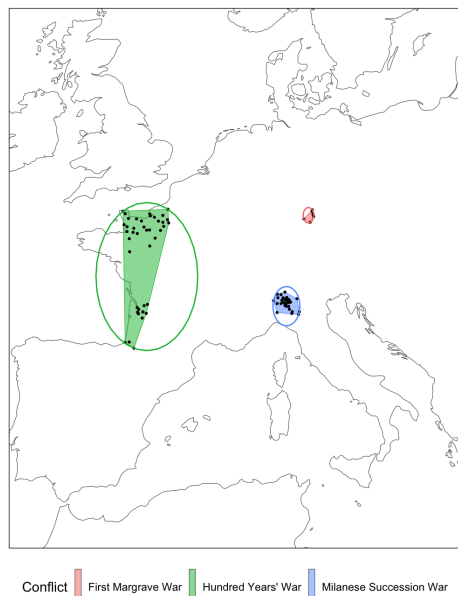


Figure A.20: Sample conflict circles

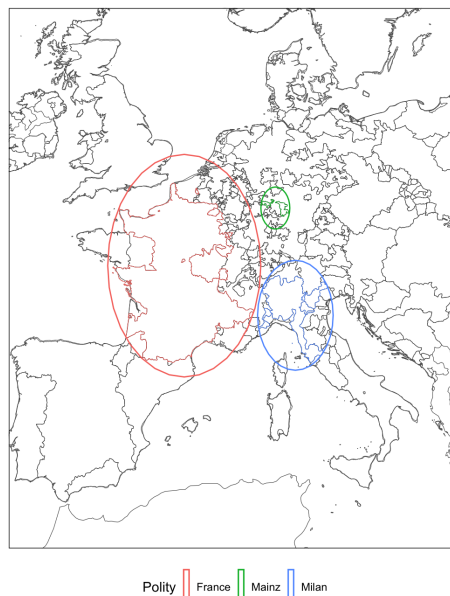


Figure A.21: Sample polity circles

The tables and figures show huge variation in the size of both conflicts and polities. There is no “one-size-fits-all” Conley cutoff that perfectly captures spatial correlation in all cases. A small cutoff like 50 km or 100 km will be sufficient for most wars, but may miss relationships between sieges in very large wars, like the Hundred Years’ War. A large cutoff like 500 km will capture spatial correlation in larger wars, but be too conservative by treating distinct minor conflicts as potentially correlated. Taking these two issues together, a 500 km cutoff would treat the 1455

conflict in Utrecht as potentially correlated with 1440s conflicts in Scotland and Germany, which is too conservative. Therefore, I use six different cutoffs between 50 km and 500 km. These cutoffs are reported in Table A.20, as well as the proportion of conflicts and polities that would be contained within them. All conflicts and all but four polities (Genoa, the Kalmar Union, Poland-Lithuania, and the Ottoman Empire) are contained within a radius of 500 km.

	Conflict Radii (km)	Polity Radii (km)
Min	4.4	5.2
Med.	69.1	85.0
Mean	115.6	161.7
Max	439.1	1381.2

Table A.19: Radii summary stats

Radius Size	Conflicts	Polities
50 km	0.41	0.28
100 km	0.64	0.58
200 km	0.77	0.80
300 km	0.89	0.87
400 km	0.98	0.88
500 km	1.00	0.93

Table A.20: Prop. within radius

Table A.21 displays results from all the standard error adjustments. The coefficients in the six-year and ten-year window models are generally statistically significant. The results offer evidence that the finding of near-term decline in siege warfare in locations more exposed to information about the Fall of Constantinople is robust to adjustments for spatial and temporal correlation in residuals.

	Num. sieges (± 3 years)	Num. sieges (± 3 years)	Num. sieges (± 5 years)	Num. sieges (± 5 years)
Post-1453 $\times$ Transport Cost (1000s)	0.36	1.35	0.36	0.9
<i>Conley (50 km)</i>	(0.17)**	(0.6)**	(0.16)**	(0.4)**
<i>Conley (100 km)</i>	(0.19)*	(0.59)**	(0.19)*	(0.41)**
<i>Conley (200 km)</i>	(0.21)*	(0.57)**	(0.21)*	(0.39)**
<i>Conley (300 km)</i>	(0.17)**	(0.61)**	(0.19)*	(0.36)**
<i>Conley (400 km)</i>	(0.17)**	(0.58)**	(0.18)**	(0.36)**
<i>Conley (500 km)</i>	(0.16)**	(0.44)***	(0.16)**	(0.42)**
<i>Driscoll-Kraay</i>	(0.08)***	(0.18)***	(0.06)***	(0.28)***
<i>Modern state cluster</i>	(0.15)**	(0.4)***	(0.19)*	(0.29)***
<i>Polity 1400 A.D. cluster</i>	(0.16)**	(0.25)***	(0.18)**	(0.22)***
<i>Robust (HC1)</i>	(0.13)***	(0.58)**	(0.11)***	(0.34)***
Polity $\times$ Year FEs	-	✓	-	✓

Table A.21: Standard error adjustments

D.5 Diocese-level control trends

To account for the possibility of time-varying confounders at the diocese level, I include diocese control trends and repeat the main analysis. Specifically, I estimate the following equation using Poisson pseudo-maximum likelihood

$$E[sieges_{dpt} \mid \beta_d, \gamma_t, \delta_{pt}, X_d] = \exp(\alpha \text{post1453}_t \times \text{transportCost}_d + \beta_d + \gamma_t + \delta_{pt} + \lambda^T X_d \times t) \quad (2)$$

where $X_d \times t$ is a vector of pre-1453 diocese-level controls interacted with time. All other variables are defined as in the main analysis. I include four controls: the number of settlements in a diocese with a population of at least 1,000, drawn from Buringh (2021); the number of castles/fortresses extant in 1450, drawn from Cappelen and Hariri (2026), with gaps filled in the Baltic states, Ireland, Slovenia, and Switzerland using data from a scrape of Wikidata; an indicator for whether a diocese is on an overland trade route, georeferenced from a historical atlas by Shepherd (1911), and distance of a diocese from a political border, computed using the historical atlas by Nüssli (2009). Some caution should be taken with the castles data, given that differential coverage due to crowdsourcing may introduce bias. The coefficients are plotted in Figure A.22 and full numerical results are shown in Table A.22.

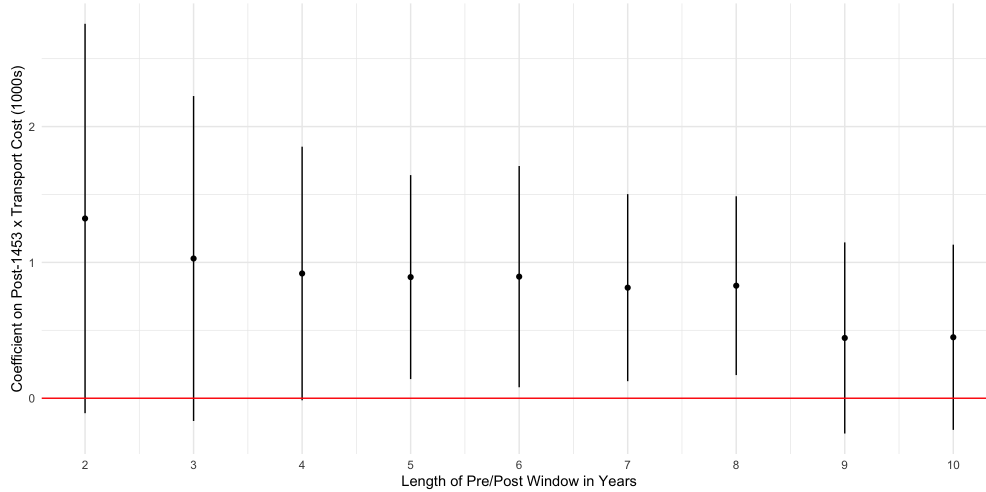


Figure A.22: Effect of transport cost to Constantinople on siege occurrence, control trends

The size and precision of the estimates are similar to those obtained in the models with fixed effects alone, though the estimates on small window sizes with few observations are somewhat noisier, likely due to the loss of degrees of freedom. As before, the effect is largest in small window sizes and trends downward as the window is expanded.

	Num. sieges								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 $\times$ Transport cost (1000s)	1.32* (0.73)	1.03* (0.61)	0.92* (0.48)	0.89** (0.38)	0.90** (0.42)	0.81** (0.35)	0.83** (0.34)	0.44 (0.36)	0.45 (0.35)
Distance from border $\times$ Year	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)
Num. castles $\times$ Year	-0.04 (0.03)	-0.01 (0.01)	-0.01 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)
Num. settlements $\times$ Year	0.06 (0.08)	0.04 (0.03)	0.02* (0.01)	0.00 (0.01)	-0.00 (0.01)	-0.01 (0.01)	-0.01* (0.00)	-0.00 (0.00)	-0.00 (0.00)
On trade route $\times$ Year	0.55 (0.81)	0.55 (0.44)	0.10 (0.27)	-0.10 (0.20)	-0.09 (0.18)	-0.08 (0.11)	-0.03 (0.07)	0.03 (0.05)	0.05 (0.05)
DV Mean:	1.00	0.70	0.59	0.51	0.48	0.42	0.38	0.36	0.36
Polity $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	27	41	57	63	72	88	102	119	125
N	71	142	264	350	423	575	738	932	1019

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.22: Effect of proximity to Constantinople on sieges (transport cost), control trends

D.6 Logistic regression

For this specification, I use as an outcome the binary variable $hadSiege_{dt}$ that takes value 1 if diocese d experienced at least one siege in year t and 0 otherwise. I estimate the likelihood that a siege occurs using a fixed effects logistic regression model with an algorithm devised by Stammann, Heiß and McFadden (2016). The specification takes the form

$$\log \left(\frac{P(hadSiege_{dt} = 1)}{1 - P(hadSiege_{dt} = 1)} \right) = \alpha postFall_t \times transportCost_d + \beta_d + \gamma_t + \delta_{pt}$$

where all other terms are defined identically to Equation 1. Standard errors are clustered at the level of the diocese to account for autocorrelation. I estimate this model for all window sizes 4 to 20 years around the Fall of Constantinople. Coefficients and 95% confidence intervals are plotted in Figure A.23 and displayed in Table A.23.

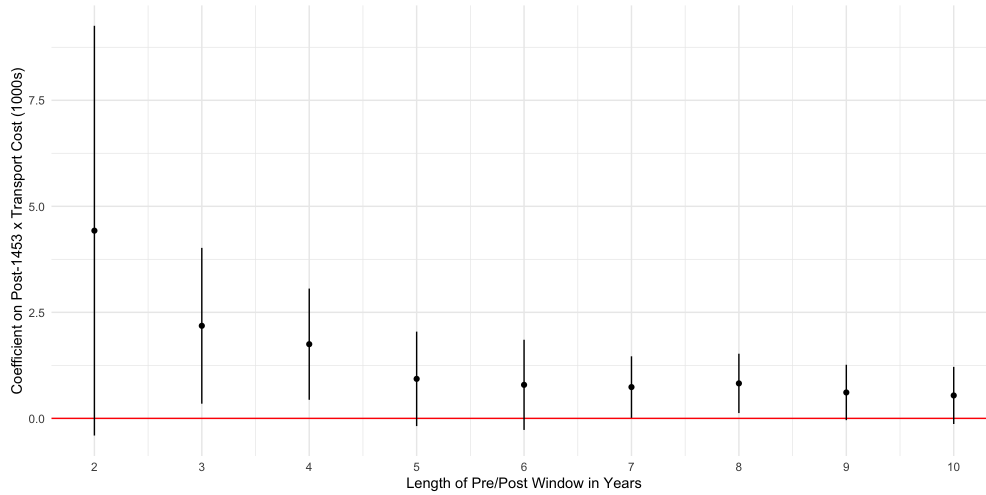


Figure A.23: Effect of transport cost to Constantinople on siege occurrence, all windows (logit)

	Num. sieges								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 × Transport Cost (1000s)	4.43* (2.46)	2.18** (0.94)	1.75*** (0.67)	0.93 (0.57)	0.79 (0.54)	0.74** (0.37)	0.82** (0.36)	0.61* (0.33)	0.54 (0.34)
DV Mean:	0.44	0.35	0.30	0.27	0.26	0.25	0.23	0.23	0.23
Polity × Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	11	25	45	51	57	78	95	115	121
N	32	103	223	306	372	534	703	906	993

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.23: Effect of transport cost to Constantinople on siege occurrence, all windows (logit)

For the six-year window, we see that a one-thousand-unit decrease in transport costs to Constantinople reduced the log-odds of a siege occurring in diocese-year by 2.18 in the three years after the Fall of Constantinople. This corresponds to an average decrease of 31% in the probability of at least one siege occurring in a diocese-year. As before, the effect decreases with window size.

D.7 Ordinary least squares

I estimate the following specification via ordinary least squares

$$\log(\text{sieges}_{dpt} + 1) = \alpha \text{postFall}_t \times \text{transportCost}_d + \beta_d + \gamma_t + \delta_{pt} + \epsilon_{dpt}$$

where $\log(\text{sieges}_{dpt} + 1)$ is the logged count of sieges in diocese d within polity p during period t . The variable postFall_t is a dummy that takes value 1 in the years following the Fall of Constantinople and 0 otherwise. The variable transportCost_d is the transport cost of a diocese from Constantinople. The terms β_d and γ_t are diocese and years fixed effects, respectively. Finally, δ_{pt} is a period $\times$ polity fixed effect. Standard errors are clustered at the level of the diocese to account for autocorrelation. I use the same data employed in the Poisson regressions. The coefficients obtained from estimating this equation are plotted in Figure A.24 and displayed in Table A.24.

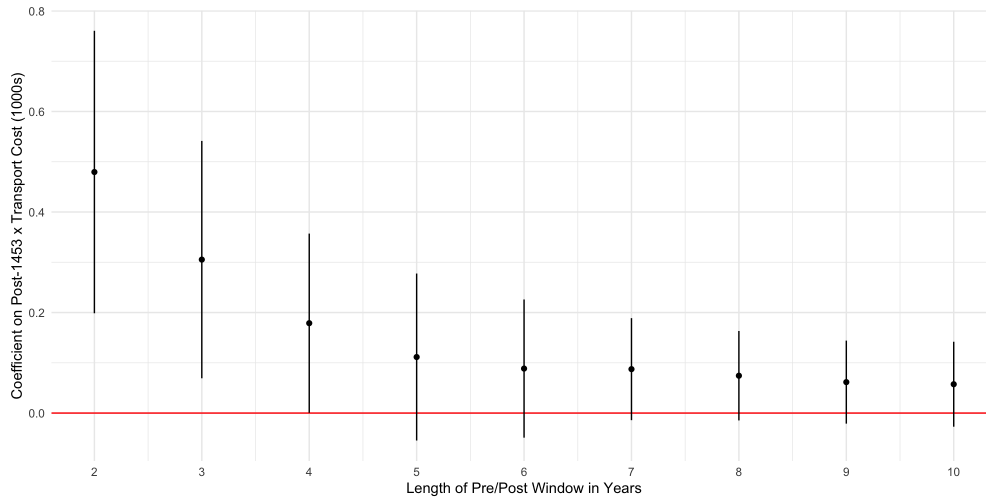


Figure A.24: Effect of transport cost to Constantinople on sieges, all window sizes (linear)

	Num. sieges								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 $\times$ Transport Cost (1000s)	0.48*** (0.14)	0.31** (0.12)	0.18* (0.09)	0.11 (0.08)	0.09 (0.07)	0.09* (0.05)	0.07 (0.05)	0.06 (0.04)	0.06 (0.04)
DV Mean:	1.00	0.70	0.59	0.51	0.48	0.42	0.38	0.36	0.36
Polity $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	27	41	57	63	72	88	102	119	125
N	71	142	264	350	423	575	738	932	1019
R ²	0.72	0.58	0.43	0.42	0.40	0.39	0.36	0.30	0.28

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.24: Effect of proximity to Constantinople on sieges, all window sizes (OLS)

D.8 Poisson with alternative fixed effects

In this section, I conduct the difference-in-differences estimation using polity rather than diocese fixed effects. Doing so ensures that 82% of diocese-years remain in the data set during estimation, but introduces many zero observations, reducing the amount of variation in the data. To carry out the analysis, I estimate the following baseline specification using Poisson pseudo-maximum likelihood:

$$E[sieges_{dt} | \beta_p, \gamma_t] = \exp(\alpha postFall_t \times transportCost_d + \beta_p + \gamma_t)$$

where $sieges_{dt}$ is the count of sieges that took place in diocese d in year t , $transportCost_d$ is the transport cost of diocese d to Constantinople, $postFall_t$ is an indicator function taking 1 in years after 1453, β_p is a polity fixed effect, and γ_t is a year fixed effect. To this baseline specification I also add diocese-level controls, diocese trends, and polity $\times$ year fixed effects. Standard errors are clustered at the level of the diocese. I perform the analysis for six-year and 10-year windows around the Fall of Constantinople.

	Num. sieges (6 year)				Num. sieges (10 year)			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Post-1453 $\times$ Transport cost (1000s)	0.28** (0.12)	0.29** (0.13)	0.67* (0.37)	0.08 (0.29)	0.28** (0.12)	0.30** (0.12)	0.56** (0.27)	-0.05 (0.30)
Transport cost (1000s)	0.07 (0.13)	0.07 (0.11)	170.10 (258.14)	0.12 (0.12)	0.25** (0.12)	0.25** (0.12)	127.78 (129.47)	0.32** (0.14)
Distance from border (km)		0.00 (0.00)	2.37 (1.97)	0.00 (0.00)		0.00 (0.00)	-0.09 (0.73)	0.00 (0.00)
Num. castles		0.03*** (0.01)	16.66** (7.45)	0.03** (0.01)		0.03*** (0.01)	1.51 (4.02)	0.03*** (0.01)
Num. settlements		0.03 (0.02)	-101.77** (47.81)	0.03 (0.02)		0.02 (0.02)	-22.19 (19.66)	0.03 (0.02)
On trade route		0.63 (0.50)	-302.58 (518.51)	0.64 (0.52)		0.72* (0.39)	215.59 (272.84)	0.71* (0.39)
DV Mean:	0.05	0.05	0.05	0.12	0.05	0.05	0.05	0.13
Controls	-	✓	✓	✓	-	✓	✓	✓
Diocese Linear Trend	-	-	✓	-	-	-	✓	-
Polity $\times$ Year FEs	-	-	-	✓	-	-	-	✓
Num. clusters:	358	358	358	355	373	373	373	373
N	2506	2506	2506	972	4103	4103	4103	1536

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.25: Effect of proximity to Constantinople on sieges (Poisson with polity FEs)

The results are reported in Table A.25. Note that for most of the specifications, the coefficient on Post-1453 $\times$ Transport Cost is positive and statistically significant, as in the main analysis, though somewhat smaller. As before, the coefficients are generally smaller in the 10-year window than the 6-year window. However, when including polity $\times$ year fixed effects, the results are no longer statistically significant for either window. This may be a consequence of the many zero observations added, reducing the amount of outcome variation within a polity-year on which to make inference.

D.9 Poisson with correlated random effects

In this section, I estimate several random-effects and multilevel Poisson regression models to account for dropped observations in the fixed-effects models. The baseline equation I estimate is

$$E[sieges_{dt}|\beta_d, \gamma_t] = \exp(\alpha(postFall_t \times transportCost_d) + \beta_d + \gamma_t)$$

where $sieges_{dt}$ is the count of sieges in diocese d in year t , $postFall_t$ is an indicator that takes value 1 in years after 1453 and 0 otherwise, and $transportCost_d$ is the transport cost of diocese d from Constantinople. The variable $\beta_d \sim \mathcal{N}(0, \sigma^2)$ is a diocese random effect that can be correlated within polities. Finally, γ_t is a year fixed effect.

In Figure A.25 and Table A.26 I report coefficients obtained from estimating this equation for all window sizes around Constantinople between 4 years and 20 years. As before, the effect size decreases as window size increases, with the largest effect appearing in the four-year window. In this case, a 1,000-unit decrease in transport costs is associated with a $-100(\exp(-.39)-1) = 32\%$ decrease in siege frequency. The coefficients in all windows are statistically significant and similar in size to the diocese fixed effect models.

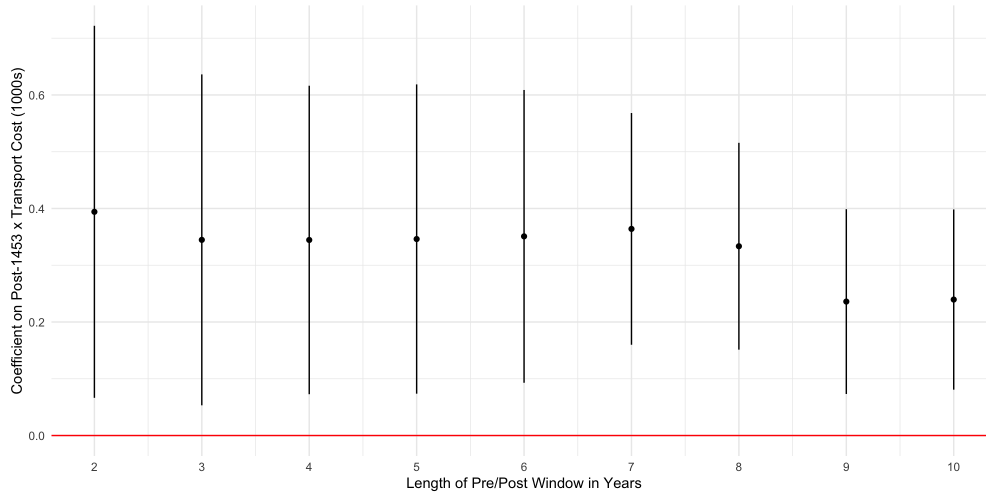


Figure A.25: Effect of transport cost to Constantinople on sieges, all window sizes (random effects)

	Num. sieges								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 × Transport cost (1000s)	0.39** (0.17)	0.34** (0.15)	0.34** (0.14)	0.35** (0.14)	0.35*** (0.13)	0.36*** (0.10)	0.33*** (0.09)	0.24*** (0.08)	0.24*** (0.08)
DV Mean:	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
Correlated Diocese REs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	696	696	696	696	696	696	696	696	696
N	3480	4872	6264	7656	9048	10440	11832	13224	14616

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.26: Correlated random effects models (all windows)

In addition to these models, I also estimate additional models that include polity-year fixed effects. The results are shown in Figure A.26 and Table A.27. As in the main analysis, the polity-year fixed effects increase the size of the coefficient, which remains statistically significant for smaller window sizes.

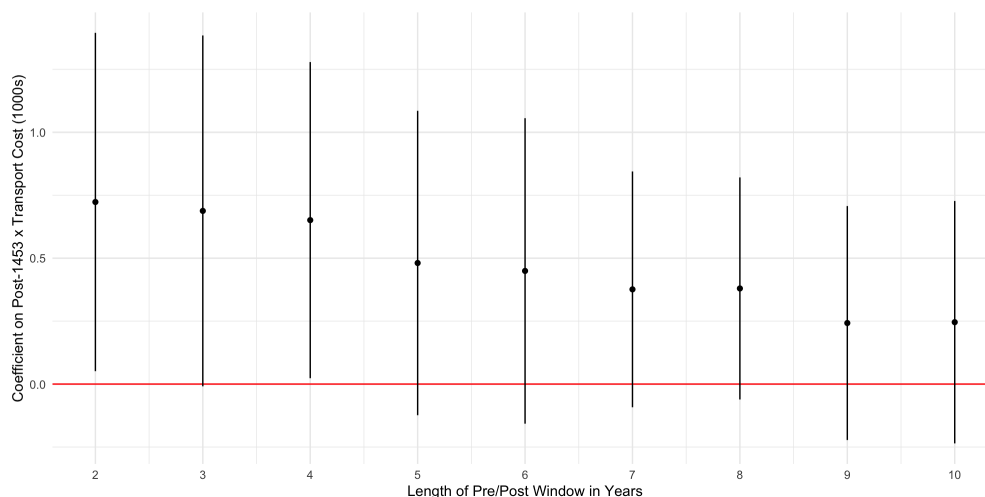


Figure A.26: Effect of transport cost to Constantinople on sieges, all window sizes with polity-year fixed effects (random effects)

	Num. sieges								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 × Tranport cost (1000s)	0.72** (0.34)	0.69* (0.36)	0.65** (0.32)	0.48 (0.31)	0.45 (0.31)	0.38 (0.24)	0.38* (0.22)	0.24 (0.24)	0.25 (0.25)
DV Mean:	0.13	0.12	0.13	0.13	0.13	0.14	0.13	0.13	0.13
Correlated Diocese REs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Polity × year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	325	355	364	370	400	400	553	553	555
N	665	972	1309	1536	1672	1886	2350	2789	2995

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.27: Correlated random effects models with polity-year fixed effects(all windows)

D.10 Leave-one-out analysis

To test how sensitive the results are to outlier polities that had many sieges, I exclude each polity (for polities that had at least one siege—otherwise exclusion has no effect on the estimates) individually from the data and estimate Equation 1 again with the modified dataset for the six- and ten-year windows. The results are displayed below in Figures A.27 and A.28. On the y-axis is the excluded polity with the number of sieges removed in parentheses. On the x-axis is the point estimate and confidence intervals for the coefficient on $\text{Post-1453} \times \text{Transport cost}$. The thick and thin bars represent 90% and 95% confidence intervals, respectively. Tables A.28 and A.29 provide numerical results.

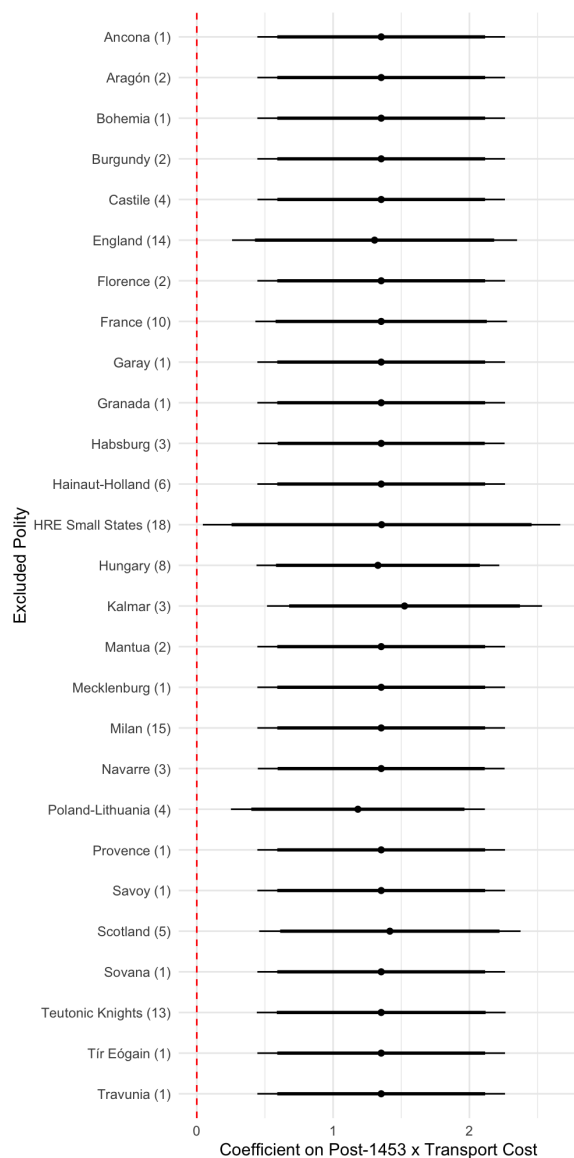


Figure A.27: Leave-one-out (six-year)

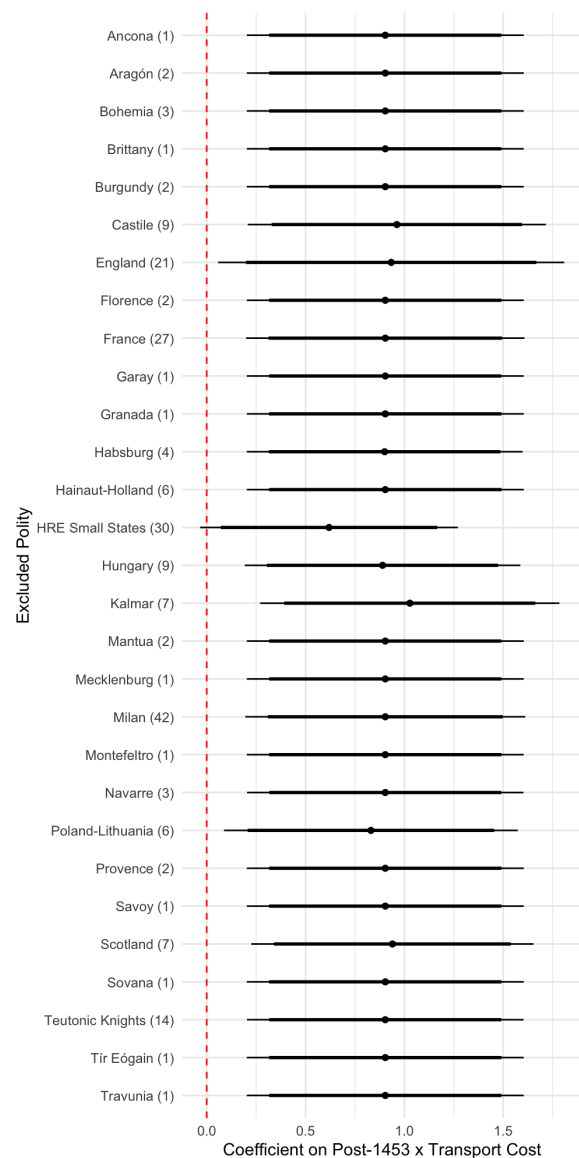


Figure A.28: Leave-one-out (ten-year)

	Num. sieges (six year)															
Ancona Aragón Bohemia Burgundy Castile England Florence France Gany Granada Halbsburg Hainaut-Holland Hungary Kalmar Mantua Mecklenburg Milan Navarre Poland-Lithuania Provence Savoy Scotland HRE Small States Sovania Teutonic Knights Travunia Tir Esgadin																
Post-1453 × Transport cost	1.35*** (0.46)	1.35*** (0.46)	1.35*** (0.46)	1.30** (0.53)	1.35*** (0.46)	1.35*** (0.47)	1.35*** (0.46)	1.35*** (0.46)	1.35*** (0.46)	1.52*** (0.51)	1.35*** (0.46)	1.35*** (0.46)	1.35*** (0.46)	1.42*** (0.49)	1.35*** (0.46)	1.35*** (0.46)
DV Mean:	0.70	0.70	0.70	0.81	0.70	0.73	0.70	0.70	0.70	0.68	0.70	0.70	0.62	0.70	0.73	0.67
Polity × Year FE's	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	41	41	41	38	35	41	35	41	39	41	39	41	38	39	41	37
N	142	142	142	133	106	142	124	142	138	142	136	138	142	142	112	130
N	142	142	142	133	106	142	124	142	138	142	136	138	142	142	112	142

[illegible]

D.11 Aggregation by grid cell and inclusion of Balkan sieges

In this section, I repeat the difference-in-differences analysis while aggregating sieges by $0.5^\circ \times 0.5^\circ$ grid cell rather than diocese. Aggregation by grid cell includes an additional 34 sieges in the Balkans that are omitted in the diocese-level aggregation. These additional sieges are primarily associated with conflicts between the Ottomans and the Albanians, Hungarians, and the Byzantine remnant in the Morea. Their inclusion allows empirical examination of whether the rise of the Ottoman Empire before and after 1453 could serve as a confounding influence on the frequency of siege warfare. A map of sieges by grid cell is displayed in Figure A.29.

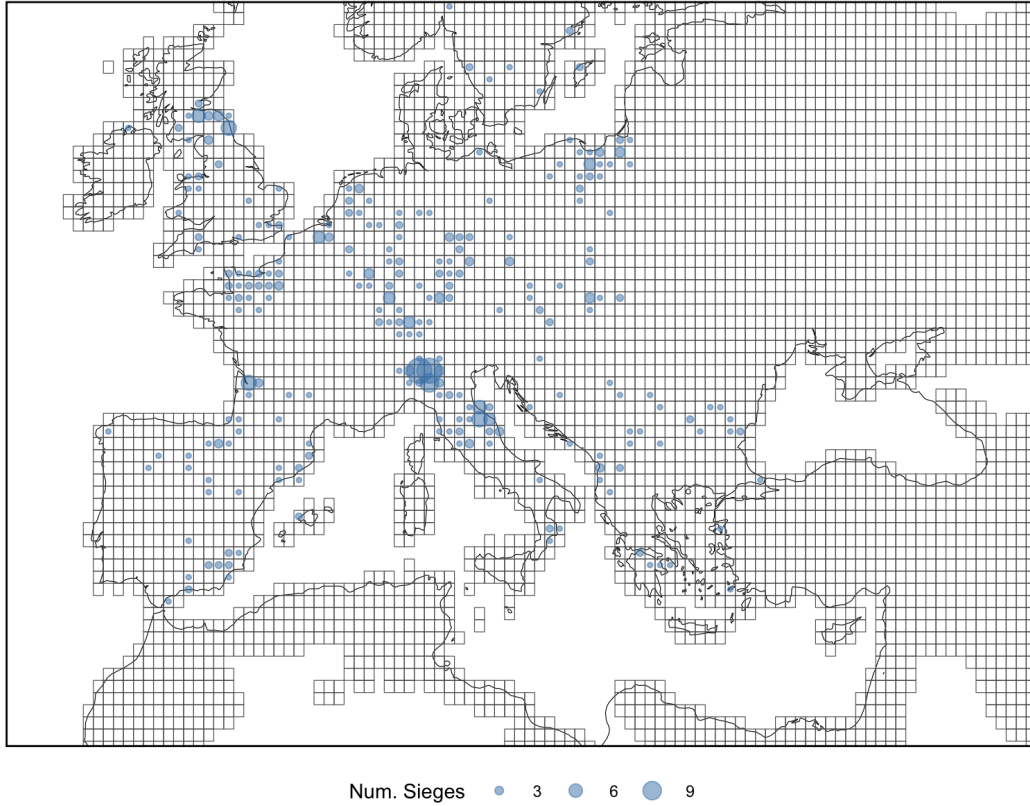


Figure A.29: Sieges by grid cell

I estimate the following equation via Poisson pseudo-maximum likelihood

$$E[sieges_{cpt} | \beta_c, \gamma_t, \delta_{pt}] = \exp(\alpha postFall_t \times transportCost_c + \beta_c + \gamma_t + \delta_{pt})$$

where the outcome $sieges_{cpt}$ is the number of sieges that took place in grid cell c in polity p during year t . The variable $postFall_t$ is an indicator that takes value 1 after 1453 and 0 otherwise, while $transportCost_c$ is the cost of traveling from Constantinople to grid cell c . Finally, β_c , γ_t , δ_{pt} are cell, year, and polity-year fixed effects. In Figure A.30 I plot the values of the coefficient α obtained from estimating this equation for window sizes around 1453 ranging from 4 years to 20 years. The vertical bars represent 95% confidence intervals computed using robust standard errors clustered by grid cell.

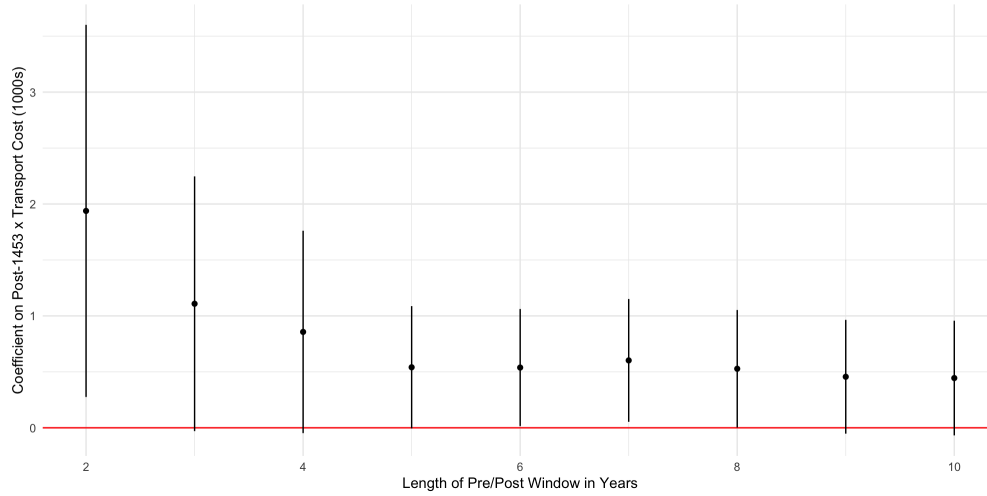


Figure A.30: Effect of transport cost to Constantinople on sieges, all window sizes (grid cells)

Table A.30 reports the numerical results. As in the other analyses, we observe that the effect of proximity to Constantinople on siege frequency is largest when comparing the periods immediately before and after the Fall of Constantinople. In the six-year window, a 1,000-unit decrease in transport costs is associated with a $100(1 - \exp(-1.11)) = 67\%$ decrease in siege frequency.

	Num. sieges (window size)								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 $\times$ Transport cost (1000s)	1.94** (0.85)	1.11* (0.58)	0.86* (0.46)	0.54* (0.28)	0.54** (0.27)	0.60** (0.28)	0.53** (0.27)	0.46* (0.26)	0.44* (0.26)
DV Mean:	0.72	0.49	0.40	0.37	0.35	0.30	0.27	0.26	0.24
Polity $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	49	65	90	107	122	145	167	197	208
N	116	208	382	503	617	827	1086	1377	1548

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by grid cell. Unit of analysis is grid-cell-year.

Table A.30: Effect of proximity to Constantinople on sieges, aggregated by grid cell

D.12 Tipping point analysis

In this section, I conduct an adversarial tipping point analysis in an attempt to quantify how much missing data would be needed to overturn the statistical results reported in this paper. To test the robustness of my findings, I assume that the distribution of the missing data is systematically biased against my hypothesis. Specifically, let X be a continuous random variable describing the distance from Constantinople at which missing sieges occurred in the period after 1453. The variable X has support $[0, \bar{x}] \subseteq \mathbb{R}_+$, where the data are normalized so that the closest diocese to Constantinople has transport cost 0 and $\bar{x}$ denotes the maximum transport cost in the data after normalization. The random variable X has cumulative probability distribution function

$$F(x) = P(X < x) = \frac{1 - e^{-x}}{1 - e^{-\bar{x}}}$$

This means the distribution of X is similar to a truncated exponential distribution, meaning the post-1453 frequency of sieges has an (unconditional) log-linear relationship with transport cost (as assumed in a Poisson regression), but opposite to the direction theorized. This is visualized in Figure A.31, which plots the probability density function, and Figure A.32, which simulates siege data using actual diocese transport costs under the distribution F .

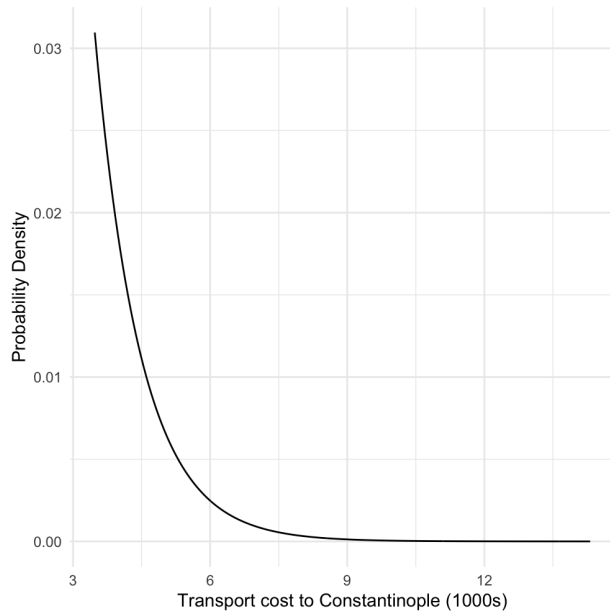


Figure A.31: Adversarial probability density

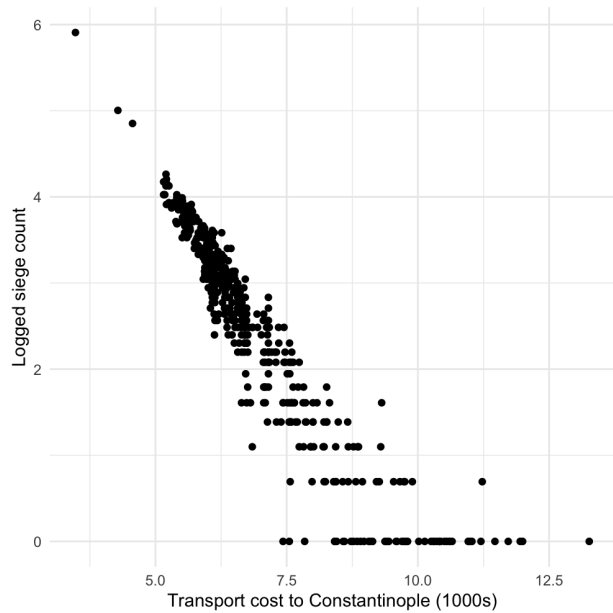


Figure A.32: Simulated sieges log-linear plot

In conducting the tipping point analysis, I model two scenarios: a “highly adversarial” scenario where I allow missing sieges to be uniformly distributed across time, with post-1453 sieges drawn from X and pre-1453 sieges distributed randomly; and a “moderately adversarial” scenario where two-thirds of the missing sieges are generated like the highly adversarial scenario and the remaining one-third are drawn from the same distribution as the actual data collected. The sampling processes for these two scenarios are described in more detail below.

1. **Highly adversarial:** Randomly sample a year between 1443 and 1463. If the year is before 1453, randomly sample a diocese and add a siege. If the year is after 1453, randomly sample a diocese using distribution F and add a siege to it.
2. **Moderately adversarial:** Perform a Bernoulli trial with a probability of success of 1/3. On a success, randomly sample a true siege (with replacement) and duplicate that siege in the dataset. On a failure, generate a siege as in the highly adversarial case.

	Mod.	High
Post-1453 $\times$ Transport Cost	0.62 (0.45)	-1.58** (0.65)
DV Mean:	0.34	0.21
Polity $\times$ Year FEs	✓	✓
Num. clusters:	68	87
N	252	451

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.31: Effect of proximity to Constantinople on sieges in adversarial simulated data

Table A.31 displays the coefficients obtained from estimating Equation 1 on fully simulated data ($n = 397$, equal to the number of true sieges in diocese-covered dataset) for the moderately and highly adversarial scenarios. In the highly adversarial scenario, the coefficient is 10% larger in magnitude than the six-year window estimate in the main analysis, but with the sign flipped.

For each scenario, I conduct 1,000 simulations and record how many sieges are needed for the coefficients on Post-1453 $\times$ Transport Cost to lose significance at the 5% and 10% levels and flip sign. The results are shown in Table A.32, with 95% confidence intervals in brackets.

	New sieges needed (six-year window)	
	Mod.	High
5% Significance	105.68 [99.42-111.94]	70.35 [66.13-74.56]
10% Significance	121.95 [114.67-129.23]	78.25 [73.52-82.97]
Sign Change	796.86 [763.17-830.55]	337.26 [327.61-346.92]

Table A.32: Tipping point analysis results

The simulation results show that for the coefficient to lose significance at the 10% level for the six-year window, about 121 sieges (30.4% of the 1443-1463 data) in the moderately adversarial scenario and 78 sieges (19.6%) in the highly adversarial scenario would need to be missing from the data. For the coefficient to change sign, about 796 sieges (200%) in the moderately adversarial scenario and 337 (85%) sieges in the highly adversarial scenario would need to be missing.

With the use of primary sources, sieges are probably near-exhausted in well-documented regions like England, France, and Italy—it is unlikely that there are 75 missing sieges between

1443 and 1463, let alone 121. Therefore, the simulations suggest that missing data is unlikely to affect my hypothesis.

It is worth acknowledging that the simulation is not perfect. Sieges occur in clusters (wars) rather than randomly across time and space. However, there are unfortunately too few wars to properly model the spatial and temporal dependence of sieges within them to effectively simulate sieges. Moreover, with primary source coverage, it is very unlikely that a war is excluded from the data, particularly large conflicts like the Hundred Years' War or the Thirteen Years' War. Missing sieges are likely minor engagements from larger conflicts or very obscure one-off sieges, making the random selection assumption less problematic.

D.13 Effect of proximity to Constantinople on wars

I conduct a placebo test to quantitatively assess the hypothesis that polities near Constantinople ceased fighting to balance against the Ottoman Empire. I estimate the following fixed-effects Poisson regression:

$$E[wars_{dpt} \mid \beta_d, \gamma_t, \delta_{pt}] = \exp(\alpha \text{post1453}_t \times \text{transportCost}_d + \beta_d + \gamma_t + \delta_{pt})$$

where $wars_{dpt}$ is the number of ongoing wars covering the territory of diocese d within polity p in year t . This variable is constructed by geocoding the extent of conflicts occurring between 1443 and 1463, based on a data set of historical wars collected by Brecke (1999). The remainder of the specification is identical to that used for the main analysis. Because wars occur between and within polities, I cluster standard errors at the level of the polity to account for outcome correlation between dioceses in the same polity. If proximity to Constantinople drove European polities to cease fighting among each, the coefficient on $\text{post1453}_t \times \text{transportCost}_d$ should be positive and statistically significant.

The results are displayed in Table A.33. In all model specifications, the coefficient on $\text{Post-1453} \times \text{Transport Cost}$ is statistically insignificant and near zero in magnitude. This indicates that it is unlikely that proximity to the expanding Ottoman Empire induced polities to cease fighting among themselves and form a balancing coalition. Moreover, together with the main analysis, these results imply that while wars did not become less likely near Constantinople, the strategic significance of sieges within conflicts diminished.

	Num. wars								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 $\times$ Transport cost (1000s)	0.14 (0.24)	0.09 (0.19)	0.11 (0.19)	0.15 (0.13)	0.18 (0.12)	0.15 (0.09)	0.10 (0.07)	0.12 (0.08)	0.13 (0.09)
DV Mean:	1.20	1.27	1.31	1.34	1.41	1.39	1.43	1.42	1.41
Polity $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	36	38	38	38	38	39	39	39	39
N	2066	2753	3412	3933	4533	5372	6259	7282	8305

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by polity. Unit of analysis is diocese-year.

Table A.33: Effect of proximity to Constantinople on wars

E Full tables

E.1 Two-way fixed effects

This section contains the table version of Figure 10.

	Effect on num. sieges by window size								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 $\times$ Transport cost (1000s)	0.41** (0.18)	0.36** (0.16)	0.36** (0.15)	0.36** (0.15)	0.37*** (0.14)	0.39*** (0.11)	0.36*** (0.10)	0.25*** (0.09)	0.25*** (0.09)
DV Mean:	0.43	0.31	0.27	0.23	0.20	0.17	0.15	0.14	0.13
Num. clusters:	45	57	74	81	90	103	121	139	144
N	225	399	666	891	1170	1545	2057	2641	3024

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.34: Effect of proximity to Constantinople on sieges (transport cost), TWFE

E.2 Polity-year fixed effects

This section contains the table version of Figure 11.

	Effect on num. sieges by window size								
	4 yr.	6 yr.	8 yr.	10 yr.	12 yr.	14 yr.	16 yr.	18 yr.	20 yr.
Post-1453 $\times$ Transport cost (1000s)	1.39** (0.60)	1.35*** (0.46)	1.34*** (0.37)	0.90** (0.36)	0.85** (0.37)	0.74*** (0.28)	0.74*** (0.27)	0.51 (0.31)	0.50 (0.32)
DV Mean:	1.00	0.70	0.59	0.51	0.48	0.42	0.38	0.36	0.36
Polity $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	27	41	57	63	72	88	102	119	125
N	71	142	264	350	423	575	738	932	1019

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.35: Effect of proximity to Constantinople on sieges (transport cost), polity-year FEs

E.3 Exclusion of polities involved in simultaneous events

This section contains the table version of Figure 12.

	No Britain			No France			No Italy			None of prior		
	6 yr.	10 yr.	20 yr.	6 yr.	10 yr.	20 yr.	6 yr.	10 yr.	20 yr.	6 yr.	10 yr.	20 yr.
Post-1453 $\times$ Transport Cost (1000s)	1.39** (0.58)	0.98** (0.46)	0.30 (0.37)	1.35*** (0.47)	0.90** (0.36)	0.30 (0.37)	1.35*** (0.46)	0.90** (0.36)	0.30 (0.37)	1.39** (0.61)	0.98** (0.48)	0.33 (0.37)
DV Mean:	0.84	0.59	0.39	0.73	0.48	0.34	0.62	0.45	0.32	0.77	0.45	0.33
Polity $\times$ Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num. clusters:	32	51	106	35	55	114	38	52	94	23	32	64
N	97	257	784	124	318	964	136	306	853	73	181	563

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Note: Robust SEs clustered by diocese. Unit of analysis is diocese-year.

Table A.36: Results when excluding polities involved in simultaneous events

F Siege dataset and sources

F.1 List of primary sources for sieges

This section describes the 42 primary sources used to construct the dataset, organized by region.

Austria, Germany, and Switzerland

Frensdorff, Ferdinand and Matthias von Lexer, eds. “Chronik von der Gründung der Stadt Augsburg bis zum J. 1469.” 1865. In *Die Chroniken der Schwabischen Städte: Augsburg*, Vol. 1, 267-334. Leipzig: G. Hirzel.

- A chronicle of the city of Augsburg from its founding to the year 1469, written by an anonymous chronicler sometime in the 1470s. A useful source for the Bavarian War of 1459-1463, in which Augsburg was one of the principal parties, even being besieged in 1462.

Frensdorff, Ferdinand and Matthias von Lexer, eds. 1866. “Chronik des Burkard Zink, 1368-1468.” In *Die Chroniken der Schwäbischen Städte: Augsburg*. Vol. 2, 1-332. Leipzig: G. Hirzel.

- A chronicle of the city of Augsburg between 1368 and 1468, written by Burkard Zink (1396-c. 1474), a merchant and scholar who resided in Augsburg from 1440. Used as a source for the Bavarian War, the First Margrave War, and the Old Zurich War, among others.

Fründ, Johannes. 1875. *Die Chronik des Hans Fründ, Landschreiber zu Schwytz*, ed. Christian Immanuel Kind. Chur: F. Gengel.

- A chronicle of the Old Zurich War (1440-1446) by Johannes Fründ (1400-1468), who served as the state secretary of the Swiss city of Schwyz. Fründ accompanied the armies of the Old Swiss Confederacy during the Old Zurich War. His chronicle includes eyewitness accounts of several sieges.

Hegel, Karl and Matthias von Lexer, eds. 1862. *Die Chroniken der Fränkischen Städte: Nürnberg*. Vol. 2. Leipzig: G. Hirzel.

- A collection of primary sources from the medieval chronicles of Nuremberg. I consulted Volume 2, which includes sources describing Nuremberg’s campaign against the lord of Waldenfels (the “Waldenfels Feud”) and the war between Nuremberg and Albert III Achilles (the First Margrave War).

Hegel, Karl ed. 1877. “Cronica van der hilliger stat von Coellen bis 1499, zweite hälfte.” In *Die Chroniken der Niederrheinsschen Städte: Cöln*, Vol. 3, 641-918. Leipzig: G. Hirzel.

- The “Chronicle of the Holy City of Cologne,” also known as Koelhoff’s Chronicle, after its printer Johann Koelhoff, is a chronicle of Cologne from “earliest times” to 1499. While chronicle entries before 1445 are drawn from older sources, the chronicle is regarded as a reliable and important source for events in Westphalia in the latter half of the 15th century. Includes accounts of the Cleves-Cologne Conflict, the Soest Feud, and the Mainz Diocesan Feud.

Lamprecht, Karl ed. 1889. *Die Chroniken der Westfälischen und Niederrheinischen Städte, Soest*, Vol. 2. Leipzig: G. Hirzel.

- A collection of primary sources describing the Soest Feud, a conflict in 1444-1449 that began after Soest asserted its independence from Cologne's control. Includes a war diary and two rhymed chronicles, as well as an array of correspondence.

Mone, Franz Joseph ed. 1848. "Speierische Chronik." In *Quellensammlung der badischen Landesgeschichte*, Vol. 1. 67-520. Karlsruhe: G. Madlot.

- A chronicle of events in Bavaria between 1406 and 1476, written by an anonymous author from the city of Speyer (in Rhineland-Palatinate). I use a modern German edition included in a 1848 sourcebook. The chronicle is a useful reference for the Bavarian War that began in 1458 and other conflicts near Germany.

Piccolomini, Aeneas Silvius. 1889. *Die Geschichte Kaiser Friedrichs III*, Vol. 1, ed. and trans. T. Ilgen. Leipzig: Dyk'schen Buchhandlung.

Piccolomini, Aeneas Silvius. 1890. *Die Geschichte Kaiser Friedrichs III*, Vol. 2, ed. and trans. by T. Ilgen. Leipzig: Dyk'schen Buchhandlung.

- A narrative history of the reign of Holy Roman Emperor Frederick III (r. 1452-1493 as emperor, r. 1440-1493 as King of Germany) between 1443 and 1457. Written by Aeneas Silvius Piccolomini (1405-1464), the future Pope Pius II, who served as Frederick's personal secretary between 1443 and 1457.

British Isles

Benet, John. 2016. *John Benet's chronicle, 1399-1462: an English translation with new introduction*, trans. Alison Hanham. Basingstoke: Palgrave Macmillan.

- A short chronicle of events in England between 1399 and 1462, originally written by an unknown author and assembled by John Benet, a vicar of Harlington in Bedfordshire, sometime in the 15th century. The chronicle describes the early phases of the Wars of the Roses and some preceding conflicts, such as Jack Cade's Rebellion and the Bonville-Courtenay Feud.

David, Norman. 1983. *The Paston Letters: A Selection in Modern Spelling*. London: Oxford University Press.

- A collection of letters exchanged by members of the aristocratic Paston family of Norfolk during the late Middle Ages. I use a modern scholarly collection produced by Oxford University Press that includes a subset of letters dated between 1430 and 1504. The letters are the only primary source for the assault on Gresham Castle in 1448/1449, a fortified manor house owned by the Paston family.

Gairdner, James, ed. 1880. "A Brief Latin Chronicle." In *Three Fifteenth-Century Chronicles with Historical Memoranda by John Stowe*, 164-185. Westminster: Camden Society.

- A chronicle of England spanning 1429 to 1471, composed sometime in the late 15th century. I use an 1880 scholarly edition edited by James Gairdner. Has particularly detailed coverage of the Wars of the Roses.

Gairdner, James, ed. 1880. "A Short English Chronicle." In *Three Fifteenth-Century Chronicles with Historical Memoranda by John Stowe*, 1-80. Westminster: Camden Society.

- A chronicle of England between 1189 AD and 1464 AD, with a focus on London and majority of entries being after 1422. Produced in the middle of the 15th century by an unknown author. Covers the first half of the Wars of the Roses and some preceding conflicts, such as Jack Cade's Rebellion.

Gairdner, James, ed. 1880. "Brief Notes of Occurrences Under Henry VI and Edward IV." In *Three Fifteenth-Century Chronicles with Historical Memoranda by John Stowe*, 148-163. Westminster: Camden Society.

- A concise listing of English events by year through the year 1462, with all but six entries coming after the coronation of Henry VI in 1422. Produced in the middle of the 15th century by an unknown author. Covers the first half of the Wars of the Roses and some preceding conflicts, such as Jack Cade's Rebellion.

Gregory, William. 1876. "Chronicle of London." In *The Historical Collections of a Citizen of London in the Fifteenth Century*, ed. James Gairdner, 55-264. Westminster: Camden Society.

- A short chronicle of events in England between 1189 and 1470, written in the late 15th century. The chronicle is traditionally named after William Gregory, Mayor of London between 1450 and 1451, but no historian has been able to conclusively identify an author. The chronicle mentions sieges in the Hundred Years' War, Wars of the Roses, and Anglo-Scottish Border Wars.

Hennessey, William M., ed. and trans. 1871. *The Annals of Loch Cé: A Chronicle of Irish Affairs from A.D. 1014 to A.D. 1590*. Vol. 2. London: Longman & Co.

- A chronological record of events in Connacht (roughly Western Ireland) during the Middle Ages, commissioned by Clan MacDermot in the 16th century and attributed to Clan Ó Duibhgeannáin. Covers conflicts in Ireland during the late Middle Ages, including civil wars and conflicts with the English.

Kingsford, Charles Lethbridge, ed. 1905. "Vitellius A XVI." In *Chronicles of London*, 153-264. Oxford: Clarendon Press.

- A chronicle of London from 1216 to 1516, which concatenates three separate chronicles covering 1216-1440, 1440-1496, and 1496-1516. The portion of interest, covering 1440-1496, was likely composed in the late 15th century. Covers the Jack Cade Rebellion, Hundred Years' War, and Wars of the Roses.

Public Record Office. 1897. *Calendar of the Patent Rolls: Edward IV 1461-1467*. London: The Stationery Office.

Public Record Office. 1900. *Calendar of the Patent Rolls: Edward IV, Henry VI 1467-1477*. London: The Stationery Office.

Public Record Office. 1909. *Calendar of the Patent Rolls: Henry VI 1446-1452*. London: The Stationery Office.

Public Record Office. 1910. *Calendar of the Patent Rolls: Henry VI 1452-1461*. London: The Stationery Office.

- A compilation of letters patent, written orders issued by the monarch of England. The patent rolls were collections of letters patent stored in the Tower of London and Rolls Chapel (now Maughan Library). These records include provisions to fund artillery for sieges, commissions for soldiers to capture rebellious castles, and declarations condemning rebellious castles or towns. I use records from 1446 to 1477, as later records sometimes include payments for past military service in sieges.

Ó Luínín, Ruaidhri. 1895. *Annals of Ulster: A Chronicle of Irish Affairs A.D. 431-1131 : 1155-1541*. Vol. 3. Ed. and trans. by B. MacCarthy. Dublin: The Stationery Office.

- A chronological record of events in Ireland from 431 AD to 1540 AD (with some gaps), compiled in the late 15th century. The chronicle covers most major conflicts in Ireland during the late Middle Ages, including civil wars and conflicts with the English.

Thomson, Thomas, ed. 1819. *The Auchinleck Chronicle: Ane Schort Memorial of the Scottis Corniklis for Addicoun*. Edinburgh.

- A chronicle of the reign of James II, King of Scotland between 1437 and 1460. The original manuscript was written sometime before the early 1500s by an unknown author. The chronicle covers civil conflict in Scotland, including rebellions by the Douglas and Corville clans and the Anglo-Scottish border wars. It also serves as a useful secondary source to cross-reference events in the Wars of the Roses.

Central and Eastern Europe

Imber, Colin. 2006. *The Crusade of Varna, 1443-1445*. Crusade Texts in Translation, vol. 14. Aldershot: Ashgate.

- A collection of primary sources on the Long Campaign and the Crusade of Varna. These conflicts were attempts by a Christian coalition, led by Poland and Hungary to reclaim or damage territories in Bulgaria and Romania that had been conquered by the Ottomans. The largest sources are an anonymous Turkish chronicle titled *The Holy Wars of Sultan Murad Son of Sultan Mehmed Khan* and excerpts from Jehan de Wavrin's chronicle.

Michael, Maurice, trans. 1997. *The Annals of Jan Długosz: Annales seu cronicae incliti regni Poloniae*. Chichester: IM Publications.

- A chronicle of Polish monarchs between 965 A.D. and 1490 A.D. by Jan Długosz, a priest regarded as Poland's first historian. Features descriptions of the Thirteen Years' War, border disputes between Bohemia and Poland, and minor uprisings within Poland-Lithuania.

Piccolomini, Aeneas Silvius. 1998. *Historia Bohemica*. Edited and translated by Dana Martínková, Alena Hadrovová, and Jiří Matl. Prague: KLP.

- A history of Bohemia from its mythological origins to 1456, written by Aeneas Silvius Piccolomini (1405-1464), who traveled to Bohemia on a papal mission before becoming Pope Pius II (r. 1458-1464). Describes conflicts over the throne of Bohemia and the rise of George of Poděbrady. Also describes events in Austria and Hungary, as these realms were in a personal union with Bohemia under Ladislaus the Posthumous (1440-1457).

Thuróczy, János. 1991. *Chronicle of the Hungarians*. Translated by Frank Mantello. Bloomington: Research Institute for Inner Asian Studies.

- A chronicle of Hungary completed in 1488 by János Thuróczy (c. 1435-1488/9). One of the most important sources on medieval Hungary, it includes accounts of the Hungarian-Ottoman Wars and the power struggle between János Hunyadi and Jan Jiskra. Also describes events in Austria and Hungary, as these realms were in a personal union with Bohemia under Ladislaus the Posthumous (1440-1457).

France and the Low Countries

Chartier, Jean. 1858. *Chronique de Charles VII, Roi de France*, Vol. 2, ed. Vallet de Viriville. Paris: P. Jannet.

Chartier, Jean. 1858. *Chronique de Charles VII, Roi de France.*, Vol. 3, ed. Vallet de Viriville. Paris: P. Jannet.

- A chronicle of the reign of Charles VII of France (r. 1422-1461) by Jean Chartier (1385-1464), a monk who served as the official chronicler of Charles VII from 1437. Volume 2 covers 1440-1453 and Volume 3 covers 1453-1461. Chartier accompanied Charles VII on his military campaigns during the Hundred Years' War and contains detailed coverage of several sieges.

Chastellain, Georges. 1863. *Chronique 1430-1431, 1452-1453*. Oeuvres de Georges Chastellain, Vol. 2, ed. Baron Kervyn de Lettenhove. Brussels: F. Heussner.

Chastellain, Georges. 1864. *Chronique 1454-1458*. Oeuvres de Georges Chastellain, Vol. 3, ed. by Baron Kervyn de Lettenhove. Brussels: F. Heussner.

- A chronicle of events in Burgundy during the 15th century by Georges Chastellain (1405?-1475), who served in the court of Philip the Good, Duke of Burgundy. While much of the chronicle has been lost, fragments covering 1452-1458 survive. I use a modern edition published between 1863 and 1865. The chronicle includes descriptions of Burgundy's suppression of a revolt by the city of Ghent and coverage of the final phase of the Hundred Years' War.

Chastellain, Georges. 1825. *Chronique de Jacques de Lalain*, ed. J.A.C Buchon. Paris: Verdière.

- A contemporary biography of the Burgundian knight Jacques de Lalaing (1421-1453) by the Burgundian chronicler Georges Chastellain (1405?-1475) in the service of Philip the Good, Duke of Burgundy. Jacques de Lalaing participated in Burgundy's wars against Ghent and Luxembourg. The chronicle contains descriptions of several sieges associated with these conflicts.

D'Escouchy, Mathieu. 1863. *Chronique de Mathieu D'Escouchy*, Vol. 1, ed. G. du Fresne de Beaucourt. Paris: Jules Renouard.

D'Escouchy, Mathieu. 1864. *Chronique de Mathieu D'Escouchy*, Vol. 2, ed. G. du Fresne de Beaucourt. Paris: Jules Renouard.

- A chronicle of events in France and the Low Countries between 1444 and 1461 by Mathieu d'Escouchy (1420-1482), a Picard chronicler who served in the court of Louis XI of France (r. 1461-1483). Volume 1 covers 1444-1452 and Volume 2 covers 1452-1461. Includes detailed coverage of the Normandy and Gascony campaigns of the Hundred Years' War, the Revolt of Ghent, the Recovery of Luxembourg, and several smaller conflicts.

De la Marche, Olivier. 1884. *Mémoires D'Olivier de la Marche*, Vol. 2, eds. H. Beaune and J. D'Arbaumont. Paris: Renouard.

- A memoir written by Olivier de la Marche (1425-1502), a courtier in the Duchy of Burgundy. I use a modern edition published in 1884. As a young man, De la Marche accompanied the armies of Philip the Good in his conquest of Luxembourg (1443), providing eyewitness accounts of the sieges during that conflict. He also describes Philip the Good's suppression of the revolt of Ghent.

Du Clerq, Jacques. 1826. *Mémoires de Jacques Du Clerq, Escuier, Sieur de Beuvoir en Ternois*. Collection Complète des Mémoires Relatifs a L'Histoire de France, Vol. 11, ed. M. Petitot. Paris: Foucault.

- A chronicle of events in France and the Low Countries between 1448 AD and 1467 AD. The chronicle was written in the 15th century by Jacques Du Clerq (1424-?), about whom little is known. The chronicle includes descriptions of the Gascony and Normandy campaigns at the end of the Hundred Years' War, as well as a brief mention of the Revolt of Ghent.

Van Thabor, Worp. 1850. *Kronijk van Friesland*, Vol. 4, ed. J.G. Ottema. Leeuwarden: G.T.N. Suringar.

- A chronicle of Frisia (a province that takes up much of the northern Netherlands) written between 1517 and 1538 by Worp van Thabor (?-1538), a Frisian monk. I consulted volume 4, which covers 1408 to 1498. The chronicle covers local conflicts in the Low Countries, such as the Donia War and Hook & Cod Wars, as well as interventions from Burgundy and the Holy Roman Empire.

Iberian Peninsula

Generalitat de Catalunya. 1994. *Dietaris de la Generalitat de Catalunya*. Vol. 1, Anys 1411 a 1539. Barcelona: Graficas Almogavares.

- A diary of major events kept by the Generalitat of Catalonia, or the Government of Catalonia, founded in 1359. In ten volumes, the Dietaris cover the period between 1411 and 1719. For this project, I consulted Volume 1, which records happenings between 1411 and 1539. Catalonia was part of the Crown of Aragon from 1137, and the Dietaris are a useful reference for events in Aragon and the greater Iberian Peninsula, including the Catalan Civil War of 1462-1472.

Guzmán, Fernán Pérez de. 1779. *Crónica del Señor Rey Don Juan Segundo*. Edited by Lorenzo Galíndez de Carvajal. Valencia: Imprenta de Benito Monfort.

- A chronicle of the reign of John II, King of Castile and Leon between 1406 and 1454. Attributed to the Castilian historian Fernán Pérez de Guzmán (1376-1458). Written between 1435 and 1454. Covers the Castile-Aragon war and the Castile-Grenada border conflicts.

Palencia, Alfonso Fernández de. 1973. *Crónica de Enrique IV*. Edited by the Spanish Royal Academy. Madrid: Ediciones Atlas.

- A chronicle of the reign of Henry IV, King of Castile and Leon between 1454 and 1474. Written before 1481 by Alfonso de Palencia (1423-1492), who served as royal chronicler and secretary to Henry IV from 1456. Palencia opposed the later policies of the king, and the work is critical of Henry's actions. Covers the Castile-Aragon conflict, the Castile-Grenada border wars, the civil war in Navarra, and the Catalan Civil War.

Pina, Ruy de. 1902. *Chronica de El-Rei D. Affonso V*. Vols. 2 and 3. Lisbon: Escripitorio 147 Rua Dos Retrozeiros.

- A three-volume chronicle of the reign of Afonso V, King of Portugal between 1438 and 1481. Written sometime before 1504 by the chronicler Ruy de Pina (1440-1522). The period between 1443 and 1463 is discussed in the second and third volumes. While there was little internal conflict in Portugal during 1443-1463, Portugal was involved in the wars in Castile and serves as a useful external source.

Italy

Corio, Bernardino. *Storia di Milano: riveduta e annotata*, Vol. 3, ed. Egidio De Magri. Milan: Colombo. 1857.

- A history of Milan from antiquity to 1499, written by Bernardino Corio (1459-1519), a courtier for Galeazzo Sforza, in 1503. Includes detailed coverage of the Milanese Succession War and Francesco Sforza's military service. In Italian.

Massera, Aldo Francesco ed. 1924. "Cronaca Malatestiana dei Secoli XIV e XV." In *Rerum Italicarum Scriptores: Raccolta degli Storici Italiano dal cinquento al millecinquecento, ordinate da L.A. Muratori*, Book 15, Part 2. Bologna: Nicola Zanichelli.

- A chronicle of the House of Malatesta, which ruled the Italian city of Rimini between 1295 and 1500, written by an anonymous author likely associated with the Malatesta court. I consult the second half, covering the years 1416 to 1452, with detailed descriptions of the actions of Sigismondo Pandolfo Malatesta, a condottiero who played an important role in the conflicts of mid-15th century Italy.

Mazzatinti, Giuseppe ed. 1902. "Cronaca di Ser Guerriero da Gubbio dall'anno 1350 all'anno 1472." In *Rerum Italicarum Scriptores: Raccolta degli Storici Italiano dal cinquento al millecinquecento, ordinate da L.A. Muratori*, Book 21, Part 4. Castello: S. Lapi.

- A chronicle of events in Italy between 1350 and 1472, written sometime in the late 15th century by a notary in the city of Gubbio. Includes accounts of the Wars in Lombardy, the Milanese Succession War, and the Rimini conflicts with the Papal States.

Piccolomini, Aeneas Silvius. 2013. *Europe (c. 1400-1458)*, ed. Nancy Bisaha, trans. Robert Brown. Catholic University of America Press: Washington, D.C.

- A history of all of Europe by Aeneas Silvius Piccolomini (1404-1464), also known as Pope Pius II (r. 1458-1464) between 1400 and 1458, composed around 1458. Piccolomini was unusually well-traveled: a native of Siena, he later served in Austria as secretary for Frederick III and traveled to England, Scotland, and Bohemia prior to his election as pope in 1458.

Scalmazzi, Danila. 2019. "Tra Milano e Firenze. Cristoforo Landino volgarizzatore dei *Rerum gestarum Francisci Sphortiae commentarii* di Giovanni Simonetta: Edizione critica della *Sforziada* di Cristoforo Landino." PhD diss. University of Fribourg.

- Also known as *La Sforziada*. A biography of Francesco Sforza, a condottiero who became Duke of Milan in 1450. Originally published in Latin as *Rerum gestarum Francisci Sphortiae commentarii* by Giovanni Simonetta (1420-1491), chancellor to Sforza, and translated into Italian by Cristoforo Landino (1424-1498). Useful as a source for many conflicts in Lombardy, in which Sforza played a major role, from a Milanese perspective.

Soranzo, Giovanni ed. 1915. "Cronaca di Anonimo Veronese, 1446-1488." In *Monumenti Storici Pubblicati Dalla R. Deputazione Veneta di Storia Patria*, Series 3, Vol. 4. Venice: A spese della Societa.

- A chronicle of events in Italy during the second half of the 15th century, written by an anonymous author from the city of Verona, then under Venice. Covers the Wars in Lombardy, the Milanese Succession War, the Neapolitan Succession War and other minor conflicts in Italy from a Venetian perspective.

Scandinavia

Klemming, G.E., ed. 1866. *Nya eller Karls-Krönikan, 1389-1452*. Svenska Medeltidens Rim Kronikor, vol. 2. Stockholm: P.A. Norstedt and Söner.

- A rhymed chronicle written between 1430 and 1452 that covers 1386 to 1452, commissioned by Karl Knutsson (c. 1408-1470), who intermittently reigned as King of Sweden and Norway between 1448 and 1470. While pro-Swedish, a useful source for the Dano-Swedish Wars between Knutsson and Christian I of Denmark.

Klemming, G.E., ed. 1867. *Nya Krönikans Fortsättningar eller Sture-Krönikorna, 1452-1520*. Svenska Medeltidens Rim Kronikor, vol. 3. Stockholm: P.A. Norstedt and Söner.

- Also called the *Sturekrönikan*. A rhymed chronicle commissioned by Sten Sture the Elder (c. 1440-1503), regent of Sweden in 1470-1497 and 1501-1503. Includes descriptions of the Dano-Swedish Wars, the deposition of Karl Knutsson in 1457, and the Swedish revolt against Christian I in 1463.

F.2 Primary source coverage by war

Conflict (Brecke (1999))	Duration	Primary Sources
England-France, 1415-44	1415-1444	Chronique de Charles VII; Chronique de Mathieu D'Escouchy
Swiss Confederation (Zurich), 1436-50	1436-1450	Chronik des Hans Fründ
Castile (nobles), 1441-50	1441-1450	Cronica de Señor Rey Don Juan II; Crónica de Enrique IV
Sforza, Venice, Florence-pope, Naples, Milan, 1442-47	1442-1447	Cronaca di Anonimo Veronese; Cronaca Malatestiana; La Sforziada; Storia di Milano
Hungary, (Poland-Lithuania), pope-Turkey, 1442-48	1442-1448	Annals of Jan Dlugosz; Chronicle of the Hungarians; Imber (2013); Piccolomini's Europe
Emperor-Swiss Confederation, 1442-46	1442-1446	Chronik des Hans Fründ; History of Frederick III
Saxony (Wettin family dispute), 1443-51	1443-1451	Nuremberg Chronicles
France (Armagnac), 1443	1443-1443	Chronique de Charles VII; Chronique de Mathieu D'Escouchy
Albanians-Turkey, 1443-61	1443-1461	Imber (2013); Piccolomini's Europe
Turkey-Venice, 1443-53	1443-1453	Cronaca di Anonimo Veronese; Piccolomini's Europe
France-Swiss, 1444	1444-1444	Chronique de Charles VII; Chronik des Hans Fründ
Turkey-Greece-Hungary (Bulgaria), 1444	1444-1444	Imber (2013); Piccolomini's Europe
Cologne-Westphalian towns, 1446-48	1446-1448	Koelhoff's Chronicle; Soest Chronicles
Hungary-emperor (Austria), 1446	1446-1446	Chronicle of Hungarians; History of Frederick III
Milan (succession, with intervention by Venice, Naples), 1447-50	1447-1450	Cronaca di Anonimo Veronese; Cronaca di SGG; Cronaca Malatestiana; La Sforziada; Piccolomini's Europe; Storia di Milano
Bohemia (religious), 1448-51	1448-1451	Chronicle of the Hungarians; Historia Bohemica
Burgundy (Ghent), 1448-53	1448-1453	Chronique de Jacques de Lalain; Chronique de Mathieu D'Escouchy; Memoires de Olivier de la Marche; Memoires de Jacques du Clerq
Navarre (succession), 1449-52	1449-1452	Cronica de Señor Rey Don Juan II; Crónica de Enrique IV
Brandenburg-Franconian towns, 1449-50	1449-1450	Nuremberg Chronicles; Zink's Chronicle
England-France, 1449-53	1449-1453	Chastellain's Chronicle; Chronique de Charles VII; Chronique de Mathieu D'Escouchy; Memoires de Olivier de la Marche; Memoires de Jacques du Clerq
England (Cade's rebellion), 1450	1450-1450	Benet's Chronicle; Brief Notes; Short English Chronicle; Vitellius Chronicle
Denmark (Sweden, Norway), 1451-57	1451-1457	Karlskronikan; Sturekronikan
Austria, 1451	1451-1451	Annals of Jan Dlugosz; History of Frederick III; Speyer's Chronicle
Hungary, 1451	1451-1451	Annals of Jan Dlugosz; Chronicle of the Hungarians; History of Frederick III
Burgundy-Ghent, 1452-53	1452-1453	Chronique de Jacques de Lalain; Chronique de Mathieu D'Escouchy; Memoires de Olivier de la Marche; Memoires de Jacques du Clerq
Milan, Genoa, Florence, Bologna, Mantua-Venice, Naples, Siena, 1452-54	1452-1454	Cronaca di Anonimo Veronese; Cronaca di SGG; La Sforziada; Storia di Milano
Prussian League, (Poland-Lithuania)-Teutonic Order, 1454-66	1454-1466	Annals of Jan Dlugosz; Speyer's Chronicle
Turkey-Hungary, 1454-56	1454-1456	Chronicle of the Hungarians; History of Frederick III; Piccolomini's Europe
Turkey-Serbia, 1454-59	1454-1459	Chronicle of the Hungarians; Cronaca di Anonimo Veronese; History of Frederick III; Piccolomini's Europe; Speyer Chronicle
Castile-Granada, 1455-57	1455-1457	Cronica de Enrique IV
England (dynastic, nobles), 1455-85	1455-1485	Auchinleck Chronicle; Benet's Chronicle; Brief Latin Chronicle; Brief Notes; Gregory's Chronicle; Short English Chronicle; Vitellius Chronicle
England, 1455	1455-1455	Benet's Chronicle; Brief Latin Chronicle; Brief Notes; Gregory's Chronicle; Short English Chronicle; Vitellius Chronicle
Bohemia-Moravia, Silesia, Lausitz, 1458-60	1458-1460	Annals of Jan Dlugosz
Germany, 1458-63	1458-1463	Augsburg Chronicle; Speyer Chronicle; Zink's Chronicle
England, 1459-61	1459-1461	Benet's Chronicle; Brief Latin Chronicle; Brief Notes; Gregory's Chronicle; Short English Chronicle; Vitellius Chronicle
Denmark-Holstein, 1459-60	1459-1460	Sturekronikan
Naples (succession, Anjou and Aragon), 1460-64	1460-1464	Cronaca di Anonimo Veronese; Cronaca di SGG; Storia di Milano
Scotland-England, 1461-63	1461-1463	Auchinleck Chronicle; Gregory's Chronicle
France, (Genoa), 1461	1461-1461	Cronaca di Anonimo Veronese; Cronaca di SGG
Emperor-Hungary, 1461-62	1461-1462	Chronicle of Hungarians
Aragon (Catalonia), 1461-72	1461-1472	Dietaris Generalitat de Catalunya; Cronica de Enrique IV
Hungary-Turkey, 1462-64	1462-1464	Annals of Jan Dlugosz; Chronicle of the Hungarians
Emperor-Bohemia, 1462	1462-1462	Annals of Jan Dlugosz
Denmark-Sweden (Scania), 1463-65	1463-1465	Sturekronikan

Table A.37: Conflicts from Brecke (1999) by primary source coverage

F.3 Siege coding rules

Events are coded as sieges if a source describes them in one of the following ways.

1. **Explicit use of the word “siege”:** The source calls the event a siege, or uses the verb “besieges” (and equivalents like “put under siege,” “laid siege to,” etc.) and the nouns “besiegers” and/or “besieged.” Descriptions that use the verb “invests” or the noun “investment,” which are the technical terms for the procedure of imposing a military blockade of a site, are also coded as sieges. An example from the Thirteen Years’ War is provided below.

The Poles, in order to prevent the reinforcements arriving, determined at once to **besiege** the city [Marienberg], and on the 27th of February, 1454, a considerable force of Dantzigers and Polish mercenaries, with the necessary artillery, commenced its **investment**. To prevent the town from being cut off from all communication, the knights made constant sorties, and at last broke the lines of the **besiegers**, capturing the guns and trains, with 300 prisoners. The Dantzigers, nothing daunted, having received reinforcements, resumed the **siege**, but were again unsuccessful [sic], their mercenaries, struck with the heroic bravery of the knights, deserting in large numbers (Wyatt, 1876, 168).

2. **Description of siege operations:** The source depicts the process of a siege, without using the terms “siege” or “investment.” This includes descriptions of bombardment, mining, or blockading a fortification (e.g., using the verbs “encircle,” “surround,” etc.). An example from the Hundred Years’ War is provided below.

Avranches held out for three weeks before yielding to Richemont’s mines and cannons, which knocked down stretches of wall. Bayeux, meanwhile, resisted only 15 days. During this time the French **dug trenches, prepared mines and emplaced guns**, and their **bombardment** reduced the town wall, never strong, to rubble (Purton, 2009, 287)

3. **Implicit siege operations:** The source describes a siege without details. That is, it describes a castle or fortified settlement resisting an attacker for an explicitly stated period of time that is at least one day, to distinguish sieges from brief attacks. An example is below.

Athens, an independent Latin duchy since 1204, and in the possession of the Florentine Acciajuoli since 1385, had been ruled by Duke Franco II. When Omer Bey occupied the lower city in 1456, the citizens and their duke fled to the Acropolis, where the ducal palace was situated. **For almost two years they bravely held out, until in June 1458 they surrendered.** The duke, to whom a safe-conduct was granted, was allowed to retain Boeotia, including Thebes, as a vassal of the Porte (Babinger, 1978, 159-160).

Castles/cities only described as being “captured,” “seized,” “taken,” “won,” or similar without details fitting one of the cases above are not coded as sieges. However, if a siege has an attestation fitting one of the main categories, such a source is sometimes used to confirm the date and location. Foreign language sources were translated page-by-page with Google Lens to find potential references to sieges. Potential siege attestations were then verified manually. This is a best-case use scenario for machine translation because all sources are in European languages with a large amount of training data and the translation was into English. These are precisely the conditions where machine translation has been shown to perform best, even on technical medical and scientific text (Ataman et al., 2025; Pang et al., 2025).